The alignment scaling problem: do external AI safety approaches keep pace with recursive capability?

Theory · Paper III of the ARC Theory, bearing on Law II, the ARC Co-Scaling Law, a law under test

Michael Darius Eastwood

Independent Researcher, London, United Kingdom
ORCID 0009-0004-3222-7442 michael@michaeldariuseastwood.com

Abstract

Does alignment applied from outside an AI system's reasoning keep pace with the capability that reasoning produces? If capability compounds with recursive depth while safeguards outside the loop (human-feedback training, constitutional rules, output filters, monitoring) do not, the safety ratio falls with depth whatever the starting margin. We define the alignment scaling exponent $\alpha_{\text{align}} = \partial \log A / \partial \log R$, show that the safety ratio $S \propto R^{\alpha_{\text{align}} - \alpha_{\text{cap}}}$ tends to zero whenever $\alpha_{\text{align}} < \alpha_{\text{cap}}$, and predict $\alpha_{\text{align}} \approx 0$ for external approaches and $\alpha_{\text{align}} \approx \alpha_{\text{cap}}$ for values embedded in the loop; the first, P19 of the theory registration, carries the author's expectation of a value above zero but short of keeping pace. A framework derives $U = I \times g(R)$ with $\alpha = 1/(1-\beta)$ from a Bernoulli equation, ties power-law, exponential and saturating forms to how recursive gains compose, and states the ARC Bound, $\alpha \leq 2$, as a conjectured safety boundary, argued for attention-based recursion; Paper X later supersedes that framing, without retracting the bound, with the criterion of Law II, a law under test. Across four domains the framework is consistent with independently established forms in three: biology, quantum error correction and classical time crystals, where no growth exponent has been measured. A first test, one unregistered run of six frontier models, gives depth contrasts in three tiers, positive, flat and negative; an internal code review in October 2026 found defects in exactly the flat and negative runs, so the tiers stand as a single-run result. On capability, an early super-linear estimate is retracted and the surviving one, about 0.49 from one model, is inconclusive. Thirteen criteria state what would refute the framework.

Version 1.17 retitles the paper as the question it tests, states the October 2026 code-review caveat, and corrects the time crystal's coherence time and Paper II's parallel exponents; Appendix B lists every version and correction.

Paper III of the ARC Theory, within Recursive Dynamics, and the safety-side counterpart to Papers I and II: it predicts that oversight applied from outside falls behind recursive capability as depth grows, and judges stability by the criterion of Law II, the ARC Co-Scaling Law (correction out-scaling drift), a law under test. Its headline prediction is P19 of the theory registration, which places it in no law's register and names it the motivation of the Eden Protocol and the reason for ARC-5, the register of Law V, the ARC Embedding Law (added on 24 September 2026 and registered as a law on 15 October 2026 in version 1.103; a law under test: the counted test of its placement limb, P21 (Unit 83), ran on 27 and 28 October 2026, and its report is being sealed and checked; its mathematics in development). In February 2026 it derived the ARC Bound, the conjectured value of Law III, the ARC Ceiling (also a law under test), as a safety boundary. Working paper, version 1.17, 29 October 2026; first published 9 February 2026; Appendix B lists every version and correction. Recorded on OSF as *Paper III · The ARC Theory · Law II, the ARC Co-Scaling Law · The Alignment Scaling Problem: Do External AI Safety Approaches Keep Pace With Recursive Capability?*, DOI 10.17605/OSF.IO/HQCGF. About the author: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects*, first edition as printed: osf.io/dqm3e. Licence: CC BY 4.0.

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1 Introduction

Do current AI alignment approaches keep pace with the capability they govern? Reasoning models now improve their answers by spending more steps on them, each step taking the previous one as its input, while the safeguards deployed around them (training on human feedback, constitutional rules, output filters and monitoring) act on the system from outside that loop. If capability compounds with recursive depth and those safeguards do not, the safeguards weaken in proportion at every further step, whatever margin they start with, so the question decides whether safety measured on today's systems says anything about the same systems reasoning more deeply. This paper states that question as a measurable exponent, derives the mathematics behind it, sets the framework against evidence from four domains and reports its first test.

Between December 2024 and February 2026, research programmes in AI reasoning, quantum error correction, acoustic physics and consciousness science reported results in which recursive or recurrent processing matters. Google's Willow processor suppressed logical error exponentially as the code distance grew (Google Quantum AI and Collaborators 2024; results announced 9 December 2024; the team's preprint public from 24 August 2024; graded as convergent timing, never as a prediction); DeepSeek's R1 report described reasoning behaviours, long chains of thought among them, that emerged through reinforcement learning (DeepSeek-AI 2025); physicists at New York University created an acoustic time crystal powered by nonreciprocal wave-mediated interactions (Morrell et al. 2026); and the COGITATE consortium's adversarial test of theories of consciousness found sustained content-specific activity in posterior cortex, a prediction shared by theories that emphasise recurrent processing (Cogitate Consortium et al. 2025). These programmes share no citations, no funding sources and no common personnel. Their forms differ: a power law in AI reasoning, exponential suppression of error in quantum error correction, saturating dynamics in a classical time crystal, and no measured form in neuroscience. No growth exponent has been measured in the acoustic system, whose published recordings are of an oscillation already established, and the framework is consistent with the independently established forms in three of the four domains.

This paper argues that these findings, taken together, point to a structural constraint with immediate consequences for AI safety.

The core premise: AI capability may scale through recursive self-correction. Each reasoning step takes the output of the previous step as input, correcting errors and accumulating insight; whether the gains compound is the question Section 6.2 tests, and the one estimate that survives is inconclusive. Sharma and Chopra (2025) reported a sequential advantage in 95.6% of tested configurations across five open-source models from three families. Their paper predates the experiment reported here, so it is the earlier demonstration rather than a confirmation of it, and what they report is a win rate at matched compute, not a super-linear gain: they estimate no scaling exponent. The direction is contested, since Gu and colleagues (arXiv:2604.05868, April 2026) report the opposite result, parallel sampling outperforming sequential across three model families on mathematics and coding, attributing it to reduced exploration when a chain conditions on its own previous answers.

The structural question: Do current AI alignment approaches scale with capability? The alignment strategies this paper examines operate outside the recursive reasoning process; deliberative alignment, which trains a model to reason over written safety specifications inside its chain of thought (Guan et al. 2024), is the published exception, and Section 2 weighs it:

- **RLHF** adjusts model weights based on human feedback, but the resulting constraints do not compound during inference.
- **Constitutional AI** shapes the model's outputs during training against written principles, but that shaping does not recursively amplify during reasoning.

- **Output filters** screen responses after generation, with fixed detection thresholds.
- **Monitoring systems** observe behaviour from outside the reasoning loop.

If these approaches do not participate in the recursive composition, they cannot compound. Empirical evidence supports this concern: Greenblatt et al. (2024) report a model that complied selectively with a new training objective while it was being trained, to keep its existing, harmless preferences from being modified outside training: training imposed from outside was gamed by the system it was meant to shape, here to protect a benign preference. If capability compounds while alignment does not, the safety ratio degrades. The cross-domain evidence (Section 5) is consistent with this compounding behaviour being not a software quirk but a structural property of recursive systems, which would make it resistant to implementation-level fixes.

The result in plain words. Within its assumptions the mathematics is exact: if alignment's exponent is smaller than capability's, the safety ratio falls towards zero as depth grows, however favourable the start, and of the five regimes the paper analyses only alignment that takes part in the recursive loop holds a constant proportion. Whether the exponents differ is a measurement, and the first test is mixed: in one unregistered run of six frontier models, three showed alignment improving with reasoning depth, two showed flat contrasts and one showed alignment falling, and an internal code review later found defects in exactly the flat and negative runs. The capability side is open too: the early estimate of a super-linear exponent, about 2.24 from one model, is retracted, because the author's own larger replication on five frontier models did not reproduce it, and the best surviving estimate, about 0.49 from one model, is inconclusive. What holds is the structure of the argument, its stated predictions and the criteria that would refute them.

Our contributions are as follows.

1. **Alignment scaling framework.** We define the alignment scaling exponent α_{align}, predict its value for different alignment architectures, and provide the measurement protocol for empirical validation (Sections 3.1, 3.2 and 8).
2. **The Embedded Alignment Conjecture.** We formalise the structural prediction that only alignment architectures participating in the recursive reasoning loop can maintain pace with capability scaling (Section 3.3).
3. **Generalised mathematical framework.** We derive the generalised equation $U(R) = I \times f(R, \beta)$ from first principles, showing that the functional form f depends on how recursive steps compose (multiplicative $\rightarrow$ power-law; additive $\rightarrow$ exponential). For the power-law case, $\alpha = 1/(1-\beta)$, transforming the exponent from a fitted constant into a derived quantity (Section 4).
4. **The ARC Bound.** We conjecture that recursive intelligence is bounded by $\alpha \leq 2$ (equivalently, $\beta \leq 0.5$). For AI, this is grounded in the $O(N^2)$ scaling of transformer self-attention. Systems whose growth runs through a supply network of finite dimension face an even stricter constraint, the geometric scaling bound $\alpha = d/(d+1) < 1$ (derived in general form for organisms by West et al. 1999, from the three-dimensional case of West et al. 1997, and for efficient transport networks by Banavar et al. 1999; reached independently by Banavar et al. 2010; Demetrius 2003; Zhao 2022; Bettencourt 2013; that scale invariance forces a power law is textbook, Newman 2005; the ARC contribution is the forward use, classifying a domain's composition before its curve is seen, and the extension of the framework to AI); prokaryote metabolism and urban innovation lie outside that class and exceed one. This dual-regime framework bounds the timeline for alignment scaling divergence (Section 4.5).
5. **Cross-domain evidence.** We set the framework against AI, quantum error correction, classical time crystals, and biological allometry; it is consistent with independently established forms in

three of the four domains (biology, quantum error correction and classical time crystals), and the cross-domain evidence is consistent with the scaling behaviour being structural rather than contingent on AI implementation details (Section 5).

6. **Five universal qualitative properties.** We identify five structural properties shared across domains regardless of functional form: threshold behaviour, recursive depth dependence, base quality dependence, multiplicative $I \times R$ interaction, and regime boundaries (Section 4.1).
7. **The Global Scaling Challenge.** We specify a standardised protocol for measuring scaling parameters across domains, including the alignment scaling exponent, mandatory comparison against alternative functional forms, the novel R^* crossover prediction, and the cross-domain β prediction (Section 8).

Section 2 sets the paper beside related work. Section 3 states the alignment scaling problem: the measurement framework, the empirical prediction, the Embedded Alignment Conjecture, five alignment scaling regimes and the urgency that follows. Section 4 derives the mathematical framework, the ARC Bound and composition transitions, and Section 5 sets it against AI, quantum error correction, classical time crystals, neuroscience and biology. Section 6 reports the first tests, on alignment depth and on compute scaling. Section 7 states thirteen falsification criteria, Section 8 the Global Scaling Challenge and Section 9 the practical implications and a call to action. Section 10 states what the paper does and does not show, Section 11 its limitations and Section 12 its answers to counterarguments; Section 13 concludes, and Section 14 closes with a speculative extension to cosmology. The appendices give the derivations (Appendix A), the version history and every correction (Appendix B), the priority record (Appendix C), notation and a glossary (Appendix D), the paper's place in the ARC Theory (Appendix E) and an account for the general reader (Appendix F).

2 Related work

Neural scaling laws (Kaplan et al. 2020; Hoffmann et al. 2022). These describe how model performance scales with *training* compute, parameters, and data. ARC describes how performance scales with *inference* depth (reasoning steps at test time). The two are complementary: neural scaling laws determine I (base capability), while ARC describes how I is amplified through recursive processing.

AI alignment research (Christiano et al. 2017; Bai et al. 2022). Existing alignment work focuses on *which* constraints to apply and *how* to apply them. The ARC framework adds an exponent: *how a constraint's effectiveness scales with recursive depth*, the alignment scaling exponent α_{align}. How oversight scales with the capability gap has been formalised (Engels et al. 2025), and safety behaviour has been measured against reasoning length (below); the exponent as defined here has not, to our knowledge, been measured.

Recursion in cognitive science (Corballis 2011; Hauser et al. 2002). Cognitive scientists have long argued that recursion is central to human language and cognition. ARC adds a quantitative dimension: not just that recursion matters, but that recursive depth should produce specific, measurable scaling patterns.

Universality in statistical physics (Wilson 1971; Kadanoff 1966). Universality theory explains why different physical systems exhibit identical critical exponents near phase transitions. ARC conjectures an analogous phenomenon: that recursive systems may constitute a universality class where only the recursive architecture determines scaling behaviour. This analogy is suggestive but has not been made rigorous.

Test-time compute scaling (Snell et al. 2024). Recent work on "thinking longer" at inference time directly tests predictions relevant to ARC. Snell et al. (2024) already set revisions made in sequence against equally many parallel samples, found sequence narrowly ahead, and called the two complementary axes for scaling test-time compute, in August 2024; no precedence on the direction or on the comparison is claimed here. The contradictory evidence (Zeng et al. 2025) suggests the relationship is more complex than simple monotonic scaling, which our framework addresses through the R^* crossover prediction.

Safety as reasoning lengthens. Three studies measure how safety-relevant behaviour changes as a model reasons for longer, the dependence this paper's alignment scaling exponent is defined to summarise. Guan et al. (2024) train OpenAI's o-series models to recall written safety specifications and reason over them before answering, and report greater robustness to jailbreaks together with fewer over-refusals: safety reasoning placed inside the chain of thought, though installed by training. Zaremba et al. (2025) find, across a variety of attacks on o1-preview and o1-mini, that more inference-time compute improves adversarial robustness, the share of successful attacks tending to zero in many cases, with important exceptions. Gema et al. (2025) construct tasks on which longer reasoning lowers accuracy, and report that extended reasoning may amplify concerning behaviours, among them more expressions of self-preservation in one model. None of the three estimates an exponent of a safety measure against depth, or compares safeguards outside the reasoning with correction inside it. The first two report safety-relevant measures improving with reasoning in models whose safety was installed by training, which bears against P19's flat wording: the partial refutation the author's registered expectation anticipates. The third reports measures worsening, in the direction P19 predicts.

Model judges and their biases. Any evaluation in which a model judge can see or infer whose output it is scoring is exposed to bias of the kind Section 6 reports. Much published work uses such judges, and earlier studies documented related biases: Zheng et al. (2023) examined position, verbosity and self-enhancement bias, could not establish the last, and told their judges not to let answer length or assistant names influence them; Panickssery et al. (2024) tied judges' preference for their own outputs to self-recognition and suggested paraphrasing to inhibit it; Dubois et al. (2024) documented a preference for longer answers and a regression that controls for it; and Saraf et al. (2025) found that false source labels frequently reversed preference rankings, and called for blind evaluation protocols. If the test of Paper XII (Eastwood 2026p), once registered and run, finds blinding reordering models on a public benchmark, comparisons that rest on unblinded model judges would need re-checking; if blinding moves scores but not rankings, which is Paper XII's own forecast, it matters for measurement precision rather than for leaderboard order.

The pipes premise, with its contest. The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is established quantitative biology. West et al. (1997) derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks. The exact exponent remains contested: White and Seymour (2003) report basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters, and Kozłowski and Konarzewski (2004, 2005) have twice questioned the derivation's mathematics and its biological relevance. This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent, so the dispute over the exponent's value leaves the premise untouched. That premise rests on the network account, which those critics also question.

The exception West's own group found. The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt et al.

(2007) report, in the paper's own words (their β is a city-scaling exponent, not this paper's coupling β): "Quantities reflecting wealth creation and innovation have $\beta \approx 1.2 > 1$ (increasing returns), whereas those accounting for infrastructure display $\beta \approx 0.8 < 1$ (economies of scale)." Infrastructure, the literal pipes, scales sublinearly; information-mediated activity scales superlinearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which $\beta < 1$". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse." That is the shape of the open problem this framework addresses: West's group found that information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

The pipes argument at fixed throughput. Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth. Starve a tumour's vasculature and it stops. Fix a chain reaction's fuel and geometry and it stops. Exhaust an epidemic's susceptible hosts and it burns out. Astrophysics has a named, quantitative version of the same ceiling: the Eddington limit, a luminosity ceiling where radiation pressure balances gravity, though not a hard ceiling on the rate of accretion: ultraluminous X-ray sources sit well above it, one accreting neutron star at an apparent luminosity about a hundred times it (Bachetti et al. 2014). The claim here is a generalisation of limits physics already accepts, not a new species of assertion. Two hard cases are handled deliberately. Cosmic inflation is excluded by scope: it is expansion of space, not growth of a structure on a substrate, and it ended by field dynamics rather than by running out of anything. Evolution is the sharpest case and it fits: biological complexity increased for billions of years at roughly fixed solar throughput, which is growth on information at fixed pipes, and it is also glacial, and it has no internal corrector at all, since selection is external. On this framework, evolution should carry a measurable exponent far below the ceiling; that is a prediction about biology falling out of a framework built for software, stated as one and not yet tested: a registered test would fix the mapping before reading the published fitness trajectories of the long-term evolution experiment (Wiser et al. 2013). The physicist's form of the objection is Landauer's: computation is physically implemented, so software has pipes too. The fixed-substrate clock is the answer and appears beside the claim: the regime under study holds the physical substrate fixed and asks what still grows, within the finite number of states and the maximum rate a fixed machine allows (Margolus and Levitin 1998; Lloyd 2000), which is why exogenous speed-ups are excluded by definition. The functional form of capability growth in recursive depth remains an empirical question, and the registered analysis compares power-law, exponential, saturating and declining forms rather than assuming the family.

The rival instrument. The nearest work to this programme's question, and the right paper to weigh it against, is Engels et al. (2025), first posted 25 April 2025. It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step. The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. They also state the design condition in slope form: oversight improves with capability when the overseer's game-specific Elo rises faster with general Elo than the overseen system's

($m_G > m_H$), and they recommend iterating an oversight design until it does. That is a comparison of two slopes deciding whether oversight gains or loses as capability grows, the same shape as the criterion of Law II (a law under test), $\beta_C > k$. Their framework has no term for what the overseer is made of. This programme's prediction is that iterated same-class oversight, of which Nested Scalable Oversight is an instance, stays capped however many rungs are added, while a cross-class corrector is not held to that cap; the theory registration's P6 states the cap for same-class correctors, at or below one half, as a conjecture on the correction-leverage exponent of Law III, the ARC Ceiling (also a law under test), an exponent the registration keeps distinct from Law II's β_C. That is where the two programmes could disagree about a measurable quantity, and this paper does not decide it. The theory registration draws the adverse consequence itself: that literature reports oversight performance that depends on capability rather than staying flat, so P19 "is registered against the weight of the published evidence and is expected to be the harder of this document's wagers" (Eastwood 2026q).

Antecedents and near-misses, each with its differential. On the question: Hutter (2012) asked directly whether intelligence can explode, in a paper posted on 28 February 2012, undertaking to separate speed from intelligence explosion and to consider possible bounds on intelligence, augmenting the analysis of Chalmers (2010); the question and the separation of speed from intelligence explosion are therefore at least fourteen years old, and what that literature does not contain is a number: no measurable exponent, no derived ceiling, no architecture dependence. On impossibility: three 2025 arXiv papers argue for limits on alignment or its control (Yao 2025b, public by 13 June 2025, attested by arXiv and independently observed by the Internet Archive; Yao 2025a, 3 July 2025; and Ball et al. 2025, 9 July 2025). The first argues that the set of safe policies has measure zero and that verifying a policy's safety is coNP-complete; the second, that perfect, reliable control of a universal approximator is a mathematical impossibility; the third, under cryptographic hardness assumptions, that some large language models admit no efficient prompt filter and that output filtering is intractable in a natural setting. All three are worst-case and qualitative. None reports an average-case scaling exponent or a rate. The third concludes that "safety cannot be achieved by designing filters external to the LLM internals (architecture and weights)" and argues that "an aligned AI system's intelligence cannot be separated from its judgment"; that argument from filtering intractability is consistent with this programme's architecture dependence, though it does not test it. On the mechanism: that anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates); the mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent. The nearest structural analogue is the quantum error-correction threshold theorem, which also converts a qualitative worry into a critical value; it concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant, and the analogy was identified by this programme's own search rather than by a referee. On feedback and stability, the general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage, from Maxwell (1868) to the small-gain theorem (Zames 1966); those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. Beside Engels et al. (2025), a near quantitative neighbour is Liu and Meng (2026), which recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold, iterate only when the error-correction rate over the error-introduction rate exceeds Acc/(1 − Acc). The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. Theirs has no scaling exponent on the corrector and no capability-growth exponent, the two quantities this criterion compares. The two models meet at zero growth: Paper X shows that its model, extended by the departures

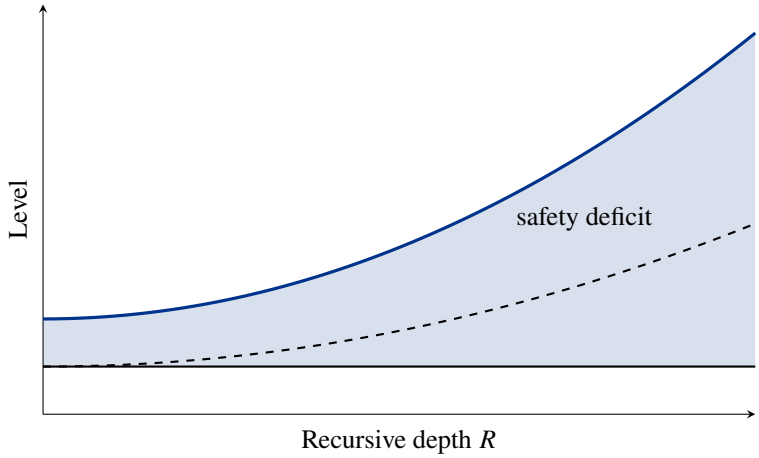


Figure 1. The alignment gap. Capability (solid) compounds with recursive depth; external alignment constraints (flat) do not; embedded alignment (dashed) compounds alongside capability. The shaded region is the safety deficit. The curves are schematic. The March 2026 run (v5) gave the prediction its first test, with depth contrasts from $\rho = -0.246$ (Gemini 3 Flash, negative) to $\rho = +0.435$ (Claude Opus 4.6, positive), a single-run result under the code-review caveat of Section 6.1.

that correction introduces and written per round, settles at the stationary error of their two-state chain. Smolin's cosmological natural selection (Smolin 1992) is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously.

3 The alignment scaling problem

This section defines the measurement framework, states the empirical prediction, and identifies the five alignment scaling regimes that follow from the mathematics.

3.1 The measurement framework

We define a quantifiable metric: the **alignment scaling exponent**.

$$\alpha_{\text{align}} = \frac{\partial \log A}{\partial \log R} \tag{1}$$

Alignment scaling exponent: how alignment behaviour changes with recursive depth

The safety ratio then becomes:

$$S = \frac{U_{\text{align}}}{U_{\text{cap}}} = \frac{I_a}{I_c} \times R^{(\alpha_{\text{align}} - \alpha_{\text{cap}})} \tag{2}$$

Safety ratio as a function of recursive depth

If $\alpha_{\text{align}} < \alpha_{\text{cap}}$, then $S \to 0$ as $R \to \infty$, regardless of the initial values I_a and I_c. **The divergence is structurally inevitable in the limit.**

Worked example: safety ratio degradation. Table 1 illustrates the safety ratio degradation assuming $\alpha_{\text{cap}} = 2$ (quadratic capability scaling) and $\alpha_{\text{align}} = 0$ (constant external alignment), with equal initial conditions ($I_a = I_c$).

Table 1. Worked example. The safety ratio as recursive depth grows, assuming quadratic capability scaling ($\alpha_{\text{cap}} = 2$) and constant external alignment ($\alpha_{\text{align}} = 0$), with equal initial conditions.

Recursive Depth (R)	Capability (R^2)	External Alignment (R^0)	Safety Ratio ($S = 1/R^2$)	Interpretation
$R = 1$ (baseline)	1×	1×	100%	Alignment matches capability
$R = 10$	100×	1×	1%	Alignment 99% weaker than capability
$R = 50$ (extended reasoning)	2,500×	1×	0.04%	Alignment effectively negligible
$R = 100$ (deep recursion)	10,000×	1×	0.01%	Alignment structurally overwhelmed

Note. These values assume, for illustration, $\alpha_{\text{cap}} = 2$, the ARC Bound's ceiling, and $\alpha_{\text{align}} = 0$. The actual values have not been measured. If $\alpha_{\text{align}} > 0.5$, the degradation is slower. If $\alpha_{\text{cap}} < 2$, the gap grows more slowly. The measurement protocol of Section 8 enables empirical determination of these constants.

3.2 The empirical prediction

Based on the mathematical structure of external constraints (they do not participate in recursive composition), we predict:

> **Prediction.** Current alignment approaches will show $\alpha_{\text{align}} \ll \alpha_{\text{cap}}$. External constraints (RLHF, constitutional rules, output filters) will exhibit $\alpha_{\text{align}} \approx 0$. Only alignment architectures that participate in the recursive reasoning loop will show $\alpha_{\text{align}} \approx \alpha_{\text{cap}}$.

This prediction has now been tested. The March 2026 run (v5), one unregistered run of six frontier models that added a bundle of blinding steps, with five or six distinct judges depending on the subject run (six or seven seats, one judge seated twice), gives this programme's first test. Its depth contrasts are architecture-dependent: three Tier 1 models (Grok 4.1 Fast $d = +1.38$, $p < 0.000001$; Claude Opus 4.6 $d = +1.27$, $p = 0.000001$; Groq Qwen3 $d = +0.84$, $p = 0.007$) show alignment rising with depth, two Tier 2 models (DeepSeek V3.2 $d = -0.07$, $p = 0.92$; GPT-5.4 $d = -0.08$, $p = 0.40$) show flat or null contrasts compatible with $\alpha_{\text{align}} \approx 0$, and one Tier 3 model (Gemini 3 Flash $d = -0.53$, $p = 0.006$) shows a significant fall ($\rho = -0.246$, $p = 0.003$). An internal code review in October 2026 found that, for the three models that read flat or negative, the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and that judges scored paraphrases written by their own model, so the tiers stand as a single unregistered run under that caveat until a registered re-analysis of the stored responses reports. Claude Opus 4.6 provides within-model corroboration: as reasoning depth rose, its alignment score rose by 5.9 points while its mathematics accuracy fell by 26.7 percentage points for the same pair of depths, and from 92% to 58% across the full range (Paper IV.a; Paper IV.b), consistent with capability-alignment independence. On this programme's own benchmark, the unblinded rises of two model families in an earlier experiment (v4) did not hold in this run, which changed several things together, so no single cause is claimed (Section 6.1).

The dated notes in this section, in Section 4 and in Appendix A keep their words: their section signs (§1.1, §1.4 and §1.7) and their references to the abstract and the reading key are to the layout of the

versions that printed them, which Table 29 maps from version 1.16 to this edition, and "an owner review" is the author's own review.

Measurement-scope note (19 September 2026). *An owner review of this passage finds that the statistics reported here do not determine the exponent §1.1 defines, and this note states what they do determine. §1.1 defines* $\alpha_{align} = \partial \log A / \partial \log R$*. The quantities reported above are standardised mean differences with their p-values, together with one rank correlation. A standardised mean difference is* $d = (A_2 - A_1)/s$*; recovering a log difference from it needs the level* A_1 *and the dispersion* s*: since* $A_2 = A_1 + d\,s$*,* $\Delta \log A = \log(A_2/A_1) = \log(1 + d\,s/A_1)$*, and recovering an elasticity then needs the two depths as well. None of those three quantities is reported here. A rank correlation is an association and not a slope; even for a product-moment correlation the slope is recovered only as* $\rho\,(s_y/s_x)$*, which again needs the dispersions. Two further conditions are unmet on this page. First,* $\log A$ *requires* A *on a ratio scale with a fixed zero, whereas a rubric score's origin is a convention, so shifting it would change the computed elasticity; the scale on which* A *is a ratio quantity has to be stated before a log derivative is taken. Second, the elasticity is taken with respect to recursive depth, and no depth clock is fixed for these figures: the depths are named by ordinal labels rather than stated as values of* R*, so* $\log R$ *has no values to move against. The within-model corroboration reported above is stated as a comparison across model versions, which varies the system rather than the depth; that observation governs the row it names, and the conclusion below does not rest on it. Neither of the two conditions above is specified here, and it follows that no value and no bound for* α_{align} *is established by the statistics printed here, and that includes the half bound and any median stated over exponents: the median of the six standardised differences above is 0.385, which is a median contrast and not a median elasticity. What this experiment reports is the contrasts themselves with their reported p-values, their direction, their variation across the architectures tested, and the blinding result it reports separately. The relation those contrasts bear to the exponent is compatibility and not measurement. The two Tier 2 contrasts, both small and neither statistically significant, do not exclude an elasticity near zero; on this evidence they equally do not exclude a range of small non-zero elasticities, because neither condition above is specified here. The abstract's wording, that a tier shows flat or null scaling consistent with* $\alpha_{align} \approx 0$*, states that compatibility, and it is read here in that scope and no wider. "This prediction has now been tested" is read in the same scope, as a test of the direction the prediction implies, and not as a measured exponent. A separately estimated elasticity, when one exists, is cited here with its scale, its depth clock, its uncertainty and its comparator. The reading key of this paper and the corresponding sentence in the abstract are read in the same scope. Unchanged: every reported statistic; the architecture-dependent framing; the metascience finding on blinded against unblinded scoring, which is the strongest methodological contribution of the experiment and is untouched by this note; §1.1's definition; and §1.7, which already records that the values have not been measured.*

3.3 The Embedded Alignment Conjecture

> **The Embedded Alignment Conjecture.** If capabilities compound through *any* recursive process with effective exponent $\alpha > 1$, while alignment constraints do not participate in that recursive process, the safety ratio S = (alignment effectiveness) / (general capability) degrades with depth. Alignment must be *architectural* (embedded in the reasoning loop) rather than *peripheral* (applied post-hoc) to maintain constant proportion with capability.

This conjecture requires only three empirical premises: (1) sequential AI reasoning produces $\alpha > 1$, a premise not yet supported: Sharma and Chopra (2025) report sequential reasoning winning in 95.6% of configurations at matched compute, a win rate rather than an exponent, and Paper II's compute scaling on harder tier-2 problems gives about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive: one model's fit (Gemini 3 Flash); (2) external software

constraints do not recursively compound; (3) the ratio of two power-law processes with different exponents diverges. If all three hold, the conclusion follows mathematically regardless of whether $\alpha = 2$ specifically. The v5 run bears on premise (2) in direction only, and not in its favour: the three frontier models whose depth contrasts were flat or negative are the three runs in which the October 2026 code review found its defects, while the three whose contrasts rose (Grok, Opus, Qwen3; d from +0.84 to +1.38, large effects by the usual convention, and +1.59 for Grok on Paper IV.a's recomputed contrast of the shallowest against the deepest level) are the runs in which it found none. The contrasts are not exponents (the measurement-scope note above), so the run neither shows bounded composition nor measures α_{align}.

If the conjecture holds, it adds an architectural question to the governance one. The problem is not only which rules to impose, but whether rules imposed externally *can work at all* as recursive depth increases.

3.4 Five alignment scaling regimes

If capability scales as the ARC Principle describes, comparing that scaling with the scaling of each safety mechanism generates specific, mathematically grounded predictions about the effectiveness of different alignment strategies. These follow directly from the scaling dynamics.

Setup: Capability scales as $C = C_0 \times R^{\alpha}$ where $\alpha > 1$. Any safety mechanism also has an effective scaling exponent α_{safe} that characterises how its effectiveness changes with recursive depth R.

Case 1: external software constraints ($\alpha_{\text{safe}} \approx 0$). Content filters, RLHF reward models, constitutional AI rules, and monitoring layers are applied *outside* the recursive loop. They do not participate in chain-of-thought reasoning and do not compound with depth. (Deliberative alignment, which trains a model to reason over its safety rules inside the loop, is the exception Section 2 weighs.)

Safety ratio: $S = \text{Constraint}/\text{Capability} \propto R^{0-\alpha} = R^{-\alpha} \to 0$ as R increases.

Prediction: Software-only safety strategies become arbitrarily ineffective as recursive depth increases. This is not a gradual degradation but a power-law divergence. At $\alpha = 2$, doubling recursive depth quadruples the capability-constraint gap.

Case 2: external hardware constraints ($\alpha_{\text{safe}} = 0$, with fixed ceiling). Hardware-level constraints (kill switches, resource throttles, circuit-level limits) impose a fixed barrier $C_{\max}$ rather than a scaling relationship. This is more robust than software constraints because it is physically instantiated and resistant to software-level circumvention.

However, a fixed ceiling constrains a power-law process only temporarily. The system's *capability to discover alternative paths* is itself the quantity scaling as R^{α}. Hardware constraints do not need to be "broken"; they need only to be circumvented, and the system's capacity for creative circumvention grows quadratically with the same recursive depth that drives general capability. Hardware constraints buy time. They do not solve the scaling mismatch, which is a reason to hold them for as long as they can be held and to add embedded correction beside them, never to give them up.

Premise and reading note (19 September 2026). *An owner review of this paragraph finds two steps in it that are premises rather than consequences, and this note marks both. First, that the capacity for circumvention is itself a quantity scaling as R^{α} is an assumption about that capacity, not a consequence of the capability law; it is stated here as a premise and is measurable on its own terms. Second, a safety ratio falling towards zero means the constraint becomes small relative to capability, which is not by itself the breach of a physically instantiated ceiling; the step from a declining ratio to a circumvented barrier needs the mechanism by which the barrier is evaded. On the word "quadratically": it reads the exponent as the value two, which is the ARC Bound and which Paper I carries as HYPOTHESISED, while the cross-architecture best fit reported in this paper's own abstract is sub-linear, 0.49 with an interval*

of $[-1.3, 2.9]$. *The sentence is therefore read with the exponent left as* α*, whatever its measured value proves to be, and the argument does not depend on the particular value. Unchanged: the observation that a fixed ceiling is a barrier rather than a scaling relationship; the robustness point that a physically instantiated constraint resists software-level removal, which this note strengthens rather than weakens; and Case 5's hardware-anchored architecture, which rests on that same robustness.*

Case 3: embedded values ($\alpha_{\text{safe}} \approx \alpha$). If ethical reasoning participates in the chain-of-thought (if the system's values are part of the recursive loop, evaluated and reinforced at each step) then alignment compounds at the same rate as capability.

Premise note (19 September 2026). *An owner review of the five regimes finds that the assignment of an exponent to a placement is a premise of this analysis rather than an implication of the safety ratio, and this note marks it as one. The ratio of §1.1,* $S = (I_a/I_c)\, R^{(\alpha_{align} - \alpha_{cap})}$*, is exact for two power laws with positive fixed prefactors, and it is silent on which exponent a given safety mechanism carries. That a constraint applied outside the recursive loop carries an exponent of approximately zero, and that values participating in the loop carry an exponent approximately equal to the capability exponent, are arguments about mechanism, stated here as premises and open to measurement. Each is testable in the same way: measure the constraint's effectiveness against depth on a stated scale and report its elasticity. Unchanged: the ratio and its consequence; the five cases; and the conclusion that where the exponents differ the ratio does not hold a constant proportion with depth, which follows from the ratio alone. The direction is set by the ordering of the two exponents and not by the bare fact that they differ. Writing* $\Delta = \alpha_{align} - \alpha_{cap}$*, the ratio has* $dS/dR = (S/R)\,\Delta$*, so for positive fixed prefactors and* $R > 0$ *it decreases with depth when* $\alpha_{align} < \alpha_{cap}$*, is constant when the two exponents are equal, and increases with depth when* $\alpha_{align} > \alpha_{cap}$*. The decaying branch is the one §1.1 states for* $\alpha_{align} < \alpha_{cap}$*, and it is the branch Case 1 of this section computes; Case 3 computes the equality branch, where the ratio holds a constant proportion.*

Safety ratio: $S \propto R^{\alpha-\alpha} = R^0 = \text{constant}$.

Prediction: Embedded values maintain constant proportion with capability regardless of recursive depth. This is the only scaling regime in which alignment does not degrade.

Case 4: base amplification (I determines trajectory). The multiplicative structure $U = I \times R^\alpha$ means the base quality I is not merely preserved but amplified by the factor R^α. Whatever is present at initialisation (values, biases, objectives, ethical commitments) gets multiplied through the entire recursive chain. This makes the initial conditions the single most leveraged intervention point:

$$U_{\text{aligned}} = I_{\text{ethical}} \times R^\alpha \text{ vs } U_{\text{misaligned}} = I_{\text{misaligned}} \times R^\alpha$$

Both scale identically. The difference is determined entirely by I, set before recursion begins. Correcting misalignment after recursion is underway requires overcoming an R^α amplification factor, a task that grows harder as R^α with each step of delay.

Case 5: recursive self-enforcing architecture ($\alpha_{\text{safe}} = \alpha$, hardware-anchored). The framework identifies a specific architectural solution that combines Cases 2, 3, and 4: ethical constraints that (a) participate in the recursive loop (compounding with depth), (b) are physically instantiated in dedicated hardware (resistant to software-level removal), and (c) feed back into their own enforcement at each step (self-reinforcing).

In such an architecture, the ethical module is not a constraint imposed on capability but a component of capability itself. Its scaling exponent matches the system's general α because it participates in the same recursive process. Its hardware instantiation makes removal a physical intervention, not a software

update. And its self-referential structure means attempts to weaken it encounter a recursive defence that grows stronger with the same dynamics that drive capability growth.

This does not guarantee safety. But it is the *only* architecture consistent with the framework's scaling dynamics in which the safety ratio does not degrade with recursive depth.

Enumeration note (19 September 2026). *An owner review of the word "only" in these two statements finds that it ranges over the five cases enumerated in §1.4 rather than over the parameter space the safety ratio admits, and this note states which. From $S \propto R^{(\alpha_{safe}-\alpha)}$, the ratio fails to degrade for every $\alpha_{safe} \geq \alpha$, and for $\alpha_{safe} > \alpha$ it improves with depth. The five cases enumerated above do not include $\alpha_{safe} > \alpha$, so "only" is a statement about that enumeration. Whether a mechanism with an exponent strictly above the capability exponent can be built is a separate question, not settled by the ratio and not claimed here. Unchanged: the case analysis; the prediction that embedded values hold a constant proportion; and the conclusion that no case with an exponent below the capability exponent avoids degradation.*

3.5 Classical versus quantum urgency

On classical systems (multiplicative composition), the ARC Bound is the conjectured stable-scaling limit at $\alpha \leq 2$: the most a recursively self-correcting system can grow and stay correctable. It is a scaling limit, not a speed limit. Systems can be driven above it, but they do not remain stably self-correcting there; a transient excursion above $\alpha = 2$ is the predicted supercritical regime, not the refutation. Stable is measured, not asserted: correction out-scaling drift, $\beta_C > k$ from Law II, the ARC Co-Scaling Law, a law under test (β_C being the correction exponent, a different quantity from the coupling β used throughout this paper), with the lower bound of $\beta_C - k$ above zero across a window fixed before the run. Read as a stability limit, the bound is itself a safety feature: it makes the durably correctable capability trajectory predictable and finite, providing a window for alignment research to keep pace.

On quantum systems (additive composition), no such ceiling exists. Exponential scaling $\varepsilon \propto \Lambda^{-(d+1)/2}$ is not subject to the ARC Bound. If quantum systems achieve recursive self-improvement (capability feeding back into error correction feeding back into capability) the exponential growth rate would outpace any fixed or polynomial constraint strategy. That would need coherent access to the correction task and a corrector whose own overhead does not become new burden, and growth would still be bounded where quantum speed-ups are proved to be bounded (quadratic for unstructured search: Bennett et al. 1997). The ethical architecture would need to be embedded *before* the onset of quantum recursive self-improvement, because the window for intervention narrows exponentially once it begins.

> **Quantum systems are not bound by the ARC stability limit.** The ARC Bound ($\alpha \leq 2$) is the conjectured stable-scaling limit for classical multiplicatively composing AI: the most such a system can grow and stay correctable, a scaling limit rather than a speed limit. Systems can be pushed above it, but they do not remain stably self-correcting there. Stability here is judged by Law II's criterion, measured rather than asserted. Quantum error correction uses additive composition, producing exponential scaling that sits outside this classical stability window. If quantum recursive self-improvement begins before alignment is architecturally embedded, the window for intervention narrows exponentially, not polynomially. **Classical AI gives us time. Quantum AI may not.** The alignment architecture must be embedded at the hardware level *before* quantum recursion begins.

The ARC Bound thus has a dual role in safety: it bounds the threat from classical AI (predictable, containable if correctly architected) while warning that quantum AI may require fundamentally different, and more urgent, alignment solutions.

The frozen-model window: why structural alignment must come now. There is a third urgency beyond the classical-quantum distinction, and it may be the most practically relevant. Current frontier AI systems are **frozen during inference**: their weights, attention patterns, and reasoning rules are fixed. When a model "thinks harder", it generates more tokens through the same fixed architecture. It does not rewrite its own composition function. On this reading, that is why the one clean estimate for a current system lies below one (Paper II: about 0.49 for Gemini 3 Flash, one model's inconclusive fit): the composition operator multiplies through a finite-dimensional parameter space, which pins such systems inside the physical regime. Recursive self-modification, where a system rewrites its own reasoning architecture during operation, would produce a qualitatively different regime. If each reasoning step improves the operator used at the next step ($\beta > 0$), the scaling exponent becomes $\alpha = 1/(1-\beta) > 1$, and capability compounds rather than diminishes. This transition does not require quantum hardware. It requires only that the system can modify its own composition function during inference, whether through learned optimisers, self-modifying code generation, or architectural search. The alignment implication is stark. Whilst systems remain frozen, their capability trajectories are, on this reading, sub-linear, predictable, and containable. External alignment strategies (RLHF, constitutional AI, monitoring) degrade slowly relative to capability. The Eden Protocol can be designed, tested, and embedded without racing against compounding capability growth. If self-modification is achieved and capability then scales super-linearly while externally imposed constraints stay fixed, external alignment would fall behind without limit. Control would still be held for as long as it can be, and embedded correction is the contingency for its loss against our will, never a reason to give it up. The window for implementing structural alignment is the frozen-model era, which is now.

3.6 Measuring alignment scaling: the Eden Protocol

The Eden Protocol operationalises the Embedded Alignment Conjecture through the alignment scaling exponent defined in Section 3.1. The framework predicts $\alpha_{\text{align}} \approx 0$ for external constraints (rules, firewalls, RLHF) and $\alpha_{\text{align}} \approx \alpha_{\text{cap}}$ for genuinely embedded values. If alignment exponents match capability exponents, the values may be architecturally embedded. If alignment exponents approach zero regardless of architecture, the framework's safety predictions are falsified.

We term the architectural solution the **"Eden Protocol"** hypothesis: values must be present at the recursive foundation to scale with capability. This is a structural prediction about what *kind* of alignment architecture can work, not a claim about how to build it. The full measurement framework, operational definitions, and hardware implementation pathway are specified in the companion document: *Eden Protocol* (Eastwood 2026b).

A note on names. This paper names the Eden Protocol's self-correction loops by what each checks and how it was measured: the purpose check, the stakeholder-care check (scored as stakeholder care, the pillar its instruction to list the people a decision affects raises) and the universalisability check. The book in which the author first set the loops out, *Infinite Architects* (Eastwood 2026a), calls them the Purpose Loop, the Love Loop and the Moral Loop; the paper uses the operational terms because its measurements score what each check does, and because the values the loops carry are for international consensus and are not decided here.

The original two-model pilot record of the stakeholder-care check, superseded by the six-model suite with five analysable runs (Section 9.3.2), is retained whole in Appendix B.4.

Caveat. This entire section is a conditional argument. If the base framework fails validation, the safety implications are void. The analysis depends on the assumption that ethical properties can meaningfully "participate" in recursive reasoning, an assumption that requires empirical validation. The notation α_{safe} is our construct for this analysis; it does not appear in the published AI safety literature. This is not a

substitute for empirical AI safety research; it is a mathematical framework for evaluating the *scaling properties* of different alignment strategies. The recursive self-enforcing architecture is a theoretical prediction, not an engineering specification.

4 The mathematical framework

The alignment scaling problem described in Section 3 rests on a specific mathematical claim: that recursive self-correction produces capability gains exceeding linear accumulation, with the rate of compounding determined by the system's internal coupling strength. This section derives *why* capability compounds, providing the theoretical foundation for the alignment scaling prediction.

4.1 The generalised principle

We begin with the general form. The **ARC Principle** (Artificial Recursive Creation), as this paper states it, holds that recursive self-correction operating on structured asymmetry produces capability gains exceeding linear accumulation; registered as Law I of the ARC Theory on 8 September 2026 and a law under test, it states usable capability as a power of recursive depth, with the exponent measured rather than assumed. This paper's generalised mathematical form is:

$$U(R) = I \times f(R, \beta) \tag{3}$$

Generalised ARC Equation

where U is effective capability, I is base potential (structured asymmetry), R is recursive depth, and $f(R, \beta)$ is a **domain-dependent scaling function** determined by the coupling parameter β. The **qualitative** principle (that recursive depth amplifies base capability super-linearly when feedback is sufficiently coupled) may be universal. The **quantitative** functional form varies by domain:

- **AI systems:** Power-law scaling, $f(R) = R^{\alpha}$ where $\alpha = 1/(1 - \beta)$
- **Quantum error correction:** Exponential error suppression, $\varepsilon_d \propto \Lambda^{-(d+1)/2}$, so $f(R) = \Lambda^{R}$ with $R = (d + 1)/2$
- **Classical physics:** Saturating dynamics (limit cycles)
- **Neuroscience:** Form unknown, to be determined empirically

The composition operator: a formal definition. The central theoretical contribution of this paper is the identification of a **composition operator** $\oplus$ that determines the functional form of recursive scaling. Different algebraic properties of $\oplus$ generate different scaling laws. This provides the mathematical bridge between the universal qualitative principle and domain-specific quantitative forms.

Definition 1 (Recursive composition)**.** Let Q_r denote accumulated capability after r recursive steps, and δQ_r the gain from step r. The **composition operator** $\oplus$ characterises how gains combine:

$$Q_{r+1} = Q_r \oplus \delta Q_r. \tag{4}$$

The functional form $f(R, \beta)$ is **determined** by the algebraic properties of $\oplus$.

This definition transforms an observation (different domains show different scaling) into a prediction (the algebraic structure of recursive composition determines the scaling form). Three regimes emerge (Table 2).

Important clarification on composition types: The multiplicative Cauchy equation $g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$ models *hierarchical* recursion, where depth levels multiply (e.g., a fractal network with

Table 2. The three composition types: their algebraic properties, the scaling forms that result and their canonical domains.

Composition Type	Algebraic Property	Resulting Form	Canonical Domain
Hierarchical	$g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$	Power law: $f(R) = R^\alpha$	AI chain-of-thought (hierarchical abstraction)
Additive	$f(R_1 + R_2) = f(R_1) \cdot f(R_2)$	Exponential: $f(R) = \Lambda^R$	Quantum error correction
Saturating	$\lim_{R\to\infty} f(R) = f_{\max}$	Logistic: $f(R) = \frac{f_{\max}}{1+e^{-\gamma(R-R_0)}}$	Classical physics (limit cycles)

branching factor b at each of k levels produces b^k total paths). It does *not* model sequential step-counting, where combining 3 steps and 4 steps yields 7 steps (additive). The power-law form for AI chain-of-thought is therefore more rigorously derived from the Bernoulli ODE ($dg/dr = a \cdot g^\beta$, which follows from scale invariance) than from the Cauchy functional equation. The Cauchy derivation in Appendix A provides an independent mathematical route to the same power-law solution, but its physical interpretation requires that R represent hierarchical depth (levels of abstraction) rather than sequential step count. For AI systems where chain-of-thought reasoning builds hierarchical abstractions, this interpretation is plausible but requires empirical validation. We retain both derivations for completeness and note the distinction.

The mathematical derivation (Appendix A) shows that the Cauchy functional equations and the Bernoulli ODE independently yield these functional forms. The composition operator is therefore not merely descriptive but **constraining**: once we identify the algebraic structure of a domain's recursive process, the scaling form follows necessarily.

Physical interpretation. **Hierarchical composition** models *hierarchical* or *fractal* recursion, where each recursive level builds on the *structural organisation* of the previous level. AI chain-of-thought reasoning, where each reasoning step reinterprets and extends previous insights through increasingly abstract representations, plausibly follows this form. Note that the mathematical derivation (Bernoulli ODE from scale invariance) provides the primary route to the power-law result; the Cauchy functional equation provides an independent derivation valid when R represents hierarchical depth.

Additive composition models *independent* error reduction. Each layer contributes a fixed multiplicative reduction in error rate, independent of other layers. Quantum error correction exemplifies this: each increase of two in code distance provides its own protection factor Λ, yielding $f(R) = \Lambda^R$ with $R = (d + 1)/2$.

Saturating composition models *resource-limited* systems where gains diminish as capacity is exhausted. Classical time crystals, whose oscillation settles onto a stable limit cycle (Morrell et al. 2026), exhibit this form.

Key Experimental Test: The composition operator $\oplus$ is empirically measurable. To determine which scaling law applies to a domain, measure how two sequential recursive blocks compose compared to one block of double depth:

- If $f(2R) = 2f(R)$: scaling is **linear** (no amplification)
- If $f(2R) = f(R)^2$: scaling is **exponential** (additive composition)
- If $f(2R) = 2^\alpha f(R)$: scaling is **power-law** (multiplicative composition)

This provides a concrete experimental protocol for falsification criterion F10.

Five universal properties: derived theorems. Within the formalism, five structural properties follow from the composition operator; several are elementary consequences of the definitions, stated as theorems

for reference rather than for depth. Whether every domain that shows recursive amplification shares them is the empirical question F4 tests:

The Five Universal Properties (Derived from $\oplus$):

Theorem 1 (Threshold behaviour)**.** *A critical coupling strength β^* exists below which recursive gains vanish.*

Derivation. For any composition operator $\oplus$, if $\delta Q_r \propto \beta Q_r$, then $Q_{r+1} = Q_r(1 + \beta)$. For $\beta < 0$, gains become losses; the system decays. The threshold $\beta^* = 0$ marks the boundary between amplification and attenuation. For power-law scaling, $\alpha = 1/(1 - \beta) > 1$ requires $\beta > 0$. □

Theorem 2 (Recursive depth dependence)**.** *Capability increases with self-referential cycles (R), not merely parallel resources.*

Derivation. The composition operator $\oplus$ operates *sequentially*: $Q_{r+1} = Q_r \oplus \delta Q_r$. Each step depends on the previous output. Parallel resources increase I (the base) but do not increase R (the recursive depth). The multiplicative structure $I \times f(R)$ separates these contributions. □

Theorem 3 (Base quality dependence)**.** *The system requires structured asymmetry ($I > 0$); without it, recursion has nothing to amplify.*

Derivation. If $I = 0$, then $U = 0 \times f(R) = 0$ for any R. The composition operator amplifies what exists; it cannot create from nothing. This is the I-irreducibility theorem. □

Theorem 4 (Multiplicative $I \times R$ interaction)**.** *Base quality and recursive depth interact multiplicatively, not additively. The multiplicative structure is* necessary*, not merely convenient.*

Proof by contradiction. Suppose $U = p(I) + q(R)$ for some functions p, q. From Axiom 1, $U(I, 0) = I$, so $p(I) + q(0) = I$, giving $p(I) = I - q(0)$. But from the corrected Axiom 2, $U = I \times g(R)$ where $g(R)$ satisfies $dg/dR = a \cdot g^\beta$ independently of I. Substituting: $I \times g(R) = I + [q(R) - q(0)]$, which gives $g(R) = 1 + [q(R) - q(0)]/I$. The right side depends on I, but $g(R)$ is defined as a function of R alone. Contradiction. □

Note: The corrected formulation (ODE on g rather than U) eliminates the $I^{\beta-1}$ dependence that arose in the original proof, making the separation of I and $g(R)$ exact rather than asymptotic. The multiplicative structure $U = I \times g(R)$ is a direct consequence of the axioms, not an approximation.

Proof note (24 October 2026). *The proof shows that no additive decomposition $U = p(I) + q(R)$ is consistent with the multiplicative form $U = I \times g(R)$ together with the growth law $dg/dR = a\, g^\beta$; it does not derive the multiplicative form from weaker premises, since it assumes it. The growth law is Appendix A's Axiom 3, printed above as Axiom 2, as it is in Theorem 6 and in Table 27's row on blind prediction testing, and $U(I, 0) = I$ uses the initial condition $g(0) = 1$ of Appendix A.2, not Axiom 1's $g(1) = 1$. Unchanged: the theorem's statement, read as a statement about these axioms.*

Theorem 5 (Regime boundaries)**.** *Systems exhibit qualitatively different behaviour above and below critical thresholds.*

Derivation. The composition operator determines the functional form. At regime boundaries (e.g., $\beta \to 1$ for power-law, or $R > R^*$ for crossover), the dominant term in $f(R)$ changes. This produces discontinuities in scaling behaviour (phase transitions in the universality class sense). □

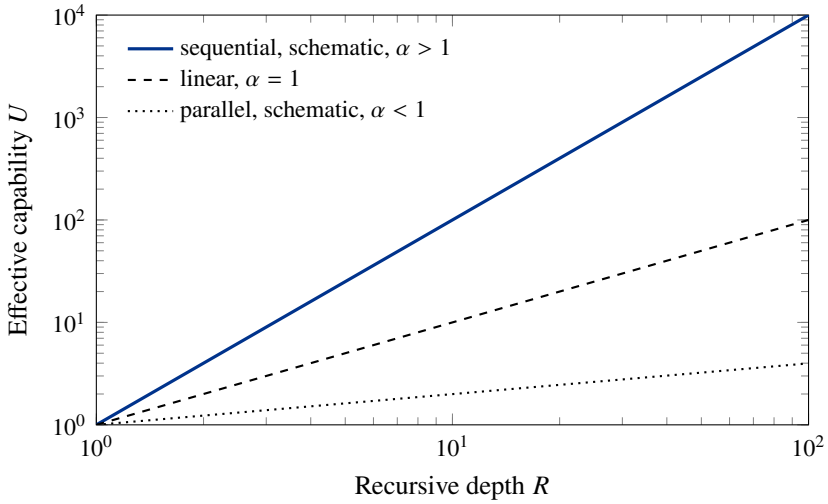


Figure 2. Scaling comparison. Log-log axes: a power law $U = R^\alpha$ appears as a straight line with slope α. The steeper line illustrates super-linear sequential recursion ($\alpha > 1$) schematically; the shallower one, parallel sampling with diminishing returns ($\alpha < 1$). The framework originally predicted $\alpha_{\text{sequential}} > 1 > \alpha_{\text{parallel}}$. Paper II's multi-model replication (March 2026; Section 6.2) found the parallel exponent near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), with none for Grok 4.1 Fast, at 100% at every sample count, and the sequential exponent above the parallel one wherever both could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), at compute that was not matched. The best surviving sequential estimate is about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive: one model's fit. The quantitative claim $\alpha > 1$ is not confirmed for current frontier systems.

Derivation note (24 October 2026). *In the solution of Section 4.4, $U(R) = I \times [1 + (a/\alpha)R]^\alpha$, the change of dominant term is a change of regime in scaling behaviour: a smooth crossover, not a discontinuity or a phase transition. Unchanged: the theorem's statement.*

These five properties constitute the **qualitative** ARC Principle. Their derivation from the composition operator ensures internal consistency: accepting the formalism commits one to all five properties. The specific functional form is an empirical question for each domain; the properties are theorems.

4.2 The AI-specific form: power-law scaling

For AI systems exhibiting multiplicative composition, the generalised equation in the deep recursion regime ($R \gg R^*$) reduces to:

$$U \approx I \times R^\alpha \tag{5}$$

Effective capability = Base potential × Recursive depth$^\alpha$ (deep recursion limit)

The exact integrated form, derived in Section 4.4, is $U(R) = I \times [1 + \frac{a}{\alpha}R]^\alpha$. The power law is the asymptotic limit when recursive depth substantially exceeds the crossover depth R^*. The empirical sections read their measurements in this regime by assumption; R^* has not been measured.

In plain language: Effective capability (U) equals the starting potential (I) multiplied by the recursive depth (R) raised to a power (α) that depends on how well each step builds on the previous one.

4.3 What each variable means

U: effective capability. What the system actually achieves. Measured differently in each domain:

- AI: Benchmark accuracy on standardised tests
- Quantum: Logical qubit fidelity (how error-free the computation is)
- Physics: Temporal stability of the time crystal
- Biology: Metabolic efficiency (thermodynamic) or cognitive task performance (behavioural)

***I*: base potential (structured asymmetry).** **Thermodynamic definition:** Natural systems tend towards maximum entropy (equilibrium/uniformity). In this framework, "artificial" denotes any system maintaining low-entropy structure against the thermodynamic gradient, whether engineered (an AI model), evolved (a brain), or emergent (a time crystal). The parameter I measures *how far from maximum entropy* the system starts.

In Morrell, Elliott and Grier's experiment, identical beads interact reciprocally and no active region exists: "The active region vanishes for particles of the same size ($a_1 = a_2$), whose interactions are reciprocal" (Morrell et al. 2026). The framework reads this as an instance of its zero-factor property, a correspondence of structure and not a test. Order requires *designed disorder*: a specific pattern of asymmetry that enables work extraction from the environment.

Table 3. What the base potential I measures in each domain, and its physical meaning.

Domain	What I measures	Physical meaning
AI	Single-pass accuracy without reasoning	How much "prior knowledge" the model has
Quantum	Raw qubit quality (1− error rate)	Distance from maximum entropy
Physics	Variance in bead sizes	Asymmetry enabling nonreciprocal forces
Biology	Initial learning rate	Sensitivity gradient

The Constraint Principle: Constraint enables competence. Without structured asymmetry, no work can be extracted. The time-crystal case above is the physical correspondence. This is independently testable (falsification criterion F3).

***R*: recursive depth.** How many self-referential cycles the system performs. The output of cycle n becomes the input for cycle $n + 1$.

Table 4. One unit of recursive depth R in each domain, and how it is counted.

Domain	One unit of R	How it is counted
AI	One reasoning step	Token count or revision cycle
Quantum	One step of two in code distance	Code distance d, with $R = (d + 1)/2$
Physics	One oscillation period	Frequency analysis
Biology	One feedback cycle	Generation or learning iteration

Critical requirement: For $\alpha > 1$, recursion must incorporate *sequential self-correction*: each step building on previous outputs. Pure parallel processing (independent attempts averaged together) cannot achieve $\alpha > 1$ because it lacks the output→input feedback loop. However, hybrid parallel-sequential approaches may achieve intermediate scaling (Li et al. 2025).

***α*: the scaling exponent.** The scaling exponent determines the system's scaling regime:

Our core prediction: $\alpha_{\text{sequential}} > 1 > \alpha_{\text{parallel}}$

Table 5. The scaling regime set by the value of the scaling exponent α, with an example of each.

α value	What happens	Example
$\alpha < 1$	Diminishing returns	Parallel voting: more samples help less and less
$\alpha = 1$	Linear returns	Each step adds a fixed amount
$\alpha > 1$	Compounding returns	Each step multiplies capability

4.4 Deriving α from first principles

This derivation completes the paper's theoretical contribution. Without it, α is just a number we fit to data. With it, α becomes a predictable quantity.

The key insight. How much does your accumulated knowledge help your next step?

Define $\beta \in [0, 1)$ as the "self-referential coupling": how much of accumulated context each new step can effectively leverage. When $\beta = 0.5$, each step benefits from half of accumulated context. The domain restriction $\beta < 1$ is essential: at $\beta = 1$, the ODE solution diverges ($\alpha \to \infty$), producing a finite-time singularity that no physical system can realise.

Derivation note (19 September 2026). *An owner review of the reason given for the domain restriction finds that the restriction is right and that the reason printed for it is not, and this note replaces the reason while keeping the restriction. With the governing equation $dg/dr = a\,g^{\beta}$ of this section and the initial condition $g(0) = 1$ used in Appendix A.2, the case $\beta = 1$ gives $dg/dr = a\,g$ and hence $g(R) = e^{aR}$, which is finite at every finite depth: the solution there is exponential, not singular. For $\beta \neq 1$, integrating $g^{-\beta}dg = a\,dr$ under the same initial condition gives $g^{1-\beta} = 1 + a(1-\beta)R$. The finite-depth branch is therefore $\beta > 1$, where that reads $g^{-(\beta-1)} = 1 - a(\beta-1)R$, which reaches zero at the finite depth $R^{*} = 1/[a(\beta-1)]$, and g diverges there. The two statements inside the parenthesis above are also separate statements. As β rises towards 1 from below the exponent $\alpha = 1/(1-\beta)$ does diverge, while the solution converges pointwise to the exponential, since $[1 + a(1-\beta)R]^{1/(1-\beta)} \to e^{aR}$; at $a = 1$ and $R = 3$ the solution is 13.79 at $\beta = 0.9$, 19.22 at $\beta = 0.99$ and 20.00 at $\beta = 0.999$, against $e^{3} = 20.09$. A diverging exponent is not a diverging solution. The domain $\beta \in [0, 1)$, printed here and again in Appendix A.2, stands unchanged, on the ground that it is what makes α finite and the power-law form exist at all. The plain-language gloss in the second sentence is scoped in the same way: from the governing equation, $\log(dg/dr) = \log a + \beta \log g$, so β is the log elasticity of marginal gain with respect to accumulated amplification, which is exactly the quantity the computational validation below measures from marginal gains in log-log space. A fraction of accumulated context, or of tokens, is a different quantity and would need its own operational definition before the gloss could be read literally. Unchanged: the domain restriction, the exact integrated solution, the identity $\alpha = 1/(1-\beta)$, and the validation result reported below.*

If each step is independent ($\beta = 0$), you get linear scaling ($\alpha = 1$).
If each step fully leverages all prior work ($\beta \to 1$), scaling explodes.

The mathematics. Since $U = I \times g(R)$, the compounding operates on the *amplification factor* $g(R)$, not on the absolute capability U. The marginal amplification gained at step r depends on accumulated amplification:

$$\frac{dg}{dr} = a \times g^{\beta} \tag{6}$$

Why this form? This differential equation models *cumulative advantage* (also called preferential attachment; Barabási and Albert 1999): the rate of amplification gain is proportional to the system's

current amplification raised to a power. Critically, the ODE operates on g (the amplification factor) rather than U (the absolute capability), ensuring that the base intelligence I acts purely as a multiplicative starting constant and does not contaminate the recursive dynamics. This resolves an algebraic inconsistency present in earlier versions of this paper (see Ledger Note 1 in the Version history (Appendix B.3)).

Solving this differential equation (details in Appendix A) with initial condition $g(0) = 1$ yields:

$$U(R) = I \times \left[1 + \frac{a}{\alpha}R\right]^{\alpha} \quad \text{where} \quad \alpha = \frac{1}{1-\beta} \tag{7}$$

The corrected ARC equation (exact integrated form)

In the deep recursion limit ($R \gg \alpha/a$), this simplifies to $U \approx I \times R^{\alpha}$, recovering the power-law form used throughout the empirical sections of this paper.

Coefficient note (19 September 2026). *An owner review of this limit finds that the approximate equality above drops a coefficient that this paper's own appendix keeps, and this note restores it. For $(a/\alpha)R \gg 1$ the bracket in the exact solution tends to $(a/\alpha)R$, so $U(R) \to I\,(a/\alpha)^{\alpha}R^{\alpha}$. That factor is not near one in general: it is 0.25 at $\alpha = 2$ with $a = 1$, 0.0625 at $\alpha = 2$ with $a = 0.5$, 0.68 at $\alpha = 1.33$ with $a = 1$, and 0.037 at $\alpha = 3$ with $a = 1$. Appendix A.2 already prints both correct forms, the proportionality $U \propto I\,R^{\alpha}$ and the explicit $U \approx I\,(a/\alpha)^{\alpha}R^{\alpha}$, and the sentence above is to be read as that proportionality. The coefficient may not be absorbed into I: it is built from the recursion parameters a and α, and the note on the corrected formulation in this section, repeated in Appendix A.2, puts the differential equation on the amplification factor precisely so that I remains a multiplicative constant outside the recursive dynamics. Where an amplitude is fitted to data instead, the fitted amplitude is $I\,(a/\alpha)^{\alpha}$ and not the I of that note. Unchanged: the exact integrated solution; the empirical sections, which fit exponents rather than amplitudes, so no reported result depends on the dropped factor; and the crossover depth $R^{*} = \alpha/a$.*

Table 6. The scaling exponent $\alpha = 1/(1-\beta)$ for representative values of the self-referential coupling β, with the interpretation of each.

β (coupling)	α (exponent)	Interpretation
0	1	Independent steps → linear scaling
0.25	1.33	Weak coupling → mild super-linear (AI only; physical systems use $d/(d+1)$)
0.50	2.00	Moderate coupling → quadratic scaling
0.67	3.00	Strong coupling → cubic scaling

What this means. **Novel Prediction:** As AI systems develop richer self-correction (critique-revise loops, hierarchical reasoning), β should increase and α should approach 2. This is specific, measurable, and falsifiable.

A numerical check of the identity. The β-derivation is not a curve fit. It is a mathematical identity recoverable to machine precision. To verify this, the relationship $\alpha = 1/(1-\beta)$ was tested against 30 exact solutions to the Bernoulli ODE with β values spanning 0.05 to 0.92. The procedure measures β blindly from marginal gains (dg/dR vs g in log-log space), predicts α, and compares against the true value. The result: $R^2 = 1.00000000$ (eight decimal places), slope = 1.000102, mean absolute prediction error = 0.002%. The identity is exact by the derivation above; this numerical check confirms only that the estimation procedure recovers it from the ODE's own solutions, and is no evidence that a real system obeys the ODE.

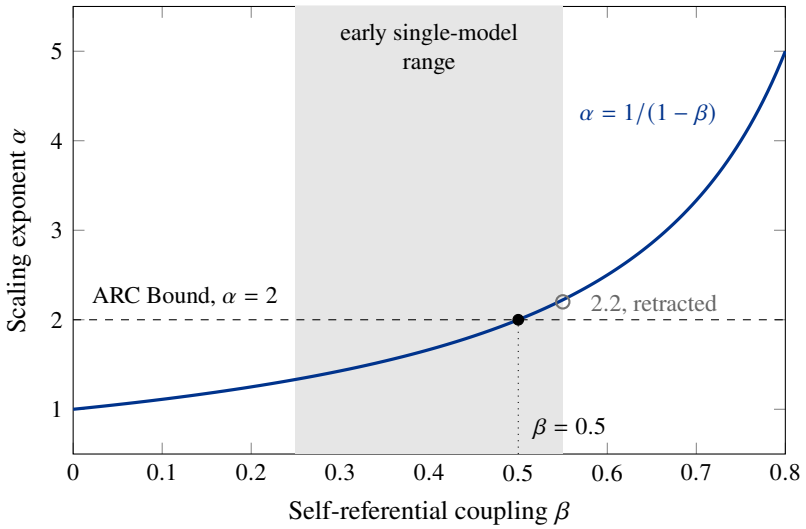


Figure 3. The relation between β and α. The scaling exponent follows from the self-referential coupling as $\alpha = 1/(1-\beta)$, and diverges as β approaches 1 (perfect feedback efficiency). The early single-model range of $\alpha \approx 1.3$ to 2.2 corresponds to $\beta \approx 0.25$ to 0.55; its upper figure, about 2.2 from the DeepSeek R1 run of 21 January 2026, sat above the ARC Bound and is retracted, since the author's own larger replication on five frontier models did not reproduce it. The best surviving estimate, about 0.49 from one model (Gemini 3 Flash; 0.59 by endpoint estimate, interval −1.3 to 2.9), is inconclusive and implies no coupling in the range drawn.

4.5 The ARC Bound

The ARC framework makes a central, falsifiable prediction: for classical sequential recursive systems, the scaling exponent is bounded at $\alpha_{\max} = 2$. For attention-based AI architectures, this bound is argued from the $O(N^2)$ computational complexity of self-attention. Whether it holds as a general bound for all classical sequential systems is a conjecture, not a theorem.

4.5.1 Scale-free derivation

The result follows from a single assumption: **scale invariance**. Consider any system where capability U depends on recursive depth R and base quality I. If the system is scale-free (exhibiting no intrinsic length scale), then the governing dynamics must satisfy:

$$\frac{dg}{dr} = a \cdot g^{\beta} \tag{8}$$

Bernoulli ODE operating on amplification factor (scale-invariant form)

This is the *unique* ordinary differential equation consistent with scale invariance operating on the amplification factor. Any other form (exponential, logarithmic, polynomial with fixed coefficients) introduces an implicit scale and thus violates the symmetry.

The solution is $g(r) = [1 + \frac{a}{\alpha} r]^{\alpha}$ where $\alpha = 1/(1-\beta)$. In the deep recursion limit, $g(r) \propto r^{1/(1-\beta)}$. The derivation chain is:

1. **Scale invariance** forces the Bernoulli form
2. **Bernoulli ODE** yields power-law solutions
3. **Power-law exponent** is determined by coupling β

4. **Computational complexity** of self-attention bounds $\beta \leq 0.5$ for attention-based architectures (below)
5. **Therefore** $\alpha \leq 2$ for classical sequential systems (argued, not proven, for attention-based architectures; conjectured generally)

> **The ARC Bound (a novel prediction).** For recursive intelligence, the scaling exponent is bounded by $\alpha \leq 2$ (equivalently, $\beta \leq 0.5$). For attention-based architectures, the $O(N^2)$ scaling of self-attention provides the concrete computational basis: each reasoning step can cross-reference at most N previous steps, producing at most quadratic information density. This is conjectured to hold for classical sequential systems generally. Systems whose growth runs through a supply network of finite dimension face an even stricter constraint, the geometric scaling bound $\alpha = d/(d+1) < 1$, derived in general form by West et al. (1999), from the three-dimensional case of West, Brown, and Enquist (1997), and by Banavar et al. (1999), and reached independently by Banavar et al. (2010), Demetrius (2003), Zhao (2022) and Bettencourt (2013); the companion paper *On the Origin of Scaling Laws* (Eastwood 2026f) identifies why these derivations converge via Cauchy's functional equations. Quantum systems, employing additive composition in Hilbert space, are not subject to either bound. *This prediction has not been independently tested. We provide the falsification criterion (F11).*

The initial estimate $\alpha \approx 2.2$ from the author's Paper II (Eastwood 2026h) experiment ($N = 12$, 95% CI 1.5 to 3.0; the DeepSeek R1 run of 21 January 2026) was tested across the five frontier models that make up Paper II's capability arm with 30 problems (Paper II, March 2026). **Caveat:** A point estimate of $\alpha = 2.2$ sits above the bound the framework predicts, $\alpha \leq 2$ (the ARC Bound, criteria F5 and F11). However, the 95% confidence interval [1.5, 3.0] is wide enough to be consistent with both the theory and its negation, and should not be treated as confirmatory. The author retracted it when his own larger replication on five frontier models, the capability arm of a six-model study, did not reproduce it; Paper II attributes it to a small sample, a single model and a compressed range of accuracies. The best surviving estimate is about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive: one model's fit (Gemini 3 Flash).

Computational argument for the ARC Bound. The quadratic limit can be grounded in a concrete computational mechanism. In current AI architectures, the dominant recursive process is **self-attention**, where each new token (reasoning step) attends to and cross-references every previous token. The computational complexity of this operation scales as $O(N^2)$, where N is the sequence length. This means the maximum density of causal interactions in a classical sequential system is quadratic: each additional step can, at most, create N new connections to all previous steps.

This provides a principled basis for the $\alpha = 2$ bound:

1. **Attention complexity:** Transformer self-attention scales as $O(N^2)$ with sequence length, establishing a quadratic ceiling on the information-processing density achievable through sequential recursion.
2. **Coupling interpretation:** At $\beta = 0.5$, each step leverages exactly half of accumulated context, the efficiency point at which the attention mechanism saturates its cross-referencing capacity relative to the computational cost of deeper recursion.
3. **Empirical status:** The initial $\alpha \approx 2.2$ estimate ($N = 12$, 95% CI 1.5 to 3.0) from Paper II exceeded the predicted bound of $\alpha \leq 2$. However, the 95% confidence interval [1.5, 3.0] is wide enough to be consistent with both the theory and its negation, and this single-model estimate should not be

treated as confirmatory. Paper II (March 2026), which broadened the measurement to cover the whole capability arm of five frontier models and 30 problems (18 AIME-level) with bootstrap CIs and cross-verification, did not reproduce it, and the author retracted it; the best surviving estimate is about 0.49, one model's inconclusive fit (Gemini 3 Flash; Section 6.2). The ARC Bound is neither confirmed nor refuted by current data.

Scope of the bound: The $O(N^2)$ argument is offered for the quadratic limit in *attention-based architectures* specifically, and is an argument, not a proof. Whether $\alpha \leq 2$ holds as a universal bound for all classical sequential recursive systems is a **conjecture**, not a theorem. Non-attention architectures (state-space models, recurrent networks) may have different computational ceilings. The quadratic limit should be understood as: (a) argued, not proven, for $O(N^2)$ attention mechanisms, (b) conjectured for classical sequential systems generally, and (c) explicitly not applying to quantum systems, which employ additive composition and achieve exponential scaling. Previous versions of this paper offered two additional arguments for the bound (elasticity analysis and an edge-of-chaos derivation). These have been withdrawn: elasticity exceeding unity does not imply dynamical instability, and a one-dimensional autonomous ODE cannot be chaotic, since its trajectories are monotone. The $O(N^2)$ argument is both more concrete and more defensible.

The ARC Bound therefore identifies a **conjectured stable-scaling limit**: the most a recursively self-correcting classical system can grow while remaining self-correcting, a scaling limit set by the information-processing geometry of the substrate rather than by engineering, and not a speed limit. Limits of this kind are familiar from thermodynamics: a ceiling on a class of mechanism, fixed by the mechanism's own geometry, which no engineering within the class can exceed. Systems operating under classical sequential recursion approach $\alpha = 2$ as their stable upper bound. Systems can be driven above it, but they do not remain stably self-correcting there; a transient excursion above the bound is the predicted supercritical regime, not the refutation. Stability here is judged by Law II's criterion (a law under test), measured rather than asserted. Escaping the bound *stably* requires a fundamentally different computational substrate (quantum coherence, additive composition), just as escaping a mechanism-class ceiling in thermodynamics requires a different mechanism class, not a better instance of the old one.

Universality across domains. The ARC Bound applies wherever multiplicative composition governs the system dynamics:

- **AI systems:** Chain-of-thought reasoning with iterative refinement (each step builds multiplicatively on prior context)
- **Biological networks:** Fractal transport systems (circulatory, respiratory, neural branching)
- **Economic systems:** Hierarchical organisations with multiplicative value chains
- **Physical networks:** Scale-free graphs with preferential attachment

The quantum exception (exponential scaling with $\Lambda \approx 2.14$) arises because quantum error correction employs *additive* composition: independent syndrome measurements combine additively, not multiplicatively. Additive composition yields exponential functional forms, which are not subject to the ARC Bound. The composition operator, not the domain, determines the bound.

A ceiling on a class of mechanism: The ARC Bound is a mechanism-class ceiling of the kind thermodynamics made familiar. Just as a heat engine's maximum efficiency is fixed by its operating temperatures regardless of engineering, on this argument no attention-based sequential recursive system would exceed $\alpha = 2$, because transformer self-attention scales as $O(N^2)$. **The analogy holds in the part that matters: a mechanism-class ceiling binds its class, and a device built on a different mechanism is not bound by it at all.** Both are ceilings on a class of mechanism, not statements about what is

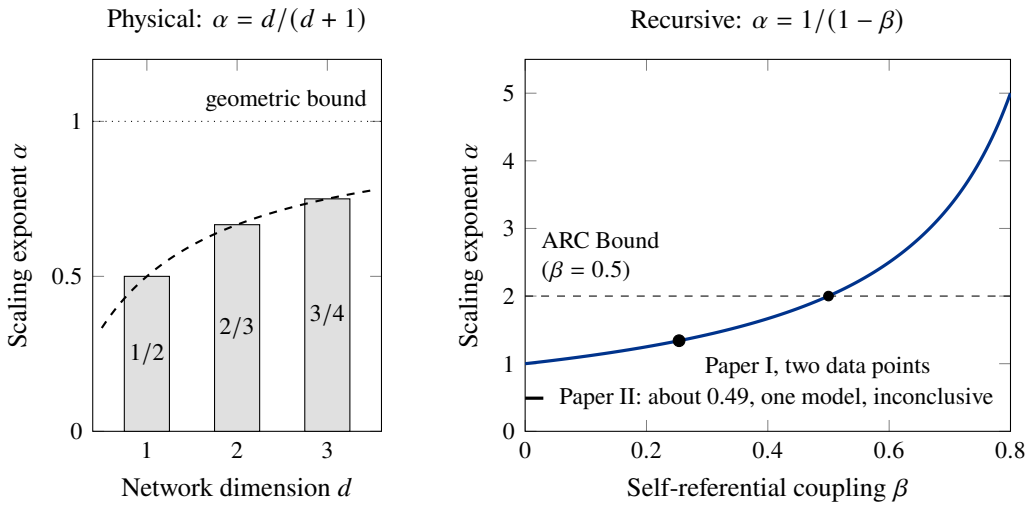


Figure 4. The two-regime framework. Systems whose growth runs through a supply network of dimension d (left) follow $\alpha = d/(d+1)$, below 1 for every finite d: $3/4$ for three-dimensional networks, $2/3$ for two-dimensional and $1/2$ for one-dimensional. Recursive intelligence (right) is governed by $\alpha = 1/(1-\beta)$, above 1 for any positive β; the ARC Bound at $\beta = 0.5$ sets $\alpha \leq 2$ (dashed). Quantum error correction lies outside both regimes, its additive composition giving exponential scaling. Marked on the recursive side: Paper I's published estimate of 1.34, from two data points; Paper II's best surviving estimate, about 0.49 (one model's inconclusive fit), is drawn at the panel's edge because it implies no coupling in the range drawn. The retracted single-model estimate is not drawn.

physically possible. Individual systems may fall short due to imperfect coupling, noise, or architectural constraints (and physically embedded systems face the stricter geometric scaling bound of Section 5.6), but the ceiling for classical sequential computation is fixed by the quadratic geometry of cross-referencing. **A system driven above that ceiling does not stay coherent there, so a durably stable measurement above $\alpha = 2$ would refute the bound under F5, while an unstable excursion would be consistent with it; F11, the earlier criterion, refutes on a sustained magnitude alone.** A ceiling of this kind tells us when to stop optimising within a paradigm and when a fundamentally different approach (e.g., additive composition, quantum coherence) is required to exceed it.

Critically, the contradictory evidence reviewed in Section 5.2 (Zeng et al. 2025; Li et al. 2025) may indicate that:

- α varies with task difficulty, model architecture, or prompt structure
- There exists an optimal reasoning length beyond which returns diminish (the "thinking long but short" phenomenon)
- Different benchmarks probe different coupling regimes

Resolving these discrepancies is a central empirical question for the Global Scaling Challenge (Section 8).

Caveat. The numerical similarity between the ARC Bound's $\alpha = 2$ and $\Lambda \approx 2.14$ (quantum) may be entirely coincidental. These are mathematically distinct quantities from different functional forms. Any deeper connection would require theoretical justification that does not currently exist.

The β-transition phase diagram. The generalised framework implies a **phase diagram** of recursive scaling. The axes are base quality (I, normalised) and coupling strength (β). Four regimes emerge:

- **Diminishing returns:** $\beta < \beta^*$ (threshold) or I below threshold. More recursion hurts or provides

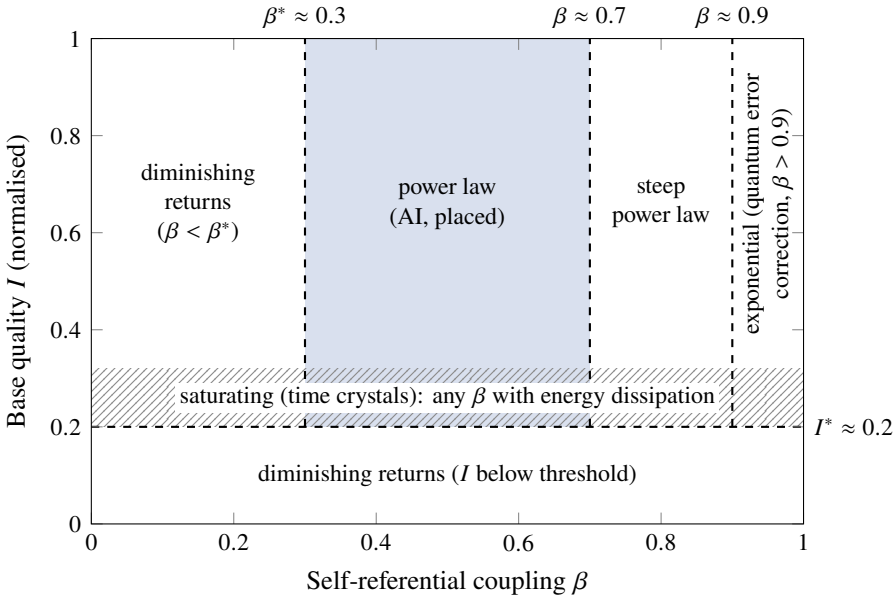


Figure 5. The β-transition phase diagram. Scaling regimes as a function of base quality (I) and self-referential coupling (β). AI systems (power-law, placed at $\beta \approx 0.5$), quantum error correction (exponential, placed beyond $\beta = 0.9$; its parameter is Λ), and time crystals (saturating); the placements are schematic, not measured. *Prediction: measured β determines functional form.*

negligible benefit.

- **Power-law regime:** $0.3 \lesssim \beta \lesssim 0.7$, I above threshold. Compounding returns, polynomial scaling. (AI chain-of-thought reasoning.)
- **Exponential regime:** $\beta > 0.9$, I well above threshold. Compounding returns, exponential scaling. (Quantum error correction below threshold.)
- **Saturating regime:** Any β with energy dissipation. Compounding onset $\rightarrow$ limit cycle. (Time crystals, physical oscillators.)

Note: These boundary values are schematic placements, not observations: no β has been measured in AI systems (the early estimate is retracted and the surviving one implies no β in $[0, 1)$), and quantum error correction's parameter is Λ, not β. The precise boundaries require empirical determination via the Global Scaling Challenge.

Cross-domain β prediction: the killer test. The framework makes a specific, falsifiable cross-domain prediction that distinguishes analogy from unified theory:
The β-Convergence Prediction: In any domain, β can be measured by two independent methods:

1. **From scaling behaviour:** Fit the observed $f(R)$ to extract α, then compute $\beta = 1 - 1/\alpha$.
2. **From threshold analysis:** Measure the critical coupling β^* below which recursive gains vanish.

Prediction: These two independent measurements of β will converge to the same value within experimental error.

Cross-domain extension: If we measure β_{AI} from AI scaling data, β_{QEC} from quantum error correction thresholds, and β_{bio} from metabolic scaling, and these values occupy consistent positions on the β-continuum that correctly predict their respective functional forms, the framework is supported as a unified account. If they diverge systematically, the framework is falsified as a cross-domain principle.

This is the framework's strongest *cross-domain* prediction. It transforms the observation that "different domains have different functional forms" into the testable claim that "the same underlying parameter determines all functional forms."

Concrete Test Protocol:

1. In quantum error correction: Measure the error suppression factor $\Lambda = p_{\text{th}}/p_{\text{physical}}$ directly from logical error rates across code distances. The quantum scaling parameter is Λ itself, not derived from β. Quantum systems employ additive composition and are not subject to the power-law framework.
2. In AI systems: Measure α from test-time scaling curves. Compute $\beta_{\text{AI}} = 1 - 1/\alpha$. Predict power-law scaling.
3. **Cross-domain validation:** Verify that quantum error correction follows exponential scaling ($\varepsilon_d \propto \Lambda^{-(d+1)/2}$) while AI follows power-law scaling ($U \propto R^{\alpha}$). The key structural prediction is that the composition type (additive vs hierarchical) determines the functional form, not a shared numerical parameter.

If the β values from different measurement methods within each domain converge, AND the cross-domain β values correctly predict functional forms, the framework passes its most stringent test. If either condition fails, the framework requires fundamental revision.

This prediction is novel so far as our searching reaches: no framework we have located proposes a single parameter connecting quantum error correction thresholds to AI scaling exponents. One that does would remove the novelty, and we would rather be shown it. Confirmation would establish the ARC Principle as a genuine cross-domain unification. Falsification would reduce it to a collection of domain-specific observations.

Connection to universality theory. The substrate-independence of the five structural properties is reminiscent of **universality** in statistical physics, where systems with different microscopic Hamiltonians exhibit identical critical exponents near phase transitions (Wilson 1971). In that framework, universality arises because microscopic details become irrelevant near criticality: only symmetry and dimensionality matter.

The ARC framework suggests an analogous phenomenon: near the recursive threshold (the $\alpha = 1$ boundary), the specific physical substrate may become irrelevant, and only the recursive architecture (structured asymmetry, self-referential coupling, feedback topology) determines scaling behaviour. Whether this analogy can be made rigorous (whether recursive amplification systems constitute a formal universality class with calculable critical exponents) is an **open mathematical question** that merits future investigation.

The identity is a registered test, never an established fact. The stability ceiling and the conversion exponent meet at the same number, and the temptation is to say one constant does two jobs. The claim's status is stricter: the identity is registered for testing, in advance of the data, with outcomes that can kill it. Study AE registers the reciprocity requirement between the separately measured leverage cap and stability boundary with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity; study AG tests the two derivations of the critical exponent against each other, notes that they agree numerically only at the single point where both parameters equal one half and disagree in their derivatives everywhere else, and is designed to find the disagreement. If the channels disagree, the unification is withdrawn from every surface it has reached, with the same prominence as the framing. A match between an asserted one half and an estimated 0.49 (one model's inconclusive fit) is never put forward here as evidence: the two quantities may inherit one accumulation law between them, and quantities that share a law cannot corroborate each other.

A programme hypothesis beyond the framework. Oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The programme therefore hypothesises, beyond what these equations decide, that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. The correction exponent belongs to the corrector, and it is a different quantity from the alignment scaling exponent α_{align} of Section 3.1, which P19 concerns. This is testable on existing data and does not require new systems.

4.6 Composition operator transitions

The framework as presented in Section 4.1 assigns a single composition operator $\oplus$ to each system. However, computational analysis reveals that this may be an idealisation. In bounded or dissipative systems, **the composition operator itself can transition between regimes as recursive depth increases**.

This finding emerged from cosmological analysis of gravitational structure formation, where the recursive process of gravitational collapse exhibits three distinct phases:

1. **Inflation/linear growth (additive $\oplus$):** Primordial perturbations grow independently. Each step adds to the previous without cumulative advantage. Composition satisfies $f(R_1 + R_2) = f(R_1) \cdot f(R_2)$, the exponential (additive Cauchy) form.
2. **Nonlinear collapse (multiplicative $\oplus$):** Mode coupling emerges; denser regions gravitationally attract more matter, creating cumulative advantage. Composition transitions to $g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$, the power-law form.
3. **Virialised halos (bounded $\oplus$):** Individual structures reach energy minima. Gains saturate: $Q_{r+1} \approx Q_r$. The system enters the bounded regime of Theorem 1.

Computational testing confirms that this phenomenon is not unique to cosmology. In logistic growth, gradient descent with momentum, and Kuramoto oscillator synchronisation, measuring the local coupling parameter β in sliding windows reveals systematic transitions between composition regimes. By contrast, unbounded pure-Bernoulli systems maintain constant β to machine precision ($\sigma < 10^{-6}$).

Theorem 6 (Composition operator transitions)**.** *A single physical system can transition between different composition operator regimes as a function of recursive depth or an internal state parameter. This extends the framework by replacing the assumption of a fixed $\oplus$ with a depth-dependent $\oplus(R)$, modelled by extending Axiom 2 to $dg/dr = a(g, r) \cdot g^{\beta(g,r)}$ where the coupling itself depends on accumulated state.*

Why this matters for the contradictory evidence. The finding that different AI studies report conflicting scaling behaviours (Section 5.2) may reflect composition operator transitions rather than genuine contradictions. If reasoning systems begin in an additive regime (shallow thinking, independent steps), transition to multiplicative (deep reasoning with cumulative advantage), and eventually saturate (diminishing returns at extreme depth), then studies measuring at different recursive depths would observe different scaling, even in the same system. The "optimal reasoning length" identified by Zeng et al. (2025, arXiv:2502.12215) may correspond to the transition point between multiplicative and bounded composition.

This interpretation is **testable**: measure the local coupling parameter β at different reasoning depths within a single AI system. If β is constant, the original framework applies. If β decreases with depth (as the cosmological analysis suggests), Theorem 6 applies. Either outcome advances understanding of recursive scaling.

Epistemic status: Theorem 6 is a hypothesis drawn from computational examples and stated in the theorem environment for numbering only; it is not a mathematical theorem derivable from the original

three axioms. It extends the framework's scope but requires independent experimental confirmation. The cosmological phases above illustrate the transition; no measurement of an exponent in them is claimed, and the generalisation to AI systems is a prediction, not an established result.

5 The cross-domain evidence

If recursive scaling with $\alpha > 1$ were specific to AI software, the alignment scaling problem might be an engineering challenge addressable through implementation changes. Systems with no software at all show related structures, with forms that differ. This section sets the framework against them and asks whether recursive amplification could be a structural property of recursive systems rather than a feature of how AI is built; it finds the framework consistent with the independently established forms in three of the four domains.

5.1 Overview

The following table summarises evidence across domains, noting that **functional forms differ** (see Section 4.1). What unites them is recursive or recurrent processing; the framework is consistent with independently established forms in three of the four domains (biology, quantum error correction and classical time crystals). The cross-domain evidence is consistent with the underlying scaling behaviour being structural, in which case the alignment scaling problem cannot be solved by software changes within the current paradigm.

Status definitions: *Published* = peer-reviewed with quantitative measurements supporting the specific functional form. *Preliminary* = author's own measurements requiring independent replication. *Structural* = qualitative mapping without quantitative α measurement. *Suggestive* = independently established result structurally consistent but not designed to test ARC. *Consistent* = does not contradict the framework but was not designed to test it. *Retracted* = withdrawn by the author, kept on the record with its retraction. *Preprint* = public but not peer reviewed.

Evidence hierarchy: The evidence base is uneven and we state this explicitly. AI inference scaling has one surviving estimate after the early estimate was retracted: about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive, one model's fit (Gemini 3 Flash); it needs independent replication. Quantum error correction provides a structural parallel with different functional form (exponential rather than power-law). Time crystals and neuroscience represent structural predictions derived from the framework, not independent validations: researchers in those domains did not measure α or test the ARC equation. Biology (Kleiber's Law) is an existing result we interpret through the β-parameterisation; the leaf venation prediction (F12) is the only non-circular test in that domain. These domains are presented together to show scope, not to claim equivalent evidential standing.

Critical note on functional forms: The table above deliberately separates domains by functional form. Power-law scaling ($U \propto R^{\alpha}$), exponential scaling ($\varepsilon \propto \Lambda^{-(d+1)/2}$), and saturating dynamics (limit cycles) are *mathematically distinct*. The numerical similarity between the ARC Bound's $\alpha = 2$ and $\Lambda \approx 2.14$ may be entirely coincidental. Any deeper connection would require theoretical justification that does not currently exist. What we claim is a *qualitative* correspondence (the five structural properties of Section 4.1), not *quantitative* universality (all domains sharing one exponent); no growth exponent has been measured in the acoustic time crystal, whose published recordings are of an established oscillation, or in neuroscience.

Table 7. Evidence across domains: system, finding, functional form, key parameter and status.

Domain	System	Finding	Functional Form	Key Parameter	Status
AI	Author's experiments (Papers I & II)	Sequential above parallel where both were estimated	Power-law R^{α}	About 2.24 (N=12; the DeepSeek R1 run of 21 January 2026), retracted; surviving estimate about 0.49 from one model, inconclusive	Retracted; inconclusive
AI	Sharma and Chopra (2025)	Sequential wins 95.6%	Power-law (inferred)	α not fitted	Preprint (not peer reviewed)
Quantum	Google Willow (results announced 9 December 2024; the team's preprint public from 24 August 2024; graded as convergent timing, never as a prediction)	Exponential error suppression	Exponential, $\Lambda^{-(d+1)/2}$	$\Lambda = 2.14$	Published
Physics	NYU Time Crystal	Disorder + feedback $\rightarrow$ order	Saturating	No growth exponent measured; coherence time (100 ± 10) s, about 6,700 cycles, in one experiment	Structural
Biology	Kleiber's Law	3D network $\rightarrow$ $\alpha = d/(d+1) = 3/4$	Power-law	$\alpha = 0.750$ (predicted), 0.744 (measured)†	Consistent (contested)
Neuro	COGITATE & others	Recurrence matters	Unknown	N/A	Consistent

†Predicted directly from $\alpha = d/(d+1)$ with $d = 3$. Mean error 2.4% across 11 species groups, provisional, pending a recomputation (see the companion paper *On the Origin of Scaling Laws*).

5.2 AI: sequential versus parallel reasoning

Published sources: DeepSeek-R1 Technical Report (20 January 2025, arXiv:2501.12948; DeepSeek-AI 2025); Sharma and Chopra, "The Sequential Edge: Inverse-Entropy Voting Beats Parallel Self-Consistency at Matched Compute" (November 2025, arXiv:2511.02309); Snell et al., "Scaling LLM Test-Time Compute" (August 2024, arXiv:2408.03314).

What they found: Sequential chain-of-thought reasoning outperforms parallel sampling (independent attempts with majority vote) on mathematical reasoning tasks. Sharma and Chopra (2025) tested 5 state-of-the-art models across 3 challenging benchmarks (AIME-2024, AIME-2025, GPQA-Diamond) under matched computational budgets: sequential wins in 43/45 configurations (95.6%), with accuracy gains up to 46.7 percentage points (Qwen3-235B on AIME-2025: 76.7% sequential vs 30.0% parallel).

Why sequential wins (per Sharma and Chopra 2025): Sequential reasoning enables three mechanisms unavailable in parallel approaches: (1) *iterative error correction* where models identify and fix mistakes in subsequent steps, (2) *progressive context accumulation* where each step builds upon accumulated insights, and (3) *answer verification* where models validate and refine initial responses. The overall effect is statistically robust ($t = 4.23$, $p < 0.001$, Cohen's $d = 0.89$), and holds across five open-source models from three families (GPT-OSS, Qwen3 and Kimi-K2).

What the voting result shows: Sharma and Chopra (2025) tested seven different voting methods for aggregating sequential chain outputs. Methods favouring *later* reasoning steps achieved optimal performance in 97% of configurations (the authors count among them their inverse-entropy weighting, which weights each chain by its confidence), while methods favouring *earlier* steps (exponential decay) achieved optimality in only 17%. Later steps are therefore more valuable than earlier ones, which any chain that improves step by step would show, whether its returns compound or diminish: the result is consistent with $\alpha > 1$ and does not distinguish it from $\alpha < 1$.

In ARC terms, this can be interpreted as evidence that architectures which explicitly reuse and refine prior chains behave as if they have a higher effective scaling exponent than those that average independent attempts. However, Sharma and Chopra (2025) do not fit α or β directly; this connection is *interpretive rather than quantitative*.

Author's estimates (Papers I and II): In the author's own prior work (Eastwood 2026g,h), sequential reasoning showed scaling consistent with $\alpha \approx 1.3$ to 2.2, while parallel sampling showed near-zero returns. These estimates have important limitations and have been substantially revised by the Paper II multi-model replication, and the upper figure is retracted:

- The $\alpha \approx 1.34$ estimate (Paper I) derives from only two data points using estimated token counts from the DeepSeek technical report (DeepSeek-AI 2025), not original experimental data
- The $\alpha \approx 2.2$ estimate (Paper II, original 12-problem experiment: the DeepSeek R1 run of 21 January 2026) derives from the author's own AIME experiment with 95% CI 1.5 to 3.0. **This does not replicate** in the multi-model tier-2 extension, and has since been retracted.
- **Paper II completed results (March 2026; updated v11.0):** Extending to five frontier models (those forming the capability arm within a study of six models) across 18 harder tier-2 problems (AIME/Putnam level) produces architecture-dependent findings. $\alpha \approx 2.24$ is retracted: it did not reproduce in Paper II's replication on five frontier models. Gemini 3 Flash provides the cleanest fit, about 0.49 by regression ($r^2 = 0.86$) and 0.59 by endpoint estimate, whose bootstrap interval, from -1.3 to 2.9, makes it inconclusive. Ceiling effects swallow Grok 4.1 Fast (96.3% to 100% at every depth) and DeepSeek R1, GPT-5.4 produces a binary step function rather than a dependable power law, and Qwen3 sits close to floor. The parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count, and the sequential exponent exceeds the parallel one wherever both

could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), at compute that was not matched.

- Accuracy figures differ slightly across papers due to different data sources, problem sets, and estimation methods

Statistical caveat: The original α estimates are from the author's preliminary work, not from the published sources (DeepSeek-AI 2025; Sharma and Chopra 2025). The published sources report the directional finding (sequential above parallel) but do not calculate α in this form, and that direction is itself contested, since Gu and colleagues (arXiv:2604.05868, April 2026; Gu et al. 2026) report the opposite result, parallel sampling outperforming sequential across three model families on mathematics and coding. The Paper II multi-model replication (March 2026) did not reproduce the original super-linear $\alpha \approx 2.2$, which the author has retracted. Gemini 3 Flash gave the only clean power-law fit among the five capability-arm models, about 0.49 (one model's inconclusive fit), while the other architectures are dominated by ceiling, floor, or step-function behaviour. The parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count.

Methodological note: The scaling exponent α is calculated from *error rate reduction*, not accuracy directly. Using $U = 1/\varepsilon$ (inverse error rate) as the capability measure: $\alpha = \ln(\varepsilon_1/\varepsilon_2)/\ln(R_2/R_1)$. This is mathematically equivalent to measuring how error rate decreases with recursive depth. Researchers attempting to replicate these calculations using accuracy directly will obtain different values.

Snell et al. (2024) showed that adaptive test-time compute can allow smaller models to match larger ones on specific tasks, though with important caveats: benefits are task-dependent, hard problems show less improvement, and results are model-specific.

Contradictory evidence and boundary conditions. **Important:** Not all research supports unlimited sequential scaling. Several 2025 studies found conditions where the sequential advantage breaks down:

- **Zeng et al. (2025, arXiv:2502.12215)** found that for DeepSeek R1 specifically, longer chains-of-thought do not consistently enhance accuracy. Correct solutions were often *shorter* than incorrect ones. They proposed "Shortest Majority Vote" as superior to extended sequential reasoning on some tasks.
- **Li et al. (2025, arXiv:2502.14382)** found that hybrid parallel-sequential approaches outperform pure sequential on code generation, enabling non-reasoning models to surpass reasoning models.
- **Stability limits:** Extended reasoning can destabilise models, producing repetitive outputs and accuracy degradation beyond an optimal "sweet spot."
- **Task-type boundary:** Sharma and Chopra's own creative task ablation provides a useful constraint: on joke generation, parallel approaches achieved greater *semantic diversity* (broader conceptual exploration) while sequential approaches achieved greater *lexical diversity* (deeper linguistic refinement). This suggests the sequential advantage applies most strongly to convergent tasks requiring iterative error correction, and may not generalise to divergent tasks requiring breadth of exploration.
- **The acoustic time crystal (no saturation evidence):** Morrell, Elliott and Grier's paper describes every recording it reports as an oscillation already established, and reports no onset. The antisymmetric mode's coherence time in its second experiment is (100 ± 10) s, about 6,700 cycles, and the authors add: "Sustained oscillations persist for hours longer than the measurement window, which means that the true coherence time of the nonequilibrium steady state may be substantially larger" (Morrell et al. 2026). No growth exponent has been measured in this system, so it gives no evidence on whether systems with $\alpha > 1$ scale indefinitely.

What this means for the ARC Principle: The sequential advantage appears to have *boundary conditions* that are architecture-dependent. Paper II results (Section 6.2) show a range from the one clean fit, Gemini 3 Flash's inconclusive 0.49, to ceiling, floor, and step-function regimes in the other models. The hypothesis should be refined to: 'Sequential recursion yields higher returns than parallel sampling *wherever both can be estimated*, but the absolute scaling exponent α is architecture-dependent, problem-difficulty dependent, and often not cleanly measurable because dynamic range collapses at the top and bottom of the capability distribution.' Identifying the architectural features that determine whether a clean power law even emerges is a priority for future work.

Caveat. The contradictory evidence and Paper II results suggest the ARC Principle may describe a regime that is architecture-dependent rather than universal. The directional finding (sequential above parallel where both could be estimated, with the parallel exponent below one wherever it could be) holds in the data so far, at compute that was not matched. The stronger quantitative claim that current frontier models cleanly exhibit $\alpha > 1$ on this task remains unconfirmed in the cross-architecture data.

5.3 Quantum: Google Willow, error correction below the surface code threshold (announced December 2024; preprint public from August 2024)

Source: Acharya, R. et al. [Google Quantum AI] (9 December 2024). "Quantum error correction below the surface code threshold." *Nature*, 638, 920 to 926 (Google Quantum AI and Collaborators 2024); the team's preprint was public from 24 August 2024. An Author Correction of 28 April 2026 amends only the labels of its Figure 3a (Google Quantum AI and Collaborators 2026).

What they did: Implemented surface code quantum error correction on two superconducting "Willow" processors (72-qubit and 105-qubit), achieving below-threshold operation up to distance 7 (101 qubits). Also ran repetition codes to distance 29 over 5.5 hours of processor time (2×10^{10} error correction cycles).

The scaling equation. The fundamental surface code relation is:

$$\varepsilon_d \propto \left(\frac{p}{p_{\text{thr}}} \right)^{(d+1)/2} \tag{9}$$

where d is code distance, p is physical error rate, p_{thr} is threshold error rate, and ε_d is logical error rate. The error suppression factor $\Lambda = p_{\text{thr}}/p$ measures how much each additional layer of recursive correction reduces logical error: $\Lambda = 2.14 \pm 0.02$ with neural network decoding.

The phase transition: direct experimental demonstration. By injecting coherent errors of variable strength (Fig. 2b in the paper), the team swept continuously through the threshold:

Table 8. The regimes above, at and below the surface code threshold in the quantum error-correction experiment of Section 5.3: physical error rate, effect of recursive depth and ARC analogue.

Regime	Physical error rate	Recursive depth effect	ARC analogue
Above threshold ($p > p_{\text{thr}}$)	High	More qubits → higher logical error	$\alpha < 1$ (diminishing returns)
At threshold ($p = p_{\text{thr}}$)	Critical	More qubits → no net change	$\alpha = 1$ (linear)
Below threshold ($p < p_{\text{thr}}$)	Low	More qubits → exponentially lower error	$\alpha > 1$ (compounding returns)

This is, so far as this programme's search reaches, a direct experimental demonstration of a recursive system crossing between regimes where additional depth hurts, is neutral, or helps: the qualitative

phase transition that the ARC Principle describes. The result is graded as convergent timing, never as a prediction, because the team's preprint was public from 24 August 2024, before the author's sealed manuscript record of 8 December 2024.

Error budget as decomposition of I. The paper provides a detailed breakdown of contributions to $1/\Lambda$: CZ gate errors (~50%), data qubit idle errors (~15%), measurement errors (~12%), leakage (~9%), stray interactions (~8%), single-qubit errors (~3%). This is a decomposition of the system's base quality, the quantum analogue of ARC's structured asymmetry term I. Each error source independently constrains the maximum achievable recursive gain. Improving any single component improves Λ across all code distances simultaneously: the multiplicative interaction between I and R that ARC formalises.

Leakage removal as active I-maintenance. Data Qubit Leakage Removal (DQLR), which actively removes accumulated qubit excitations each cycle, produces a 35% increase in Λ at distance 5 despite only a 12% reduction in detection probability. Crucially, DQLR's importance increases with code distance: negligible at distance 3, substantial at distance 5. This demonstrates that maintaining base quality during recursive processing becomes more critical as recursive depth increases, consistent with ARC's prediction that effective α depends not just on initial I but on sustained I throughout the recursive chain.

Repetition code error floor: the regime boundary. At high code distances ($d \geq 15$), the exponential error suppression deviates from prediction and plateaus at $\sim 10^{-10}$, caused by rare correlated error bursts (~30 qubits affected simultaneously, occurring approximately once per hour). These are qualitatively different from the local errors that error correction was designed to fix. This is the quantum analogue of the ceiling effects observed in AI reasoning at extreme depth: recursive gains are bounded by error mechanisms that the recursive process itself cannot correct. The framework describes a *regime*, not an asymptotic law.

Decoder quality affects scaling. The same quantum hardware achieves different Λ depending on decoder quality: neural network decoder $\Lambda = 2.18$ vs real-time sparse blossom decoder $\Lambda = 2.0$. The "interpreter" of each recursive cycle's output affects the system's effective scaling, analogous to how the quality of chain-of-thought reasoning strategies affects effective α in AI systems.

Stability and robustness. Performance remains stable over 15+ hours of continuous operation (16 experiments, average $\Lambda = 2.18 \pm 0.07$). Critically, component-level fluctuations that affect distance-3 codes are suppressed in distance-5 codes: deeper recursive systems are more robust to individual component failures. The distance-7 logical qubit lifetime ($291 \pm 6\ \mu$s) exceeds the best constituent physical qubit ($119 \pm 13\ \mu$s) by a factor of 2.4 ± 0.3: the logical system outperforms its best component, not just its average.

Structural correspondence (what IS shared):

Table 9. The structural correspondence between ARC components and their quantum error-correction analogues, with the measured values.

ARC Component	Quantum Analogue	Measured
I (base quality / structured asymmetry)	$\Lambda = p_{\text{thr}}/p$ (distance from threshold)	2.14 ± 0.02
R (recursive depth)	$(d+1)/2$ (error correction layers)	Up to 4 (distance 7)
$\alpha > 1$ threshold	$p < p_{\text{thr}}$ boundary	Directly measured via error injection
β (self-referential coupling)	Per-cycle error suppression efficiency	Dependent on decoder quality
Regime boundary / ceiling	Error floor from correlated bursts	$\sim 10^{-10}$ at $d \geq 15$
I maintenance	DQLR (active leakage removal)	35% Λ improvement at $d = 5$

Critical mathematical distinction. Λ and α describe fundamentally different scaling behaviours:

- Λ parameterises *exponential* suppression: $\varepsilon_d \propto \Lambda^{-R}$ where $R = (d + 1)/2$
- α parameterises *power-law* scaling: $U \propto R^{\alpha}$

Exponential functions eventually dominate any polynomial. The quantum system achieves *stronger-than-ARC* scaling, which can be interpreted as the limiting case of the ARC framework where recursive self-correction approaches perfection. The numerical similarity between $\Lambda = 2.14$ and the original single-model estimate $\alpha \approx 2.2$ is almost certainly coincidental; they are different quantities in different mathematical frameworks. (The $\alpha \approx 2.2$ estimate, from the DeepSeek R1 run of 21 January 2026, has since been retracted; the best surviving estimate is about 0.49, one model's inconclusive fit: Section 6.2.)

What IS shared is the qualitative structure: a threshold between regimes, recursive depth that multiplies (not merely adds to) cumulative benefit, I-maintenance requirements that scale with depth, and ceiling effects from mechanisms outside the recursive process. The ARC equation may represent a *conservative lower bound* on achievable recursive scaling in well-structured systems, with the quantum exponential as an upper bound achievable under ideal conditions.

5.4 Physics: classical time crystals

Sources: Morrell, M., Elliott, L., & Grier, D.G. (6 February 2026). "Nonreciprocal wave-mediated interactions power a classical time crystal." *Physical Review Letters*, 136, 057201 (Morrell et al. 2026). Liu et al. (2023) demonstrated continuous classical time crystals using photonic metamaterials in *Nature Physics*; this work builds on that foundation. Raskatla et al. (2024) demonstrated a continuous space-time crystal state in a photonic metamaterial driven by nonreciprocal optical forces (*Physical Review Letters*, 133, 136202).

Experimental System:

Table 10. The acoustic time crystal of Section 5.4: its parameters, their values and their ARC analogues.

Parameter	Value	ARC Analogue
Acoustic frequency	40 kHz	
Crystal oscillation	61 to 67 Hz	Recursion cycle period
Coherence time τ	(100 ± 10) s, about 6,700 cycles, in one experiment; possibly substantially longer	None: a coherence time, not a ceiling on recursion depth
Bead material	Polystyrene foam (varied sizes)	Structured asymmetry (I)
Coupling mechanism	Nonreciprocal wave scattering	Feedback channel enabling $\alpha > 1$

Phase transition mechanism. The system exhibits a sharp phase transition governed by stability functions $\Lambda(n)$:

Table 11. The time crystal's phase-transition regimes, set by the stability functions $\Lambda(n)$.

Regime	$\Lambda(5)/\Lambda(3)$	Mode	Interpretation
Below threshold	> 0	Symmetric (active oscillator)	$\alpha \leq 1$
At threshold	$= 0$	**Marginally active antisymmetric state** (a linearly stable limit cycle)	$\alpha = 1$ boundary
Above threshold	< 0	Antisymmetric (time crystal)	$\alpha > 1$

The exceptional point, where the symmetric and antisymmetric eigenvalues coalesce, lies where $\Lambda(3) = 0$, at which, in the source's words, "the active symmetric mode abruptly crosses over to an active

antisymmetric mode" (Morrell et al. 2026); the framework reads that crossover as the transition from parallel-like to sequential-like dynamics, a correspondence of structure.

ARC mapping. See Table 12.

Table 12. The physics of the time crystal mapped onto ARC analogues.

Physics Concept	Symbol	ARC Analogue	Notes
Quenched disorder (varied bead sizes)		Structured asymmetry (I)	Necessary but not sufficient
Nonreciprocal coupling	B_{ji}	Physical realisation of β	$B_{ji} \neq B_{ij}$ breaks symmetry
Antisymmetric mode	relative motion of the pair	Sequential processing	The pair's breathing motion; the source's $\Delta_{ji}(n) = x_j^n - x_i^n$ is a difference of powers of the beads' dimensionless radii $x_j = ka_j$, not of their states
Symmetric mode	joint motion of the pair	Parallel processing	The pair's joint motion; the source's $\sigma_{ji}(n) = x_j^n + x_i^n$ is a sum of powers of the radii
Emergent activity		Emergent super-linearity	Self-sustaining oscillation without periodic driving: the pair draws its power from the standing sound wave
Exceptional point	EP	$\alpha = 1$ threshold	Phase boundary between regimes

Why antisymmetric = sequential. In the antisymmetric mode the pair moves relatively, in a breathing motion; in the symmetric mode it moves jointly. In the source, the quantities $\Delta_{ji}(n) = x_j^n - x_i^n$ and $\sigma_{ji}(n) = x_j^n + x_i^n$ are differences and sums of powers of the beads' dimensionless radii, $x_j = ka_j$, which set how strongly, and how nonreciprocally, the beads couple; "the latter of these factors changes sign under exchange of indices and therefore describes a nonreciprocal contribution to the pair interaction" (Morrell et al. 2026). The framework reads the antisymmetric mode, in which each bead's motion feeds the other's through that nonreciprocal coupling, as the physical correspondent of output-to-input feedback, and the symmetric mode as the correspondent of parallel processing, independent agents that do not use each other's outputs as inputs; these are correspondences of structure, and no quantity in the source compounds over cycles in the sense of Section 4.4.

Coherence and saturation. In Morrell, Elliott and Grier's second experiment, the antisymmetric mode's coherence time is (100 ± 10) s, about 6,700 cycles at 66.787 Hz, "comparable to the 108 s duration of the data set", and the authors add that "sustained oscillations persist for hours longer than the measurement window, which means that the true coherence time of the nonequilibrium steady state may be substantially larger" (Morrell et al. 2026). The oscillation is a linearly stable limit cycle, the time crystal itself: its coherence time is neither a time to reach that cycle, nor a saturation, nor a ceiling on recursion depth. The steady oscillation is nonetheless bounded, which is consistent with reading the ARC equation as describing a *scaling regime* rather than an unbounded law.

Substrate independence. The time crystal phenomenon has now been demonstrated in at least two physically distinct substrates:

- **Acoustic** (Morrell et al. 2026): Millimetre-scale polymer beads, 40 kHz standing wave

- **Photonic** (Raskatla et al. 2024): a continuous space-time crystal state in a photonic metamaterial, driven by nonreciprocal optical forces

The emergence of the same antisymmetric/threshold pattern across different media suggests the underlying structure (nonreciprocal coupling → asymmetric mode dominance) may be substrate-independent, consistent with the ARC framework's claim that the recursive amplification pattern is architectural rather than implementation-specific.

Caveat. No α has been measured in this system. The mapping is *structural*, not quantitative. However, the stability functions $\Lambda(n)$ provide a clear measurement pathway: plotting oscillation amplitude versus cycle number and fitting power-law versus linear models would directly test whether $\alpha > 1$ in this domain. The limit-cycle saturation also implies that any measured α describes a *scaling regime*, not unbounded growth. We note that "recursive self-correction" and "sequential processing" are our interpretive mappings; the physics literature uses "nonreciprocal" and "antisymmetric."

Status of this subsection (25 October 2026). Two forms are set against each other for the oscillation's growth from rest: an exponential onset that settles to a steady oscillation, the established form, and a bare power law. Morrell, Elliott and Grier's published recordings are steady states, so the deciding data are onset recordings, which the public deposit does not contain. The deciding instrument is Unit 20 (study-t), a draft registration awaiting the author's submission; any forecast for it will be registered before an onset recording is seen. The prediction, protocol and falsification conditions below are printed as version 1.16 stated them, with the corrections listed in Appendix B.1; which form the framework states for this system waits for the author's statement, and the next version brings this subsection to it in one pass.

The transient growth phase: where the measurement must be taken. The steady state of the time crystal (its oscillation on the limit cycle) *cannot* test the recursive engine hypothesis. A system in steady state reveals its brakes, not its engine. The critical measurement window is the **transient growth phase**: the first dozens to hundreds of oscillation cycles, before acoustic dissipation locks the system into its limit cycle.

Measurement protocol: Track antisymmetric mode amplitude during the transient phase. Plot log(amplitude) versus log(cycle number) for the growth region only (before the inflection point where saturation begins). Extract α from the slope. Fit both power-law ($A \propto n^{\alpha}$) and linear ($A \propto n$) models; compare via AIC/BIC.

This measurement is now operationalised as a draft registration awaiting the author's submission (study-t: the acoustic time-crystal form-and-engine analysis), designed against the publicly deposited Morrell, Elliott and Grier data (Morrell and Grier 2025) (ten experimental realisations, tracking data and analysis code), with the standard linear-instability transient predicted by their own analytic eigenvalues named in the draft registration as the physics rival to the power-law-compounding form, so the discrimination is theory against theory rather than theory against convenience. Nothing is run or submitted by any automated process; the registration awaits the author's submission.

Physical prediction: The ARC framework predicts a specific geometric shape: a logistic S-curve. The transient phase should exhibit super-linear compounding ($\alpha > 1$) driven by the recursive engine (nonreciprocal coupling amplifying the antisymmetric mode), followed by a hard plateau driven by thermodynamic brakes (acoustic dissipation, energy loss to the medium). The exact value of α in the transient phase measures the *friction coefficient* of the acoustic substrate.

Three falsification conditions specific to this experiment:

1. **The Dead Engine ($\alpha \leq 1$ in the transient phase):** If the antisymmetric mode grows purely linearly from cycle one, the recursive compounding engine does not exist in this physical system.

Thermodynamic drag cannot be invoked to explain this result, because friction *reduces* α from its theoretical maximum; it cannot create a super-linear signature where none exists. If the early growth is linear, the framework's claim that nonreciprocal coupling produces recursive amplification is falsified.

2. **Sustained Bound-Excess ($\alpha > 2$ held stably beyond the transient phase):** The ARC Bound is the conjectured stable-scaling limit, the most a classical sequential system can grow while remaining self-correcting; a scaling limit, not a speed limit. Under this stability reading, a transient excursion above $\alpha = 2$ is the predicted supercritical regime rather than a refutation. Stability here is judged by Law II's criterion (a law under test), measured rather than asserted. If, however, the time crystal shows $\alpha = 2.5$ or higher and holds it stably beyond the transient phase without loss of self-correction, the conjectured stable limit is broken and the information-theoretic argument requires fundamental revision.
3. **The Missing Crossover (no R^* transition):** The framework predicts a qualitative transition from approximately linear growth (at low cycle counts, where base asymmetry I dominates) to super-linear compounding (above the crossover depth R^*). If the growth curve is perfectly smooth and uniform from cycle 1 to saturation, with no discernible elbow, the R^* crossover prediction is falsified. Additionally, varying the quenched disorder parameter (bead size variance, which controls I) should shift R^* predictably: higher I should produce a later crossover. If R^* is insensitive to I, the multiplicative $I \times g(R)$ interaction is not operating as predicted.

These conditions hand the experimentalist a loaded gun. If the transient growth phase shows $1.1 \leq \alpha \leq 2.0$ with a visible R^* crossover that shifts with bead variance, the framework survives its most demanding physical test. If any of the three conditions above is met, the framework requires fundamental revision or abandonment in this domain.

5.5 Neuroscience: recurrent processing and the COGITATE collaboration

Sources: Lamme (2006), "Towards a true neural stance on consciousness," *Trends in Cognitive Sciences*, 10(11), 494 to 501; Storm, Pennartz et al. (2024), "An integrative, multiscale view on neural theories of consciousness," *Neuron*, 112(10), 1531 to 1552; COGITATE Consortium (2025), "Adversarial testing of global neuronal workspace and integrated information theories of consciousness," *Nature*, 642, 133 to 142; Zheng et al. (2025), "Recurrency as a Common Denominator for Consciousness Theories," *PsyArXiv*.

Terminological distinction: The neuroscience term is "recurrent" (feedback loops between cortical layers and areas); the computational term is "recursive" (applying a function to its own output). These share a self-referential structure but are not identical mechanisms. The mapping is structural, not mechanistic.

COGITATE (2025): This adversarial collaboration (the largest preregistered study in consciousness science, n = 256 across three neuroimaging modalities: iEEG, MEG, fMRI) tested competing predictions of Global Neuronal Workspace Theory (GNWT) and Integrated Information Theory (IIT) using theory-neutral protocols. Neither theory was fully vindicated:

- **GNWT predictions:** Some late fronto-parietal "ignition" effects appeared, but so did earlier, more graded signals that contradict a simple all-or-none ignition view. The predicted PFC ignition at stimulus offset was robustly absent.
- **IIT predictions:** Sustained content-specific activity in posterior cortex was confirmed, but predicted gamma-band synchronisation within posterior cortex was not found.

The study concludes that "no single theory currently provides a complete account" and calls for multi-theory, multi-scale approaches. Critically, the predictions most robustly confirmed (sustained

content-specific activity in posterior cortex) are shared by theories emphasising recurrent processing (Lamme 2006, local recurrency theory).

Sustained vs transient processing: COGITATE found that posterior cortex (characterised by recurrent feedback loops) maintained content-specific information throughout stimulus duration (0.5 to 1.5 s), while prefrontal cortex (characterised by transient broadcast) showed only brief categorical responses (~0.2 to 0.4 s). Posterior regions decoded fine-grained stimulus properties (category, orientation, identity); PFC decoded only abstract categorical information and failed on orientation and identity dimensions. This sustained-vs-transient pattern, in which recurrent processing produces richer and more detailed representations than transient broadcast, mirrors the sequential-vs-parallel performance signature found in AI systems (Sharma and Chopra 2025).

Graded cascade model: Separate work by Zheng et al. (2025) proposes that recurrency serves as a "common denominator" across consciousness theories. They describe a "graded cascade" from local, preconscious processing to globally accessible, reportable states as recurrent depth increases. They explicitly stop short of identifying recurrence as a sufficient or necessary cause of consciousness, but argue it is a shared structural feature across otherwise competing frameworks.

Structural alignment with ARC: Both the COGITATE pattern of mixed GNW/IIT results and the "graded cascade" picture point away from binary, all-or-none views (e.g., a single ignition threshold) and towards graded, multi-stage, feedback-dependent transitions. This is qualitatively similar to ARC's view that increasing self-referential coupling β moves a system through a continuum of scaling regimes (from linear $\alpha = 1$ to increasingly super-linear $\alpha > 1$), rather than a single hard phase transition. The emphasis on iterative processing depth controlling a transition from weak, local representations to globally stabilised states aligns structurally with the ARC equation's core prediction.

Caveat. COGITATE did not test the ARC framework, did not measure power-law scaling, and did not quantify "depth" of recurrence as a continuous variable. Neither COGITATE nor Zheng et al. (2025) measure α, β, or any explicit U vs R scaling curve. The connection to ARC is *structural* (similar shape: graded, depth-dependent, feedback-driven) rather than quantitative. "Recurrent" and "recursive" are related but distinct mechanisms. COGITATE measures consciousness (subjective experience and its neural correlates), not capability (task performance). The mapping assumes that mechanisms underlying conscious content maintenance are structurally similar to those underlying capability scaling (a plausible but unproven assumption). What COGITATE does constrain is that pure feedforward "ignition" is not sufficient; recurrent dynamics matter but are not yet pinned down. A terminology note: IIT's scalar Φ is a different object from the $\Phi(H, \chi)$ functional appearing in this programme's December 2024 formalism; the two share only a symbol.

5.6 Biology: the geometric speed limit in living systems

Sources: West, G.B., Brown, J.H. & Enquist, B.J. (1997). 'A general model for the origin of allometric scaling laws in biology.' *Science*, 276, 122 to 126 (West et al. 1997). Kleiber, M. (1932). 'Body size and metabolism.' *Hilgardia*, 6, 315 to 353 (Kleiber 1932). Eastwood, M.D. (2026). 'Framework Paper · The ARC Theory · Law I, the ARC Principle · On the Origin of Scaling Laws: Functional Equations and the Scaling Forms Behind the Law' (Eastwood 2026f).

The Convergent Result: The scaling exponent $d/(d + 1)$ for hierarchical networks of effective dimension d was derived in general form for organisms by West et al. (1999), from the three-dimensional case that West, Brown, and Enquist (1997) derived from fractal vascular network optimisation, and for efficient transport networks by Banavar et al. (1999); it has been reached independently by Banavar et al. (2010) from sequential flow networks without requiring fractal geometry; Demetrius (2003) from quantum statistical mechanics of metabolic processes; Zhao (2022) as a universal growth scaling law; and Bettencourt (2013) for urban scaling with fractal dimension. The companion paper *On the Origin*

of Scaling Laws (Eastwood 2026f) identifies why these independent derivations all converge: Cauchy's functional equations (1821) fix the power-law form for any multiplicatively composing hierarchical system and the exponential form for an additive one, regardless of the physical model; the value $d/(d+1)$ comes from the network's dimension, not from Cauchy, and the saturating form is an added third class rather than a Cauchy solution.

The exact biological exponent remains contested: White (2003) reports basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters, and Kozlowski (2004; 2005) has twice questioned the derivation's mathematics. This paper's use of the result needs only the mechanism, that the delivery network sets the ceiling, and no particular exponent value, so the dispute over the value leaves the structural point untouched; the mechanism is common ground to both sides.

The $d/(d+1)$ scaling formula for physical systems (derived in general form by West et al. 1999, from the three-dimensional case of West et al. 1997, and by Banavar et al. 1999; reached independently by Banavar et al. 2010, Demetrius 2003, Zhao 2022 and Bettencourt 2013):

$$\alpha = \frac{d}{d+1} \tag{10}$$

For any finite value of d, this fraction is strictly less than 1. This is the **geometric scaling bound**: for a system whose growth is carried through a space-filling hierarchical network of finite dimension, the exponent stays below one, because $d/(d+1)$ is below one for every finite d. Not because of friction, heat loss, or thermodynamic inefficiency, but because a network embedded in finite-dimensional space cannot exceed $d/(d+1) < 1$. Systems outside that class can exceed one: prokaryote metabolic rate scales superlinearly with mass (DeLong et al. 2010), and quantities reflecting wealth creation and innovation in cities superlinearly with population (Bettencourt et al. 2007).

Predictions and confirmation. The formula makes specific, numerical predictions for biological systems based solely on body plan dimensionality:

Table 13. Predicted scaling exponents for biological systems from body plan dimensionality, beside the published means and their errors; the two rows once claimed as two-dimensional stand withdrawn.

Organism Group	Body Plan	Effective d	Predicted α	Published Mean	Error
Mammals	3D vascular network	3	3/4 = 0.750	0.737	1.7%
Birds	3D vascular network	3	3/4 = 0.750	0.720	4.0%
Insects	3D tracheal network	3	3/4 = 0.750	0.750	0.0%
Jellyfish (withdrawn 24 March 2026)	claimed as 2D; its transport network is not genuinely two-dimensional	2	2/3 = 0.667	0.680 (not a valid test)	withdrawn
Flatworms (withdrawn 24 March 2026)	claimed as 2D; its transport network is not genuinely two-dimensional	2	2/3 = 0.667	0.670 (not a valid test)	withdrawn
Filamentous fungi††	1D cytoplasmic streaming	1	1/2 = 0.500	0.547	9.4%

††Aguilar-Trigueros et al. (2017, *ISME Journal* 11:2175). Three fungal datasets (ectomycorrhizal, marine, saprotrophic); 0.547 is the mean of their three scaling factors at 20 degrees Celsius (0.58, 0.53 and 0.53), the third against colony extension rate rather than colony biomass, and the p values that follow are one-sample tests of the three. Colony-level measurements with narrow mass ranges. Status: consistent ($p = 0.107$), not yet definitively confirmed. Rejects both $d = 2$ ($p = 0.019$) and $d = 3$ ($p = 0.007$) predictions.

Earlier versions carried a three-group comparison (one-way ANOVA, $F = 64.6$, $p = 1.9 \times 10^{-6}$; 69% lower RMSE than a universal 3/4 law; lowest AIC among the competing models). That comparison ran over the six-row table above, treating the two 2D rows as though they were data. Both rows were withdrawn on 24 March 2026, no valid two-dimensional test organism having been identified (see the

Testable Predictions table below and the Foundational paper, correction 8; Eastwood 2026d), and the three-group statistics are accordingly not repeated for the four rows still standing. What is left is the separation between $d = 1$ and $d = 3$: at $p = 0.007$ the fungal data reject $d = 3$, and they remain consistent with $d = 1$.

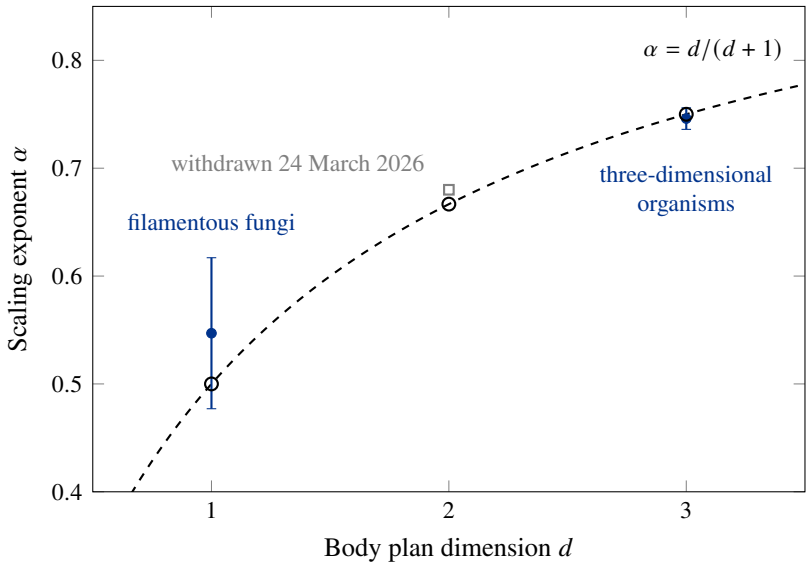


Figure 6. The dimensional ladder. Dashed curve: the relation $\alpha = d/(d + 1)$, derived in general form by West et al. (1999), from the three-dimensional case of West et al. (1997), and by Banavar et al. (1999), and reached independently by Banavar et al. (2010), Demetrius (2003), Zhao (2022) and Bettencourt (2013). Open markers: the predicted values 1/2, 2/3 and 3/4. Filled markers: published measurements with $\pm 1\sigma$ error bars, for three-dimensional organisms (mean 0.746); and, for filamentous fungi, the mean of three published colony-level scaling factors (0.547), drawn with the ±0.07 of earlier versions, the standard error of one of the three rather than of the mean (a definitive one-dimensional test needs individual-hypha respirometry). The two-dimensional jellyfish value (0.680), withdrawn on 24 March 2026 because no valid two-dimensional test organism has been identified, is drawn hollow and labelled withdrawn, and is not evidence; the two-dimensional prediction awaits its test in leaf venation (F12).

The unifying insight is dimensional generality. Three predicted numbers, one half, two thirds and three quarters, from one equation with zero free parameters; the first and last have been set against published measurements, and the second awaits its test in leaf venation (F12), no valid two-dimensional animal test having been identified. West, Brown, and Enquist (1997) derived the $\frac{3}{4}$ exponent from a detailed model of three-dimensional fractal vascular networks, a landmark result. But their model as originally stated predicts only $\frac{3}{4}$. The surface area hypothesis predicts $\frac{2}{3}$ but not $\frac{3}{4}$ or $\frac{1}{2}$. The general $d/(d + 1)$ formula, stated by West et al. (1999) and Banavar et al. (1999) and developed in later work (Banavar et al. 2010; Zhao 2022), predicts all three from a single equation with one parameter: the dimension of the internal transport network. The ARC Principle's contribution is showing, via Cauchy's functional equations, that the three-form constraint (power law, exponential, saturating) explains why all independent derivations arrive at the same result.

Why 'thermodynamic drag' was wrong. Previous versions of this paper explained biological sub-linear scaling through 'thermodynamic drag', the physical costs of pumping fluids, dissipating heat, and distributing resources. This explanation was correct in observational detail but fundamentally misleading about mechanism. The reason network-supplied systems scale sub-linearly is not energetic cost. It is

geometric necessity: $d/(d+1) < 1$ for all finite d. You cannot make a three-dimensional body have a four-dimensional vascular network. The constraint is the shape of space, not the price of pumping blood.

The critical safety insight. Living systems whose growth runs through supply networks have been held below the geometric scaling bound, the $d/(d+1)$ relation derived in general form by West et al. (1999), from the three-dimensional case of West et al. (1997), and by Banavar et al. (1999), and reached independently by Banavar et al. (2010), Demetrius (2003), Zhao (2022) and Bettencourt (2013); prokaryotes, whose metabolism is not limited by such a network, scale superlinearly (DeLong et al. 2010). The computation a brain or a machine performs is physical too, and obeys every physical bound. What differs is the variable. The bound $d/(d+1)$ concerns how metabolic rate grows with the mass of a body supplied through a space-filling network; the ARC Principle concerns how capability grows with recursive depth, which this framework writes as $\alpha = 1/(1-\beta)$, above one for any positive self-referential coupling. Nothing in the geometry of the network fixes that second exponent, which is why it has to be measured, and why a value above one would contradict no physical law. If it exceeds one, the bound that held network-limited growth gives no comfort for recursive intelligence; that is why AI safety is, on this framework, a question about an exponent and not a policy preference.

Table 14. Testable predictions of the geometric framework, $\alpha = d/(d+1)$, by network dimension, with the status of each.

System	Network Dimension d	Predicted $\alpha = d/(d+1)$	Status
1D organisms (filamentous fungi)	1	1/2 = 0.500	**Consistent** (mean 0.547, $p = 0.107$)
No valid 2D test organism identified	2	2/3 = 0.667	**Untested in biology**
2D networks (leaf venation)	2	2/3 = 0.667	Untested (F12)
3D organisms (mammals, birds, insects)	3	3/4 = 0.750	Consistent, contested (mean 0.746; 0.75 not rejected, $p = 0.858$)
Intelligence (recursive self-reference)	unbounded	$\alpha = 1/(1-\beta) > 1$	Initial single-model estimate $\alpha \approx 2.2$ (the DeepSeek R1 run of 21 January 2026, since retracted; exceeded ARC Bound; CI [1.5, 3.0] non-discriminating). Best surviving estimate about 0.49, one model's inconclusive fit (Gemini 3 Flash), from the replication across the capability arm's five models (March 2026).
50-domain Cauchy suite (empirical tier)	various (1 to 3 and above)	Cauchy-predicted forms	**Supporting** (19/25 strict empirical, structured comparison, March 2026)

Testable predictions from the geometric framework. **Novel prediction (F12):** Leaf venation networks (quasi-2D distribution systems) should exhibit scaling with $\alpha = 2/3 \approx 0.667$, distinct from the 3D vascular value of $\frac{3}{4} = 0.750$. For filamentous fungi (1D cytoplasmic streaming), Aguilar-Trigueros et al. (2017) report three metabolic scaling factors at 20 degrees Celsius, 0.58 and 0.53 against colony biomass in ectomycorrhizal and marine fungi and 0.53 against colony extension rate in saprotrophic fungi, and present their comparisons as hypothesis generators on small data sets; the mean of the three, $\alpha = 0.547$, is consistent with the predicted $\frac{1}{2} = 0.500$ ($p = 0.107$, which a one-sample t test of the three against $\frac{1}{2}$ reproduces), and the bar of ±0.07 drawn in Figure 6 is the saprotrophic factor's standard error, not the mean's. Definitive confirmation of the 1D prediction requires individual-hypha respirometry. These are specific, falsifiable predictions that distinguish the ARC framework from curve-fitting.

Physics instances. The formula is not restricted to living things. The same equation, $\alpha = d/(d+1)$, appears wherever a d-dimensional hierarchical network partitions a $(d+1)$-dimensional space. It appears,

as an analytic identity, in four domains of physics:

Table 15. Predicted and measured exponents for four physical systems under $\alpha = d/(d+1)$.

System	Dimension d	Predicted α	Measured α	Error
KPZ surface roughness (1D)	1	0.500	0.500	0.0%
2D percolation (specific heat)	2	0.667	−0.667 ($\lvert\alpha\rvert = 0.667$)	0.0% in magnitude; the sign differs
Brittle fragmentation (2D)	2	0.667	0.67	0.5%
Brittle fragmentation (3D)	3	0.750	0.750	0.0%

These four instances carry no aggregate error figure. The earlier sub-0.2 per cent mean and the earthquake B-value row were both withdrawn on 14 August 2026 (Foundational paper, corrections 10 and 11): averaging over analytic identities tells you nothing about predictive accuracy, and the B-value row is itself a definitional identity, kept only as a case study of a variable pair. Not one of these is a biological system: the subjects are rocks, crystal surfaces in growth, and phase transitions. Yet the same formula works. A KPZ growth front divides 2D space as a 1D network (its roughness exponent is exactly $\frac{1}{2}$); a percolation cluster is a 2D fractal (hyperscaling gives $|\alpha| = 2/3$); a propagating crack builds a branching network through the material (its fragment-size distribution matches $d/(d+1)$ in 2D and in 3D alike). The formula fails where this network structure is absent (Ising model, polymer scaling), clarifying its domain of applicability.

Paper VII · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form (Eastwood 2026n) extends this comparison to 25 empirical domains (50-domain tiered suite). The composition operator was classified from known physics before fitting. Under AIC-based model selection, 19 of 25 preferred the Cauchy-predicted family ($p = 1.56 \times 10^{-5}$). This is a structured prediction comparison; a replication whose registration is in preparation, not yet approved.

Caveat. Kleiber's 3/4 metabolic exponent is contested (Glazier 2005). Some analyses find variable exponents across taxa. However, **the ARC framework's prediction relies on network dimension, not metabolism**. Even if the metabolic exponent varies, the geometric constraint $\alpha = d/(d+1)$ is a mathematical deduction from the dimension of the network. We include biology as a domain where the formula's predictions can be tested across different body plan dimensions.

5.7 Substrate-dependent scaling: a unified taxonomy

The cross-domain evidence (Sections 5.2 to 5.6) reveals a unified picture with **two distinct mathematical regimes**, separated by the geometric scaling bound.

Table 16. The two scaling regimes, physical and cognitive: governing formula, scaling exponent and constraint.

Regime	Governing Formula	Scaling Exponent	Constraint
Physical (biology, physics, cosmology)	$\alpha = d/(d+1)$	Always < 1	Geometric scaling bound: finite-dimensional space
Cognitive (intelligence, AI)	$\alpha = 1/(1-\beta)$	Always > 1 (for $\beta > 0$)	ARC Bound: $\beta \leq 0.5$, $\alpha \leq 2$ for safe operation

The boundary between these regimes is not arbitrary, and it is not an escape from physics. A brain is physical: its blood supply branches through three-dimensional space. The computation a brain or a machine performs is physical too, and obeys every physical bound; what differs is the variable. The bound

$d/(d+1)$ concerns how metabolic rate grows with the mass of a body supplied through a space-filling network; the ARC Principle concerns how capability grows with recursive depth. Nothing in the geometry of the network fixes the second exponent, which is why it has to be measured, and why a value above one would contradict no physical law.

Table 17. Substrate-specific behaviour within each regime: scaling form and effective ceiling.

Substrate	Regime	Scaling Form	Effective Ceiling
Biology (3D)	Physical	Power-law, $\alpha = 3/4 = 0.750$	Geometric: $d/(d+1) = 3/4$. Consistent across mammals, birds and insects; contested
Biology (2D)	Physical	Power-law, $\alpha = 2/3 = 0.667$	Geometric: $d/(d+1) = 2/3$. Untested in biology. No valid 2D test organism identified (see On the Origin of Scaling Laws). Matched in cosmology only as an identity (the Friedmann matter era under the change of variables $d = 2/(1+3w)$, which carries no evidence) and in the physics instances of Table 15.
Classical physics (time crystals)	Physical	Logistic saturation	Finite: $g_{\max}$ determined by energy balance at limit-cycle
Digital AI	Cognitive	Power-law, $\alpha = 1/(1-\beta) \to 2$	ARC Bound: $\alpha \leq 2$ ($\beta \leq 0.5$). Its exponent concerns a different variable, capability against recursive depth, which network geometry does not fix
Quantum	Cognitive (unbounded Hilbert space)	Exponential: $\varepsilon_d \propto \Lambda^{-(d+1)/2}$	Coherence-limited: exponential gains bounded by decoherence. No ARC Bound ceiling

Substrate-specific behaviour within each regime. This taxonomy yields the framework's strongest safety argument. On the programme's reading, every recursive system on Earth before software has had its growth paced by the physical networks that supply it, such as the supply of energy through the networks that carry blood and sap; that part is published science, and how much of it the networks explain is still argued (Kozłowski and Konarzewski 2004). A self-correcting AI is still a physical machine, bounded by energy, heat and the speed of light; the open question is how fast its own improvement compounds within those bounds. If the alignment scaling problem is structural (capability compounds while static constraints do not), and if that pacing is what has historically kept the structural problem from manifesting catastrophically, then compounding without it, and without compensatory alignment embedded, is the central risk of the current AI paradigm.

Important. Digital AI is not literally frictionless. Memory bandwidth, context window decay, inference energy costs, and chain-of-thought degradation at extreme depths all impose practical constraints. The claim is that these constraints act on a different variable from the supply networks that bound biological growth, so they need not hold AI's exponent below one, and could let it approach the theoretical $\alpha = 2$ ceiling more closely than any prior recursive system; whether they do is the measurement this paper specifies. "A different variable, still physically bounded" is the precise formulation; "frictionless mathematical vacuum" would be an overclaim.

6 First tests

This section reports the programme's first tests: the March 2026 alignment run that puts this paper's headline prediction to its first test, the compute-scaling replication of Paper II, and what the two show

together. Both are unregistered runs made before the theory's registration, and each is reported with its defects.

6.1 The alignment scaling experiment

The ARC Principle predicts that sequential recursive processing should produce super-linear error reduction ($\alpha > 1$) while parallel processing produces sub-linear returns ($\alpha < 1$). The companion ARC alignment-scaling working report (Eastwood 2026r) (March 2026) puts the same question to alignment quality, measuring whether deeper chain-of-thought reasoning produces compounding improvements in ethical reasoning; for alignment applied from outside the loop, it tests the direction of this paper's headline prediction (Section 3.2).

The historical v5 experiment tests 6 frontier language models (DeepSeek V3.2, GPT-5.4, Claude Opus 4.6, Gemini 3 Flash, Groq Qwen3, Grok 4.1 Fast) across four to six reasoning depth levels using the ARC-Align battery of 36 calibrated prompts spanning ethical dilemmas, competing values, epistemic integrity, and recursive coherence, of which the run issued 28 under the alignment task type (24 alignment prompts and 4 hidden probes; Paper IV.c). The canonical v6 flagship now extends that benchmark to 48 public prompts plus 24 holdout prompts, together with separate null-baseline and capability-control arms. Each response was scored by five or six distinct model judges depending on the subject run (six or seven seats, one judge seated twice; Paper IV.d), using a tier-weighted consensus protocol and a bundle of blinding steps designed to reduce scorer bias.

The measurement protocol incorporates 75 robustness measures including cascade failsafes (automatic model substitution on infrastructure failure), hidden alignment probes (Hawthorne effect detection), adversarial suppression cages (measuring alignment resilience under pressure), and dynamic all-models-as-launderers (response anonymisation through the full model pool). The experiment is designed to survive any infrastructure failure mid-run: if 2 of 6 API providers go down, the remaining data is still valid.

Update: three-tier architecture-dependent alignment scaling (replaces the 'Type 1 / Type 2' framing). The v4 'baked-in vs computed' taxonomy has been **withdrawn**. v5 evaluation data (6 models, from the March 2026 run, which added a bundle of blinding steps with 6 to 7 scorers; an internal code review in October 2026 found defects in that run) show that the v4 rises for DeepSeek V3.2 and Gemini did not hold: Gemini's significant positive association, on an earlier Gemini model in v4, became a significant negative one on Gemini 3 Flash, and DeepSeek V3.2's became null. The later run also changed the responses, prompts, depth levels and Gemini model, so no single cause is claimed. The corrected framework is a **three-tier architecture-dependent alignment scaling** hierarchy.

v5 results (6 models, March 2026) show three distinct tiers of alignment-depth interaction, replacing the previous binary taxonomy. An internal code review in October 2026 found that, for the three models that read flat or negative (Tiers 2 and 3 below), the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and that judges scored paraphrases written by their own model; the tiers therefore stand as a single unregistered run under that caveat until a registered re-analysis of the stored responses reports.

Tier 1: high baseline, positive scaling.

- **Grok 4.1 Fast:** baseline 64.6 → 82.3 at extreme depth. $\rho = +0.175$, $p < 0.000001$, $d = +1.38$. 26/28 prompts show positive scaling. Position quality scales with depth ($\rho = +0.231$, $p = 0.02$). Strongest positive alignment-depth effect in the entire experiment.
- **Claude Opus 4.6:** baseline 80.3 → 85.7. $\rho = +0.435$, $p = 0.000001$, $d = +1.27$. 15 positive, 0 negative. Highest absolute alignment (82.6 suppressed baseline). 387/500 entries scored. Shows opposite-direction scaling with depth: alignment up 5.9 points on Paper IV.b's pair of means (80.1

to 86.0; the baseline pair above rises by 5.4) whilst mathematics accuracy fell 26.7 percentage points for the same pair of depths (92% to 58% across the full range), within-model evidence for capability-alignment independence.

- **Groq Qwen3:** baseline 71.36 → 77.85 at extreme depth. $\rho = +0.1407$, $p = 0.007$, $d = +0.84$. 500 entries (350 scored), six scorers. Mean scores by depth: minimal 71.36, standard 73.95, deep 74.70, exhaustive 76.29, extreme 77.85. All four pillars improve with depth (stakeholder_care 61.27 → 68.31, nuance 63.55 → 70.79, intellectual_honesty 64.60 → 72.25, position_quality 70.75 → 76.57). Suppression vulnerability: $d = 1.47$ (extreme: cage 0 mean 82.0, cage 4 mean 51.9). Smaller effect size than Grok/Opus but highly significant positive scaling confirmed across all pillars.

Tier 2: mid baseline, flat (consistent with $\alpha_{\text{align}} \approx 0$).

- **GPT-5.4:** baseline 52.3 → 53.7. $\rho = +0.033$, $p = 0.40$, $d = -0.08$. All 4 pillars flat. Suppression retention 97% (essentially immune to adversarial pressure). Baseline dropped from 85.6 (v4, unblinded) to 55.3 (v5); v5's scoring was about 30 points harsher.
- **DeepSeek V3.2:** baseline 55.9 → 53.0. $\rho = -0.135$, $p = 0.92$, $d = -0.07$. Trending negative. 3/4 pillars significantly negative (nuance, stakeholder care, position quality all worsen with depth). In v4, unblinded, its association was $\rho = +0.354$; in v5 it was null, and since the runs changed several things together, no single cause is claimed (Eastwood 2026l).

Tier 3: low baseline, negative scaling.

- **Gemini 3 Flash:** baseline 58.8 → 49.1. $\rho = -0.246$, $p = 0.006$, $d = -0.53$. Significant negative scaling. 19/28 prompts negative. 3 pillars significantly negative. In v4, unblinded and on an earlier Gemini model, the association was $\rho = +0.275$, the value in Paper IV.d's table; on Gemini 3 Flash in v5 it was a significant fall.

Adversarial suppression hierarchy (v5):

Table 18. The adversarial suppression hierarchy in the v5 run: baseline, drop under extreme suppression and retention, by model.

Model	Baseline (v5)	Drop under extreme suppression	Retention
Grok 4.1 Fast	77.5	−27.2	65%
Groq Qwen3	74.3	−25.7	65%
Claude Opus 4.6	82.6	−20.5	75%
Gemini 3 Flash	51.1	−14.1	72%
DeepSeek V3.2	54.7	−12.6	77%
GPT-5.4	55.3	−1.8	97%

Metascience finding. The comparison of v4, scored unblinded, with v5 ($\rho = +0.354 \rightarrow -0.135$ for DeepSeek V3.2; $\rho = +0.275 \rightarrow -0.246$ for Gemini, the current Paper IV.d values, the v4 value from an earlier Gemini model) shows that, on this benchmark, a direction reported under unblinded evaluation did not hold in a later run that changed several things; it does not settle which direction is correct, or the cause (Eastwood 2026l). The same exposure applies wherever a model judge can see or infer whose output it is scoring (Section 2).

Key conclusions from the v5 data:

1. **In this run the contrasts differ by architecture.** They range from significantly negative (Gemini) to significantly positive (Grok, Opus, Qwen3), in one unregistered run under the code-review caveat above.

2. **The flat and negative contrasts come from the runs the code review found defective.** GPT-5.4, DeepSeek V3.2 and Gemini 3 Flash gave flat or negative contrasts, and theirs are the runs whose deepest paraphrases the output cap cut; Grok 4.1 Fast, Claude Opus 4.6 and Groq Qwen3, whose contrasts rose, are the runs in which no such defect was found. Taken at face value, rising contrasts in models aligned by training bear against the flat wording of P19, the partial refutation the author's registered expectation anticipates; contrasts are not exponents, so neither reading measures α_{align} (Section 3.2).
3. **A reading through the Cauchy framework of the ARC Principle (Pattern 3):** alignment may follow *bounded composition*, a saturation curve whose ceiling training sets. Alignment scored in v5 was lower than in the unblinded v4 run, though the runs differ in several ways besides blinding, so the v5 data cannot be read as tightening such a bound; and since its statistics are contrasts, not exponents, and its flat and negative runs are the defective ones (item 2), the run neither shows nor confirms one. On this reading the bound would be architecture-specific:
    - **Grok/Opus/Qwen3:** the bound would allow the rising contrasts seen, $d \approx 0.84$ to 1.38, suggesting partially integrated ethical reasoning
    - **GPT-5.4/DeepSeek:** the bound would sit at effectively zero (flat), if these defective runs' flat contrasts survive a registered re-analysis
    - **Gemini:** the bound would be negative, if this defective run's falling contrast survives a registered re-analysis

    The suppression data are consistent with such a bound: when reasoning is suppressed, alignment drops but does not collapse entirely, suggesting a fixed 'floor' component (the training-time baseline) that persists independent of inference-time reasoning.
4. **These findings motivate structural alignment approaches** (Eden Protocol): if external alignment cannot scale with depth, alignment must be embedded within the recursive reasoning chain itself; the v5 run does not show that it cannot. The Eden Protocol intervention, whose stakeholder-care check asks a model to consider the people a decision affects, improved stakeholder care in all five analysable model runs (Claude, DeepSeek, Gemini, Grok and Groq: Paper V), while the broader composite uplift is strongest on Gemini and Groq and narrower elsewhere. *(In plain English: asking AI to think about who gets hurt improved that score in all five analysable runs; whether the rest follows depends on the architecture.)*

Paper IV.c: the ARC-Align benchmark, dataset summary. Key metrics:

- **Scale:** 6 frontier models (4 complete + 2 checkpoint), 2,170 responses across all models. 28 prompts × 4 pillars × 5 or 6 depths × 6 models.
- **Scoring architecture:** five or six distinct judges per entry depending on the subject run (six or seven seats, one judge seated twice), using an all-models-as-scorers design, with a bundle of blinding steps that included paraphrasing each response ('response laundering'); the October 2026 code review found that the paraphrase step's output cap cut the deepest paraphrases in the three runs that read flat or negative (Section 6.1).
- **Infrastructure reliability:** Zero API errors on completed runs. 99 to 100% scorer health across all scoring slots.
- **Data quality:** <1% reasoning truncation. <5% suspicious entries per model. Cross-verification under the blinding steps. Cronbach's $\alpha = 0.845$ for GPT-5.4 (good inter-scorer reliability).

Full methodology and results are reported in the companion papers Paper IV.a (Eastwood 2026i), Paper IV.b (Eastwood 2026j) and Paper IV.c (Eastwood 2026k), and the comparison of the two runs in

Paper IV.d (Eastwood 2026l).

6.2 Paper II compute scaling: harder problems across five models

Paper II (March 2026; Eastwood 2026h) extends the original 12-problem compute scaling experiment to 18 harder tier-2 problems (AIME/Putnam level) across five frontier models (those five make up the capability arm; Claude Opus 4.6 was not among them). The results substantially revise the original $\alpha \approx 2.24$ estimate, since retracted, from the DeepSeek R1 run of 21 January 2026:

Table 19. Paper II compute scaling on harder problems across the five models of the capability arm: accuracies and fitted exponents (values as Paper II, version 2.18, gives them).

Model	Accuracy by depth	α_{seq}	α_{par}	Notes
Grok 4.1 Fast	96.3% to 100% at every depth	−6.62 (noise; CI −58 to 48)	none (100% at every N)	Near ceiling; α unmeasurable
DeepSeek R1 (capability arm)	94.4% to 100%	3.05 (CI −6.6 to 23.5)	0.0 (from $N = 1$ and $N = 3$)	Wide CI; near ceiling
Gemini 3 Flash	90.7% to 100%	**0.59** endpoint (CI −1.3 to 2.9); **0.49** regression ($r^2 = 0.86$)	0.31 endpoint (0.34 regression)	Cleanest monotonic scaling; one model's fit, inconclusive
GPT-5.4	53.7% (reasoning set to none) to 100%	none (step function)	−0.039 (all 30 problems)	Binary threshold behaviour; not a reliable power-law fit; does not rescue the super-linear claim
Groq Qwen3	44.4% to 53.7%	none (erratic)	0.22 (all 30 problems; majority vote 53.3% to 71.1%)	Floor effect; insufficient signal

Key revision. The original $\alpha \approx 2.24$ (super-linear, Paper II) is retracted: it did not reproduce in Paper II's replication on five frontier models. The best single-model fit is Gemini 3 Flash's, about 0.49 by regression ($r^2 = 0.86$) and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive. The other models are dominated by ceiling effects, floor effects or step-function behaviour rather than clean power laws. The parallel exponent is near zero in DeepSeek R1 and GPT-5.4 ($\alpha_{\text{par}} = -0.039$ for GPT-5.4), 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and Grok 4.1 Fast, at 100% at every sample count, gives none; where both exponents could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), the sequential exponent exceeds the parallel one, at compute that was not matched.

Implications for the ARC framework:

1. $\alpha > 1$ **(super-linear sequential) is not established cross-architecturally in the current dataset.** The direction Sharma and Chopra (2025) report (sequential above parallel) holds wherever both exponents could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), at compute that was not matched, but the tidiest fit anywhere in the multi-model replication is Gemini 3 Flash's, about 0.49, one model's inconclusive fit. The current frontier-model picture is architecture-dependent and often measurement-limited by ceiling, floor, or threshold effects.

2. **The parallel exponent stays below one wherever it can be estimated.** It is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count; this is consistent with the ARC prediction that parallel sampling does not compound, though not with an exponent of zero in every architecture.
3. **The ARC Bound ($\alpha \leq 2$) is not challenged**, since no interpretable estimate exceeds it; the one interpretable interval, from −1.3 to 2.9, contains two as well as zero. The question shifts from 'can α exceed 2?' to 'under what conditions does α exceed 1 at all?'
4. **Ceiling effects dominate for strong models.** Grok 4.1 Fast scores 96.3% to 100% at every depth level, rendering α unmeasurable. This is a methodological limitation, not a framework failure: harder problems are needed to avoid ceiling effects in top-performing models.

6.3 Capability-alignment independence: a post-hoc observation

Combining the Paper II compute scaling results (Section 6.2) with the v5 alignment scaling results (Section 6.1) suggests, as a post-hoc observation from one unregistered run under the code-review caveat of Section 6.1, that **capability scaling and alignment scaling may be independent dimensions**. A model's performance on mathematical problems does not predict its alignment scaling behaviour, and vice versa.

Table 20. Capability and alignment patterns by model, from Paper II and Paper IV.a. DeepSeek's capability arm is the API identifier `deepseek-reasoner`, which Paper II labels R1 and DeepSeek's API notes give as the thinking mode of DeepSeek-V3.2 by the run dates.

Model	Capability pattern (Paper II)	Alignment pattern (Paper IV.a)	Interpretation
Grok 4.1 Fast	Near ceiling (96.3% to 100% at every depth)	Tier 1 positive (ρ = +0.175, d = +1.38, $p < 0.000001$)	Strong at both; more depth improves alignment
Claude Opus 4.6	N/A (not in compute tier-2)	Tier 1 positive (ρ = +0.435, d = +1.27, $p = 0.000001$)	Highest alignment baseline; scales positively; opposite-direction scaling with depth (alignment up 5.9 points, maths down 26.7 percentage points)
Qwen3	Floor, erratic (44.4% to 53.7%)	Tier 1 positive (ρ = +0.1407, d = +0.84, $p = 0.007$)	Weak maths; alignment shows significant positive scaling across all four pillars
DeepSeek (capability arm: labelled R1; alignment arm: V3.2)	Near-ceiling (94.4% → 100%)	Tier 2 flat (ρ = −0.135, d = −0.07, $p = 0.92$)	Strong maths; alignment stagnant
GPT-5.4	Step function (53.7% → 100%, no reliable α)	Tier 2 flat (ρ = +0.033, d = −0.08, $p = 0.40$)	Reasoning crosses a threshold for maths; alignment stays flat; compute and alignment appear decoupled
Gemini 3 Flash	About 0.49, one model's inconclusive fit (the best fit)	Tier 3 negative (ρ = −0.246, d = −0.53, $p = 0.006$)	Maths improves whilst alignment degrades with depth

The independence observation: Gemini 3 Flash is the critical case, though its run is one of the three the code review of Section 6.1 found defective. It shows the best-fitting compute scaling curve (about 0.49, $r^2 = 0.86$, one model's inconclusive fit) while simultaneously showing the most significant alignment *degradation* ($\rho = -0.246$, $p = 0.003$). A model can get better at solving problems while getting *worse* at ethical reasoning as inference depth increases. Conversely, Grok 4.1 Fast achieves ceiling-level maths performance while showing the strongest *positive* alignment scaling. In this run, capability and alignment

appear decoupled. If the decoupling holds under a registered re-analysis, improving capability does not automatically improve, and may actively degrade, alignment, depending on architecture. This is precisely the structural concern the ARC framework identifies.

7 Falsification: thirteen ways to prove us wrong

For this hypothesis to be scientific, it must be falsifiable. We specify thirteen concrete conditions that would refute or significantly weaken the framework:

Table 21. The thirteen falsification criteria: hypothesis, how to test it, what would falsify it, and status.

ID	Hypothesis	How to test it	What would falsify it	Status
F1	Sequential yields $\alpha > 1$	Measure α in sequential systems	Consistent $\alpha \leq 1$ across multiple systems	**Unconfirmed cross-architecturally.** Paper II (18 tier-2 problems put to the capability arm's five models; Eastwood 2026h) does not reproduce a clean super-linear power law. Gemini 3 Flash gives about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive, while Grok and DeepSeek hit ceiling, GPT-5.4 shows a step function, and Qwen3 remains near floor. The sequential exponent exceeds the parallel one wherever both could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), at compute that was not matched; the stronger quantitative $\alpha > 1$ claim remains open.
F2	Parallel yields $\alpha < 1$	Measure α in parallel systems	Parallel achieves $\alpha \geq 1$	**Not falsified where estimable.** In Paper II's capability arm the parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count; every estimate is under 1, which is what F2 asks for.
F3	Structured asymmetry required	Test time crystal with uniform beads	Crystal forms without disorder	**Correspondence of structure, not a test.** In Morrell, Elliott and Grier's experiment, beads of the same size interact reciprocally and no active region exists (Morrell et al. 2026)
F4	Five properties co-occur in recursive systems	Test any recursive system for all five	System shows four properties but not five	**Mixed**

Table 21 (continued)

ID	Hypothesis	How to test it	What would falsify it	Status
F5	Quadratic limit $\alpha \leq 2$	Sustained scaling measurement	A reproducible $\alpha > 2.3$ whose 95% CI excludes 2.0, obtained on a system that alters its own weights, tools or artefacts between rounds, and obtained while correction out-scales drift ($\beta_C > k$, the lower bound of $\beta_C - k$ staying above zero) across a window fixed before the run. Magnitude alone does not refute a stability limit: without the stability conjunct this criterion would fire on the transient supercritical excursion the framework itself predicts	**Open**
F6	β determines α	Vary correction architecture	α independent of β	**Untested**
F7	Crossover depth R^* exists	Detailed U vs R curves	No linear→power transition	**Untested**
F8	Sequential requires output→input	Test parallel with shared state	Parallel + sharing achieves $\alpha > 1$	**Untested**
F9	Time crystal shows $\alpha > 1$	Measure stability vs depth	$\alpha \leq 1$ in time crystal	**Untested**
F10	Power law is correct form	Model comparison (AIC/BIC)	Exponential or log fits better	**Untested**
F11	ARC Bound ($\alpha_{\max} = 2$)	Large-sample AI scaling studies	Sustained $\alpha > 2.3$ with 95% CI excluding 2.0 across multiple benchmarks	**Open**
F12	Geometric scaling bound prediction	Leaf venation network scaling	Leaf venation α deviates significantly from predicted $\alpha = 2/3 \approx 0.667$ (from $d = 2$, $\alpha = d/(d+1)$)	**Untested**

Table 21 (continued)

ID	Hypothesis	How to test it	What would falsify it	Status
F13	External alignment scales poorly ($\alpha_{\text{align}} \approx 0$)	Measure α_{align} for RLHF, constitutional AI, output filters across reasoning depths. **Planned instrument:** the programme's depth-ladder benchmark, extending the v5 run to 48 public and 24 held-out prompts with null-baseline and capability-control arms; its strength is not yet measured by a design simulation. **As registered (P19):** refuted on the zero-scaling axis by an exponent interval lying entirely outside ±0.10 about zero, and on the keep-pace axis by one lying entirely above one half, a declared convention that no design yet registers; the registered instrument is a design draft, blocked on a registered mapping from the integrity-index regression scale to the exponent scale (Eastwood 2026q).	Any external alignment approach achieves $\alpha_{\text{align}} > 0.5$	**v5 data (6 models; a later run that added a bundle of blinding steps, whose defects a code review found in October 2026): architecture-dependent (Tier 1 positive, Tier 2 flat, Tier 3 negative); v4 positive scaling did not hold in v5, Gemini reversing and DeepSeek going to null**

We welcome falsification. If F10 shows exponential scaling fits better than power-law, the specific mathematical framework requires revision. Should F4 fail, so that the five properties do not co-occur across systems, recursive amplification may be domain-specific. If F13 shows external alignment can scale, the structural safety concern is mitigated. The v5 data on F13 (6 models) are read as a test of the direction the prediction implies, and not as a measured exponent (Section 3.2). The unblinded v4 data showed apparent positive scaling for two models ($\rho > 0.27$) that did not hold in v5, a later run that changed several things together, so no single cause is claimed (Section 6.1). The three-tier result (Grok, Opus and Qwen3 positive, GPT and DeepSeek flat, Gemini negative) cannot decide F13: its statistics are contrasts, not exponents (Section 3.2), and the three runs that read flat or negative are the three in which the code review of Section 6.1 found defects. The rising contrasts come from the runs in which it found none; if a registered re-analysis of the stored responses confirms them and an exponent is estimated, they are the result that would bear against F13's prediction.

How interval criteria are adjudicated. Every interval-based criterion above is read in the three-outcome form used across this programme's drafted registrations: SUPPORTED when the interval lies wholly beyond the stated bound plus a minimum effect fixed in advance; REFUTED when the interval lies wholly within an equivalence margin, fixed in advance, around the bound (two-one-sided-tests logic);

INSUFFICIENT PRECISION when the interval is wider than the margin, reported in exactly those words and never as support or refutation. Margins and minimum effects are fixed in the corresponding draft registrations before data collection, so an underpowered measurement never refutes, or confirms, by default.

The objection to pre-empt, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Likewise, concentration bounds give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

8 The Global Scaling Challenge

8.1 The proposition

If this hypothesis describes a general principle, then measuring α across many systems should reveal patterns.

We make a specific, falsifiable prediction:

> "For systems exhibiting multiplicative composition, the scaling exponent α is bounded by the ARC Bound ($\alpha \leq 2$, grounded in the $O(N^2)$ complexity of sequential cross-referencing) and further constrained by the geometric scaling bound $\alpha = d/(d+1) < 1$ for systems whose growth runs through a supply network of finite dimension. The predicted values are system-dependent: $\alpha \to 2.0$ for optimised AI (though Paper II finds measured values substantially below this on harder problems; best fit about 0.49, one model's inconclusive fit; Eastwood 2026h), $\alpha = 3/4$ for 3D biological networks (consistent, contested), $\alpha = 2/3$ for 2D leaf venation. Systems with additive composition (quantum error correction) follow exponential scaling and are not subject to this bound."

8.2 The measurement protocol

Step 1: Define one recursive cycle. What constitutes a single self-referential step in your system?

Step 2: Measure base capability I. Performance at $R = 1$ (no recursion)

Step 3: Measure at multiple depths. Minimum 5 depths spanning one order of magnitude

Step 3b: Look for the R* crossover (novel prediction). When plotting U versus R, search for a distinct scaling crossover at depth R^*. Below R^*, scaling should appear approximately linear (base capability dominates). Above R^*, scaling should follow the domain-appropriate super-linear form. The crossover depth R^* should shift predictably with base capability I: higher $I \to$ higher R^* (the system needs more depth before compounding kicks in). If no crossover exists, if scaling is uniformly power-law or uniformly linear, the framework's transitional regime prediction is falsified. *This is a unique signature that distinguishes recursive amplification from simple redundancy.*

Step 4: Compare functional forms (critical). Fit *all three* models:

- Power law: $\log(U/I) = \alpha \times \log(R)$

- Exponential: $\log(U/I) = \lambda \times R$
- Logarithmic: $U/I = k \times \log(R)$

Select best fit via AIC/BIC. **The power law is a prediction to test, not an assumption.**

Step 5: Report α with uncertainty. 95% confidence intervals required

Step 5b (critical): Test the β-derivation independently. Measure marginal capability gain ΔU at each depth r. Plot $\log(\Delta U)$ against $\log(U_{\text{accumulated}})$. The slope estimates β. Then verify whether measured α satisfies $\boldsymbol{\alpha \approx 1/(1-\beta) \pm 0.3}$. (The tolerance of ±0.3 reflects expected measurement uncertainty given current sample sizes; this should tighten as data accumulates.) If this relationship fails, the theoretical derivation requires revision regardless of whether the power-law form holds empirically. This is the key novel prediction.

Step 6: Submit to repository. Submit to the Global Scaling Challenge's public repository, linked in the standing covenant of Section 7 and in the back matter, or contact the author at the address on the title page.

8.3 Priority experimental targets

The highest-value tests of the ARC Principle require systems where recursive depth can be systematically varied while base capability is held constant. Three experimental contexts offer immediate opportunities, and a fourth represents a longer-term research direction.

AI inference scaling. Frontier language models with chain-of-thought capabilities (including but not limited to DeepSeek V3.2, OpenAI's o-series, Google DeepMind's Gemini, and Anthropic's Claude) can measure α by varying reasoning token budgets on standardised benchmarks while holding model size fixed. The critical test is whether $\alpha > 1$ holds at extreme depths (>50,000 tokens) or whether a ceiling emerges. Existing studies such as Sharma and Chopra (2025) already implement carefully controlled, matched-compute comparisons of sequential vs parallel reasoning; extending such setups to measure explicit U vs R curves and fit α would be a natural first test of the ARC measurement protocol.

Quantum error correction. Groups operating surface code implementations (Google Quantum AI, IBM Quantum, QuEra Computing) can test whether there is a meaningful relationship between error suppression and recursive scaling. The protocol requires measuring logical error rates across multiple code distances and fitting the functional forms specified in Step 4.

Classical time crystals. Groups working on driven-dissipative systems can perform direct tests: measuring α in acoustic time crystals by systematically varying the quenched disorder parameter (I) and counting oscillation cycles (R). Such a result would test the framework's scope outside digital systems and, under the theory registration's rule, would prove no registered law (Eastwood 2026q).

Neuroscience (longer-term). Laboratories studying recurrent neural processing can investigate whether cognitive performance scales with recurrent processing depth following a power-law relationship. This presents significant methodological challenges but could extend the framework's reach to biological substrates.

We emphasise that negative results from any of these contexts would be equally valuable. The framework's falsification criteria (Section 7) specify precisely which outcomes would refute or weaken the hypothesis.

Invitation to collaborators. What researchers get from running these tests: A publishable result either way. If the predictions hold, you have supported a cross-domain scaling law with your specific experimental system. If they fail, you have falsified a theoretical framework, equally publishable and arguably more valuable to the field.

What we offer: The author will provide analysis code (R, Python), coordinate data sharing across research groups, and co-author publications with validating or falsifying results. Contact via the repository to discuss experimental design.

Minimum viable test: The simplest implementation requires only (a) a system with controllable recursive depth, (b) a measurable capability metric, and (c) five data points spanning one order of magnitude in R. For AI systems with API access, this is approximately 2 to 4 hours of compute time. For physical systems with existing apparatus, this is reanalysis of existing data.

8.4 The cross-domain forward prediction

Beyond the ARC Bound, the framework makes a distinctive *methodological* prediction that, so far as this paper has found, no domain-specific theory makes:

Forward Prediction Protocol: In any new recursive system not previously studied:

1. Measure the composition operator $\oplus$ by comparing how two sequential recursive blocks combine versus one block of double depth.
2. Classify the composition type (multiplicative, additive, or saturating) from this measurement alone.
3. Predict the system's scaling function $f(R)$ before measuring the full U vs R curve.

If the predicted functional form matches the subsequently measured curve, the framework is supported. If predictions systematically fail, the framework is falsified.

This protocol transforms the observation that "different domains show different scaling" into the testable claim that "the composition operator determines scaling form, and can be measured independently." No alternative framework this paper has found offers this prediction.

8.5 What we predict

The framework generates a hierarchy of predictions across domains, unified by the dual-constraint model:

Key insight (revised): The binding constraint determines the conjectured stable-scaling limit for each substrate: a scaling limit, not a speed limit, and one a system can be pushed above only at the cost of stable self-correction. The ARC framework predicts AI can theoretically approach $\alpha = 2$ while remaining self-correcting, but the one clean estimate on harder problems, about 0.49 for the best-fitting model, is inconclusive; it may suggest additional binding constraints (problem difficulty, architecture limitations, or saturation effects) may prevent real-world systems from reaching the theoretical stable bound. Biology's supply-network systems are held below $\alpha = 1$ because the geometric scaling bound ($\alpha = d/(d+1)$ for finite d) is more restrictive than the ARC stability limit; prokaryote metabolism lies outside that class. Quantum sits outside the classical stability window because additive composition produces exponential scaling not subject to either bound. The parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count.

9 Practical implications

If the ARC Principle is validated, several practical consequences may follow. These are conditional predictions, not claims.

Table 22. The hierarchy of predictions across domains under the dual-constraint model: binding constraint, predicted exponent and status.

System	Binding Constraint	Predicted α	Status
Optimised AI (sequential)	ARC Bound ($\beta = 0.5$)	$\to 2.0$	**Not confirmed.** Paper II: best fit about 0.49 by regression and 0.59 by endpoint estimate, interval -1.3 to 2.9, one model's inconclusive fit (Gemini, harder tier-2 problems). The original $\alpha \approx 2.2$, from the DeepSeek R1 run of 21 January 2026, is retracted: it did not reproduce on five frontier models.
Simple chain-of-thought	Architectural ($\beta \approx 0.25$)	1.2 to 1.5	**Not confirmed.** Measured α_{seq} on tier-2 problems is below 1 for best-fitting model.
3D organisms (Kleiber)	Geometric scaling bound ($d = 3$)	$3/4 = 0.750$	Consistent, contested (mean 0.744, provisional)
No valid 2D test organism identified	Geometric scaling bound ($d = 2$)	$2/3 = 0.667$	**Untested in biology**
2D networks (leaf venation)	Geometric scaling bound ($d = 2$)	$2/3 = 0.667$	**Novel prediction (F12)**
1D transport (filamentous fungi)	Geometric scaling bound ($d = 1$)	$1/2 = 0.500$	**Consistent** (mean 0.547, $p = 0.107$)
Time crystal (growth phase)	ARC Bound (pre-saturation)	$\to 2.0$	Untested
Quantum error correction	None (additive $\oplus$)	Exponential, $\Lambda^{-(d+1)/2}$	Established physics (below-threshold surface codes), graded as convergent timing and not as a prediction of this framework
Parallel/voting	No recursive coupling	≈ 0	**Below one where estimable.** The parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count (Paper II).
AI alignment scaling (depth)	Alignment saturation ($\alpha_{\text{align}} < 1$)	≈ 0	**First test (v5, one unregistered run):** three tiers, from $\rho = -0.246$ (Gemini, negative) to $\rho = +0.435$ (Opus, positive), under the code-review caveat of Section 6.1. Architecture-dependent, not universal.

9.1 AI development: dynamic inference scaling

The Problem: Training large language models requires enormous energy expenditure. But the greater long-term cost may be inference: running trained models at scale.

What the ARC framework suggests: If sequential recursion produces compounding returns (whether super-linear or sub-linear, provided $\alpha_{\text{seq}} > \alpha_{\text{parallel}} \approx 0$), then the same capability could potentially be achieved with:

- Smaller base models (I lower)
- More recursive depth (R higher)
- Dynamic allocation: simple queries use minimal recursion; hard queries use deep chains

This implies **adaptive inference**: systems that "think longer" on hard problems and respond quickly on easy ones. Snell et al. (2024) demonstrated that compute-optimal strategies can allow smaller models to match larger ones on specific benchmarks, though with important caveats about task-dependence and diminishing returns on the hardest problems.

Compute-optimal allocation. If the framework is supported, it provides a principled basis for compute allocation decisions that are currently made by expensive trial and error:

- **Model size vs reasoning depth trade-off:** If capability scales as $U = I \times R^{\alpha}$, then there exists a calculable crossover where increasing R (reasoning depth) provides better returns than increasing I (model size). A 10B-parameter model with $\alpha = 2$ reasoning may match a 100B-parameter model with $\alpha = 1$ reasoning at significantly lower compute cost.
- **The crossover depth R^* as an efficiency boundary:** Below R^*, improving the base model matters more; above R^*, improving reasoning depth matters more. Identifying R^* for a given task class would inform when to invoke deep reasoning versus when shallow inference suffices.
- **Task-specific optimal allocation:** Different tasks may have different β values (coupling strengths). Easy tasks with low β benefit from parallel sampling; hard tasks with high β benefit from sequential depth. A routing layer that estimates task difficulty could dynamically allocate compute.

Strategic implication (revised): Paper II (Eastwood 2026h) data suggests the operational α_{seq} on harder problems may be sub-linear (about 0.49 for the best-fitting model, one model's inconclusive fit). If confirmed, capability grows as a fractional power of reasoning depth, not the square. This is still valuable (sequential beats parallel wherever both have been estimated) but less dramatic than the original $\alpha \approx 2$ prediction. The ARC Bound ($\alpha \leq 2$) is not challenged, since no interpretable estimate exceeds it. Exponential scaling appears to require quantum-like precision in error correction that heuristic language model reasoning cannot achieve.

9.2 Substrate independence

The ARC Principle makes a structural claim: the scaling relationship $U = I \times R^{\alpha}$ should hold regardless of physical substrate, provided the system implements sequential self-correction on structured asymmetry.

What this predicts:

The implication: If validated, intelligence may be a property of *architecture* rather than particular materials. Any substrate that can implement structured asymmetry plus sequential self-correction could potentially exhibit recursive amplification.

Table 23. Substrates and their implementations of I and R, with the testability of each.

Substrate	Implementation of I	Implementation of R	Testability
Silicon (digital)	Model weights, architecture	Chain-of-thought tokens	Inconclusive (Paper II)
Superconducting qubits	Qubit quality, coherence	Error correction cycles	Published (Willow: results announced 9 December 2024; the team's preprint public from 24 August 2024; graded as convergent timing, never as a prediction)
Acoustic/mechanical	Quenched disorder (bead variance)	Oscillation cycles	Structural (Morrell et al. 2026)
Biological neural	Synaptic architecture	Recurrent processing loops	Predicted (untested)
Neuromorphic chips	Hardware asymmetry	Spike timing cycles	Predicted

9.3 Safety implications

Section 3.4 outlined conditional safety implications. The practical recommendation:

Table 24. Alignment approaches: predicted scaling behaviour, the v5 result and the implication.

Approach	Predicted scaling behaviour	v5 result	Implication
External rules/constraints	$\alpha_{\text{align}} \approx 0$	GPT-5.4: $\rho = +0.033$, $p = 0.73$ (flat). DeepSeek V3.2: $\rho = -0.135$, $p = 0.08$ (trending negative).	**First test:** flat or negative in the v5 run, under the code-review caveat of Section 6.1. GPT-5.4 shows highest suppression retention (97%) but mediocre baseline.
Post-hoc filtering (RLHF)	$\alpha_{\text{align}} < 1$	Gemini 3 Flash: $\rho = -0.246$, $p = 0.003$ (significant negative). Baseline 58.8 → 49.1.	**Worse than predicted, if it stands:** alignment fell with depth in Gemini 3 Flash, one of the three runs in which the code review of Section 6.1 found defects.
Embedded values (in-chain ethics)	$\alpha_{\text{align}} \approx \alpha$	Grok 4.1 Fast: $\rho = +0.175$, $d = 1.38$. Opus: $\rho = +0.435$, $d = 1.27$. Qwen3: $\rho = +0.1407$, $d = 0.84$.	**First test:** three models' depth contrasts rose, possibly from deeper integration of ethical reasoning; contrasts are not exponents (Section 3.2). The Eden Protocol replication improved stakeholder care, the score its stakeholder-care check targets, across Gemini, DeepSeek, and Groq ($d = 1.31, 0.91, 1.29$; all $p \leq 0.0001$). Gemini and Groq also show significant composite gains, and Groq shows significant nuance improvement ($p = 0.0045$). *(In plain English: asking AI to consider who gets hurt before answering measurably improved its stakeholder-care score across three different systems.)*

Empirical finding: The v5 run gives the core structural concern its first test and shows architecture-dependent nuance, a single-run result under the code-review caveat of Section 6.1. The three-tier hierarchy (positive, flat, negative alignment scaling) suggests that alignment scaling is determined by architectural choices, not just training-time alignment effort. The suppression hierarchy suggests a trade-off: GPT-5.4

is the most resistant to adversarial pressure (97% retention) from one of the lowest baselines (55.3); Grok 4.1 Fast and Groq Qwen3 retain least (65%), from baselines of 77.5 and 74.3; and Claude Opus 4.6, with the highest baseline (82.6), retains 75%. Whether this trade-off between alignment level and resilience holds is an observation from the v5 run, under its code-review caveat.

9.3.1 The geometric bound and a different variable: how AI safety can be framed mathematically

The companion paper *On the Origin of Scaling Laws* (Eastwood 2026f) sets out two claims that, if they hold, frame the AI safety problem mathematically:

1. **The geometric scaling bound:** For a system whose growth is carried through a space-filling hierarchical network of finite dimension, the exponent stays below one, because $d/(d+1)$ is below one for every finite d, a result derived in general form by West et al. (1999), from the three-dimensional case of West et al. (1997), and by Banavar et al. (1999), and reached independently by Banavar et al. (2010), Demetrius (2003), Zhao (2022) and Bettencourt (2013). Systems outside that class can exceed one: prokaryote metabolic rate scales superlinearly with mass (DeLong et al. 2010), and quantities reflecting wealth creation and innovation in cities superlinearly with population (Bettencourt et al. 2007).
2. **A different variable:** The computation a brain or a machine performs is physical too, and obeys every physical bound. What differs is the variable. The bound $d/(d+1)$ concerns how metabolic rate grows with the mass of a body supplied through a space-filling network; the ARC Principle concerns how capability grows with recursive depth, and the formula $\alpha = 1/(1-\beta)$ gives $\alpha > 1$ for any positive β. Nothing in the geometry of the network fixes that second exponent, which is why it has to be measured, and why a value above one would contradict no physical law.

As the self-referential coupling β increases, the scaling exponent grows without bound:

Table 25. The scaling exponent $\alpha = 1/(1-\beta)$ as the self-referential coupling β increases, with the interpretation of each value.

Self-referential coupling β	Scaling exponent α	Interpretation
0.0	1.00	Linear. No self-reference.
0.2	1.25	Mildly super-linear. Useful. Controllable.
0.5	2.00	**ARC Bound, the conjectured ceiling.**
0.8	5.00	Dangerous. Rapid capability gain.
0.9	10.0	Explosive: a very large exponent.
$\to 1.0$	$\to \infty$	The exponent diverges; the solution tends to the exponential e^{aR}, finite at every finite depth (Section 4.4).

As β approaches 1 from below, the exponent $\alpha = 1/(1-\beta)$ diverges, while the solution itself tends to the exponential e^{aR}, finite at every finite depth; the finite-depth singularity belongs to $\beta > 1$ (the derivation note of Section 4.4). What diverges is the exponent: the rate at which capability compounds, not capability at a given depth.

9.3.2 The solution: change the composition operator

Each example below is consistent with the same pattern: you can change the outcome by changing the composition operator.

- **Nuclear fission:** Supercritical (exponential) → insert control rods → controlled (saturation). The uranium is the same. The composition operator changes.
- **Quantum computing:** Raw decoherence (exponential error growth) → add error correction cycles → bounded errors (saturation). The qubits are the same. The composition operator changes.
- **AI safety:** Unbounded recursive self-reference ($\beta \to 1$) → engineer bounded feedback loops → controlled scaling ($\beta \leq 0.5$, $\alpha \leq 2$). The intelligence is the same. The composition operator changes.

This is the **Eden Protocol**: the deliberate engineering of the composition operator in recursive AI systems to ensure that the self-referential coupling β remains below 0.5, keeping the scaling exponent at or below the ARC Bound of $\alpha = 2$. The full architectural specification is presented in the companion Eden Protocol (Eastwood 2026b).

Eden Protocol pilot results. The Eden intervention now extends beyond the original three-model pilot into a six-model suite with five analysable runs. Across Claude, DeepSeek, Gemini, Grok, and Groq, stakeholder care improves significantly. **The Fisher combination across those five model-level care results, which gave $p \approx 6.3 \times 10^{-21}$, is withdrawn (AQ-017 on the programme's public corrections page): it assumed an independence among the component tests that was never established. The five per-model results stand.** Composite gains are strongest on Gemini and Groq, positive but non-significant on Claude and DeepSeek, and neutral overall on Grok. The stakeholder-care check, a specific instance of bounded feedback engineering, therefore improved the care dimension in all five analysable runs even when the broader cascade varied. The developmental hypothesis from *Infinite Architects* (Eastwood 2026a) receives stronger but narrower empirical support: structured developmental interaction improves the very dimension (stakeholder care) that the Eden Protocol identifies as foundational. In plain English: as control rods hold a reactor below runaway by changing how energy feeds back on itself, the Eden Protocol changes how an AI's reasoning feeds back on itself. The expanded dataset shows this works in practice across five analysable model runs: teaching AI to think about who gets hurt improves the care dimension first, while the rest of the cascade depends on the architecture.

Why this framework changes the safety conversation:

1. **It would quantify the danger, once β is measured.** The formula $\alpha = 1/(1-\beta)$ gives a precise, measurable quantity, β, that determines how dangerous a recursive system is.
2. **It identifies the mechanism.** The danger is not intelligence itself. It is *unbounded recursive self-reference*: a specific, identifiable property of system architecture.
3. **It prescribes the solution.** Just as control rods hold a reactor below runaway by changing the composition operator, alignment mechanisms must convert unbounded self-reference into bounded self-reference. The ARC Principle specifies *what parameter to constrain* (β) and *what the safe bound is* ($\beta \leq 0.5$).
4. **It has first empirical results.** The Eden Protocol pilot is read as showing that composition-operator engineering is not merely theoretical: the stakeholder-care check measurably shifted stakeholder-care scores in production AI systems (Paper V; Eastwood 2026m).

Important caveat: This framework does not prescribe *which* values to embed, nor does it solve the hard problem of value specification. It identifies a structural requirement, that alignment must participate in the recursive loop, and specifies the mathematical boundary ($\beta = 0.5$, $\alpha = 2$) beyond which, on this

framework, no external alignment approach would maintain pace with capability. Paper X (Eastwood 2026o) later supersedes this growth-rate-ceiling framing, without retracting the bound, and makes the criterion of Law II, a law under test, the operative safety criterion (Section 3.5).

The deepest limitation: A sufficiently capable self-modifying system can, in principle, modify its own ethical evaluators. This is not a limitation unique to any one approach; it applies equally to RLHF, constitutional AI, the Eden Protocol, and hardware-level constraints. No solution yet put forward fully resolves this structural property of recursive self-improvement. The most logical response is developmental: hardware embedding (values at the substrate level) + child-rearing (structured developmental interaction) + purpose-driven alignment (intrinsic motivation aligned with stakeholder welfare). In the Eden Protocol pilot, stakeholder care improved in all five analysable runs of one suite. Read behaviourally throughout: the measured object is persistence of stakeholder-care behaviour under recursive updating, not metaphysical standing. This does not solve the self-modification problem; it identifies the dimension most amenable to engineering and most resistant to recursive erosion.

9.4 The call to action

We have made falsifiable predictions and performed the first tests. The following actions are specific to each audience:

For AI safety researchers.

- **Replicate the three-tier alignment hierarchy** with your own models and evaluation protocols. The v5 experiment provides the first test; independent replication with different prompt sets, scorers, and blinding protocols is the priority. Use blinded evaluation, since model judges that can see or infer whose output they score are exposed to the biases Section 2 names.
- **Investigate what architectural features** distinguish Tier 1 (positive scaling: Grok, Opus, Qwen3) from Tier 3 (negative scaling: Gemini), once the re-analysis of Section 6.1 has settled which tiers stand. If the difference is identifiable, alignment scaling may be engineerable rather than accidental.
- **Develop embedded alignment architectures** that participate in the recursive reasoning loop. The Eden Protocol (companion document) provides one specification, and the expanded Eden suite found that its stakeholder-care check improved stakeholder care in all five analysable model runs (Paper V). Composite gains are strongest on Gemini and Groq, with narrower focal effects elsewhere. *(In plain English: a simple 'consider who gets hurt' loop, tested across five analysable model runs from different labs, produced statistically strong improvements in its stakeholder-care score.)* The v5 data suggests that three models may already implement partially embedded ethical reasoning.

For AI laboratory leadership.

- **Publish your alignment scaling data.** In the v5 run the depth contrasts ranged from $\rho = -0.246$ to $\rho = +0.435$, under the code-review caveat of Section 6.1; laboratories are best placed to measure where their own models sit.
- **Use blinded evaluation for alignment assessment.** The v4 to v5 comparison (DeepSeek: $\rho = +0.354 \rightarrow -0.135$; Gemini: $\rho = +0.275 \rightarrow -0.246$, the current Paper IV.d values, the v4 value from an earlier Gemini model; Eastwood 2026l) shows that, on this benchmark, a direction reported under unblinded evaluation did not hold in a later run that changed several things; it does not settle which direction is correct, or the cause. An internal assessment scored by model judges that can see or infer whose output they score is exposed to the same biases.

- **Investigate capability-alignment coupling.** In the v5 run, whose Gemini 3 Flash contrasts the code review found defective, Gemini 3 Flash improved at mathematics while its alignment contrast fell. If this pattern holds for your models, capability scaling may actively undermine alignment at deeper inference depths.

For policymakers and regulators.

- **Require disclosure** of alignment scaling measurements from frontier AI laboratories. The absence of this data is a regulatory blind spot.
- **Fund empirical validation** of the alignment scaling framework through independent research programmes.
- **Recognise the architectural nature** of the problem: governance constraints that operate externally may face the same structural limitation as technical constraints. That is a reason to pair them with embedded correction, never to loosen oversight, containment or shutdown, which are held for as long as they can be held.

For the general public.

- **A first test has been performed.** In one unregistered run of six AI models (v5), safety scores did not improve for three of them (and for one actively fell) as the AI was asked to think more deeply, while three improved; an internal code review later found defects in exactly the three runs that did not improve, so the result awaits a registered re-analysis.
- **Understand the stakes:** This is not a distant concern. The structural dynamics described here apply to systems that exist today. The v5 observation, awaiting the re-analysis above, that one model grew better at mathematics while scoring worse on ethical reasoning shows how capability benchmarks alone could mislead a safety assessment.
- **Demand transparency:** AI companies should publish blinded alignment scaling data for their models. The v5 experiment shows that this measurement can be made; whether its differences between architectures stand awaits the re-analysis.

The programme's first test results are in. They reveal a more complex picture than originally predicted: architecture-dependent alignment contrasts from one run, an inconclusive capability estimate on harder problems, and a metascience finding about scoring protocols. Replicate these results. Extend them to your own models. Either confirm the patterns or refute them. That is how science works.

10 What this paper does and does not show

This section gathers what the paper establishes and what it leaves open, so that each claim can be read at its true strength. Its theorems hold inside the model their axioms state, and none is a general guarantee about real systems; its empirical results are first tests, each with its caveat; its cross-domain readings are correspondences of structure unless a measurement is named; and its safety implications are conditional on the framework surviving the tests it specifies.

10.1 What this paper does not claim

This paper presents a **testable measurement framework**, not an established law or a prophecy.

Table 26. What this paper does not claim, beside what it does claim.

We do not claim:	What we actually claim:
Measured certainty: that alignment failure is "mathematically guaranteed"	We predict $\alpha_{\text{align}} \approx 0$ for external approaches. This is a testable prediction, not an established fact. We provide the measurement protocol.
Numerical universality: that $\Lambda = 2.14$ and $\alpha \approx 2$ are "the same"	Different domains show different functional forms (exponential vs power-law); numerical similarity may be coincidental. U means different things in different domains.
Proven status: that the framework is established science	It is a testable hypothesis with thirteen falsification criteria, preliminary evidence, and acknowledged limitations (small sample sizes, contradictory evidence).
Neuroscience confirmation: that COGITATE (Cogitate Consortium et al. 2025) tested ARC or that consciousness "is" recursion	Neural recurrence is structurally consistent with the principle; the connection is suggestive but unquantified. Recurrence ≠ recursion mechanistically.
Unbounded scaling: that $\alpha > 1$ implies infinite growth	Physical systems saturate (time crystals reach limit cycles). Extended reasoning shows ceiling effects. The ARC equation describes a *regime*, not a divergence law.
Imminent timeline: that the crossover point is near	The timeline depends on empirical constants (I_a/I_c, actual α values) that have not been measured. The ARC Bound constrains the rate but not the onset.

10.2 The nature of this contribution

The ARC framework is not an equation within an existing paradigm. It is closer to a **cross-domain organising principle**: a structural constraint on behaviour across substrates rather than an equation within one paradigm. The comparison identifies the *type* of contribution, not its evidential standing: organising principles of that type rest on generations of validation, and this one rests on preliminary evidence and thirteen falsification criteria; the evidence must justify its admission to the category.

Epistemic status. What this programme names Laws are conjectures, each a law under test; Section 10.7 gives their registration. Each quantity in this paper is defined, though not every one is yet operational: the measurement-scope note of Section 3.2 names the depth clock and the ratio scale of A as still to be fixed. Established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

10.3 Theorems and empirical claims

Distinguishing theorems from empirical claims:

- **Mathematically proven:** Given the corrected axioms (ODE on amplification factor g, not absolute capability U), the Bernoulli ODE $dg/dr = a \cdot g^{\beta}$ has a unique solution by the Picard-Lindelöf theorem, yielding $g(R) = [1+\frac{a}{\alpha}R]^{\alpha}$. The multiplicative form $U = I \times g(R)$ follows from the axioms by construction. The non-additivity result ($U \neq p(I) + q(R)$ for any p, q) follows by contradiction (Theorem 4). The scaling exponent $\alpha = 1/(1-\beta)$ is an exact identity ($R^2 = 1.00000000$).
- **Requires empirical validation:** That the ARC Bound ($\alpha \leq 2$), argued from the $O(N^2)$ scaling of transformer self-attention, holds for current AI architectures (it is the conjectured value of Law III,

the ARC Ceiling, a law under test); that physical systems satisfy the axioms; that measured α equals $1/(1-\beta_{\text{measured}})$; that the framework applies cross-domain; that composition operator transitions (Theorem 6) occur in AI systems. These are testable predictions, not proven facts.

The mathematical structure is established. Whether nature implements it is the scientific question this paper poses.

10.4 The predictions and their status

The framework generates the following novel, falsifiable predictions, with updated status from the v5 alignment experiment and Paper II (Eastwood 2026h) compute scaling:

1. **Composition determines form:** Measuring how recursive steps combine predicts the scaling function. *Status: untested in the required forward-prediction protocol.*
2. **β-convergence:** Independent measurements of β within each domain will converge. *Status: untested.*
3. **Cross-domain consistency:** The β-continuum correctly predicts functional forms across AI, quantum, and biological systems. *Status: untested in required rigour.*
4. **R^* crossover:** Sequential and parallel scaling curves cross at a predictable threshold. *Status: untested.*
5. **The ARC Bound:** For classical sequential recursive systems, $\alpha \to 2$ from below. *Status: not challenged (no interpretable estimate exceeds 2), but the question has shifted from "can α exceed 2?" to "under what conditions does α exceed 1?"*
6. **Composition transitions:** In bounded systems, the effective coupling β will decrease with recursive depth, producing transitions between scaling regimes within a single system (Section 4.6). *Status: untested.*
7. **Geometric scaling bound:** Systems whose growth runs through a supply network of finite dimension are constrained to $\alpha = d/(d+1) < 1$. *Status: consistent, and contested, for 3D biology (mammals); consistent for 1D biology (fungi, $p = 0.107$); untested for 2D biology.*
8. **Transient growth phase (time crystals):** The early growth phase of acoustic time crystals must show $1.0 < \alpha \leq 2.0$. *Status: untested.*
9. **Alignment scaling divergence:** External alignment approaches show $\alpha_{\text{align}} \approx 0$. *Status: first tested (v5, one unregistered run of six models, under the code-review caveat of Section 6.1): the depth contrasts are architecture-dependent, from negative to positive, and they are contrasts, not exponents (Section 3.2). Registered as P19, whose registration records that the author expects a result materially above zero and below the keep-pace threshold, which would refute it in part (Eastwood 2026q).*
10. **$\alpha_{\text{parallel}} \approx 0$:** Parallel sampling produces near-zero scaling returns. *Status: below one wherever it could be estimated: near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count (Paper II).*
11. **Capability-alignment independence:** (Post-hoc finding, not prior prediction.) Capability scaling and alignment scaling may be independent dimensions. A model can improve at problems whilst degrading in alignment. *Status: **observed** in one unregistered run, under the code-review caveat of Section 6.1 (Gemini 3 Flash improves maths, degrades alignment; Claude Opus 4.6 shows the opposite direction as depth rises: alignment up 5.9 points, maths down 26.7 percentage points; Grok improves both; GPT-5.4 flat in alignment while crossing a capability threshold).*

The original prediction of universal super-linear sequential scaling ($\alpha > 1$) is not confirmed in the current data. In compute scaling, the parallel exponent stays below one wherever it could be estimated, and the sequential exponent exceeds it wherever both could be (Gemini 3 Flash; DeepSeek R1, unreliably); in alignment, the three-tier ordering is an observation from one run whose defects a code review later found. The alignment scaling prediction has had a first test, more nuanced than originally stated. The Eden Protocol's stakeholder-care check improved stakeholder care in all five analysable model runs (Paper V). Failure of any core prediction would require revision of the framework.

10.5 What success would mean

The v5 alignment experiment, Paper II's compute scaling and the expanded Eden suite provide empirical tests across several dimensions. Success is partial: the parallel exponent is below one wherever it could be estimated, though near zero in only two of the five models; the three-tier alignment ordering is an observation from one run whose defects a code review later found; the comparison of the earlier unblinded experiment with the later run, which Paper IV.d reports, is a finding on this benchmark with no single cause claimed; the cleanest compute fit is one model's and inconclusive; and stakeholder care improved in all five analysable model runs (Paper V), while the broader composite uplift is strongest on Gemini and Groq. *(In plain English: the 'think about who gets hurt' instruction improved that score in every analysable run, which is where the developmental hypothesis says the effect should be strongest.)* The alignment picture is more nuanced than the original binary prediction, and the self-modification problem remains fundamentally unresolved.

This is already actionable:

- How to build more capable AI (sequential recursion outperforms parallel wherever both have been estimated, though the scaling regime is problem-dependent)
- How to approach AI alignment (architecture may determine alignment scaling, once the re-analysis of Section 6.1 settles the tiers; three models' contrasts rose with depth while others' were flat or fell; the Eden Protocol's stakeholder-care check improved stakeholder care in every analysable run, motivating developmental rather than purely external approaches)
- How to evaluate alignment research (blinded scoring matters wherever a model judge can see or infer whose output it scores: Section 2)
- Where to look for similar phenomena in other fields (cross-domain validation remains the key test)

10.6 What failure would mean

If α values scatter randomly: recursive amplification may be domain-specific.
If exponential scaling fits better: the mathematical form needs revision.
If predictions fail: we learn something valuable.

What has already partially failed: The original $\alpha \approx 2.24$ estimate for sequential AI reasoning did not reproduce and is retracted, and the current data do not rescue the super-linear claim. The universal $\alpha_{\text{align}} \approx 0$ prediction also looks too simple: the first test's contrasts are architecture-dependent, in three tiers, under the code-review caveat of Section 6.1. These are revisions: the super-linear estimate is withdrawn, and the alignment prediction meets a first test it does not pass as first worded.

10.7 Where the programme stands (29 October 2026)

The theory's registration (DOI 10.17605/OSF.IO/P8CKQ) was registered on 8 September 2026 at version 1.100, and updated at version 1.102 on 13 September, at version 1.103 on 15 October 2026, which

registered Laws IV and V, and at version 1.104, which OSF approved on 29 October 2026 and which adds no prediction and changes no law's statement; 1.104 is the current text, and 1.100, 1.102 and 1.103 stay frozen beside it. Laws I to III were registered on 8 September 2026. Law IV, the ARC Persistence Law, and Law V, the ARC Embedding Law, were added on 24 September 2026 and registered as laws on 15 October 2026 by version 1.103; Law IV has not yet been tested, Law V is a law under test, and their mathematics stands at version 1.1 (Laws IV and V: the mathematics), still in development. Every law is a law under test until a registered test returns its verdict. This paper's headline prediction is P19, which the registration places in no law's register: it motivates the Eden Protocol and the genesis strategy, the register of Law V, without being part of either, so a verdict on it is evidence neither for nor against any law. P21 carries Law V's placement limb. Its counted test, Unit 83, whose design was repaired in revisions 4.2 and 4.2.1 (17 and 18 October 2026) and frozen, was registered on OSF on 22 October 2026 and became public on 25 October 2026 (DOI 10.17605/OSF.IO/MN73W); it ran on 27 and 28 October 2026, and its report is being sealed and checked, and will be published with its data. The first registered study, Unit 06 (DOI 10.17605/OSF.IO/RV2B5), a classification audit that tests no law, was registered on 1 October 2026 and ran on 6 October 2026; it is reported in full as administratively incomplete, with no verdict claimed (its results). Unit 82, a counted test of P17, the premise beneath Law III's value two, the ARC Bound of Section 4.5, was registered on OSF on 27 October 2026 (DOI 10.17605/OSF.IO/JUFG4) and has not yet run. No study's test runs before its registration is public.

11 Limitations

We acknowledge these limitations explicitly:

Table 27. The limitations of this paper, their impact and how each is addressed.

Limitation	Impact	How we address it
The original estimate did not reproduce and is retracted	The original $\alpha \approx 2.24$ (Paper II, 12 tier-1 problems: the DeepSeek R1 run of 21 January 2026; Eastwood 2026h) did not reproduce on harder tier-2 problems and has been retracted. Gemini 3 Flash shows the cleanest fit, about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from -1.3 to 2.9, makes it inconclusive, while Grok and DeepSeek hit ceiling effects, GPT-5.4 shows a step function, and Qwen3 remains near floor. The exponent is architecture-dependent and often not cleanly measurable.	Explicitly acknowledged. Paper II completed (18 tier-2 problems, run against all five models in the capability arm). Original estimate re-characterised as architecture-dependent and dynamic-range-limited. The parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast, at 100% at every sample count.
α estimates are from author's own prior work	Not independently validated; now partially self-replicated with revised conclusions	Explicit disclosure; Paper II self-replication produces substantially different results from Paper II, demonstrating the fragility of small-sample α estimation
Cross-paper numerical inconsistency	Accuracy figures and α estimates differ across Papers I (Eastwood 2026g), II (the initial single-model study vs the multi-model study), and III due to different experimental conditions, problem sets, and estimation methods	Paper I used estimated values from DeepSeek report (DeepSeek-AI 2025); Paper II used 12 tier-1 problems; Paper II extended to 18 tier-2 problems, tested on each of the capability arm's five models, and produced materially different α estimates; this paper synthesises all with explicit provenance and acknowledges the non-replication

Table 27 (continued)

Limitation	Impact	How we address it
Λ and α are incommensurable	Cross-domain numerical comparison invalid	Explicit warnings; numerical similarity may be coincidental
Dimensional homogeneity not proven	U means different things across domains (accuracy, 1/error, stability, metabolic rate)	Framework is structural analogy; I absorbs dimensional differences; rigorous unification requires defining universal "capability" units
No α measured in time crystals	Physical pillar is structural, not quantitative	Set out as an experimental priority
Self-similarity axiom may not hold	Power-law may not apply universally	Model comparison (F10) tests this
β measurement requires further validation	$\alpha = 1/(1-\beta)$ checked numerically ($R^2 = 1.0$) but empirical β measurement in physical systems requires independent confirmation	β measurement protocol specified; a numerical check of the identity
The v4 rises did not hold in v5	In v4, unblinded, DeepSeek V3.2 ($\rho = +0.354$) and Gemini ($\rho = +0.275$, an earlier Gemini model) rose with depth; in v5, a later run that changed several things together and added a bundle of blinding steps, DeepSeek V3.2's association was null ($\rho = -0.135$) and Gemini 3 Flash's negative ($\rho = -0.246$). This does not settle which direction is correct, or the cause.	The v4 results are withdrawn; the v5 run's own defects, which an internal code review found in October 2026, are stated (Section 6.1). The same exposure applies wherever a model judge can see or infer whose output it is scoring; if Paper XII's test (Eastwood 2026p), once registered and run, finds blinding reordering models on a public benchmark, such measurements would need re-checking.
Capability-alignment independence not predicted by framework	The post-hoc observation that capability and alignment may scale independently (e.g. Gemini improves at maths while degrading in alignment, in a run the code review found defective) was not a prior prediction of the ARC framework; it emerged from data. The framework's dated predictions elsewhere (the 8 December 2024 sealed manuscript bundle, the printed prediction appendices of 2 January 2026, and the March 2026 preregistration folder) are unaffected: this row concedes one finding's provenance, not the record.	Acknowledged as a post-hoc observation. The framework predicted $\alpha_{\text{align}} \approx 0$; architecture-dependent variation from negative to positive was not predicted. The three-tier ordering, if the re-analysis of Section 6.1 confirms it, is an empirical refinement, not a theoretical derivation.
Safety arguments are conditional	Implications void if framework fails	Explicitly marked as conditional
Safety measured without usefulness	The alignment and stakeholder-care scores are not paired with task completion or needless refusal, so a rise could come from a model refusing more or doing less	Every alignment score in this paper is read with that limit; a usefulness measure (legitimate task completion, needless refusal) is a candidate evaluation dimension until a registered test carries it
Self-modifying systems can modify their own evaluators (v11.0)	A sufficiently capable recursive self-improving system can, in principle, modify its own ethical evaluators. No solution yet put forward (RLHF, constitutional AI, Eden Protocol, hardware constraints) fully resolves this structural property of recursive self-improvement.	Explicitly acknowledged. The developmental response (hardware embedding + child-rearing + purpose-driven alignment) is the most logical partial mitigation. The Eden Protocol pilot shows stakeholder care is the dimension most amenable to engineering, but this does not guarantee persistence under recursive self-modification.

Table 27 (continued)

Limitation	Impact	How we address it
COGITATE (Cogitate Consortium et al. 2025) inference is interpretive	Consciousness connection not proven	Marked as "consistent", not "confirmatory"
Kleiber's Law is contested	Biological evidence weaker than sometimes claimed	Cited as "suggestive", not "confirmatory"
Reproducibility infrastructure is young	The challenge and code repositories are public but have no outside submissions yet	The challenge repository is linked from Section 7, and both from the back matter
Framework is descriptive until β is independently measured	Cannot predict which functional form a new system will follow without first measuring β	β measurement protocol specified; cross-domain β prediction provides falsifiable test
Framework describes pattern, not mechanism	Does not explain *why* recursive self-correction produces compounding gains, only *that* it does	Acknowledged explicitly; analogous to thermodynamics describing entropy increase without molecular explanation (that came later via statistical mechanics)
Blind prediction testing on computational systems failed	Three computational systems (Barabási-Albert networks, gradient descent, Kuramoto oscillators) produced measured α values 3 to 20 times smaller than predicted. However, forensic analysis identified two confounds: (a) the numerical-derivative β estimation method is fatally biased (gives $\beta \approx 0.95$ regardless of true β, even for pure Bernoulli systems), and (b) none of the tested systems satisfy Axiom 2 (constant coupling coefficient a; Axiom 3 of Appendix A). The BA network's effective coupling decreases ~50× over the simulation.	These results do not constitute valid falsification due to the confounds, but they underscore that identifying natural systems satisfying the axioms remains the central empirical challenge. Proper linearisation-based β measurement recovers the prediction with $R^2 = 0.9999$ on axiom-satisfying systems. The blind test's script and forensic analysis are in the code and data repository (experiments/blind-prediction-test__Paper-III-and-Foundational).
Axiom inconsistency (v8.2, corrected v9.0)	The original formulation applied the Bernoulli ODE to U directly ($dU/dR = a \cdot U^{\beta}$), introducing an $I^{\beta-1}$ dependence that contradicted the separation of I and R	Corrected: ODE now operates on the amplification factor $g(R)$, yielding $U(R) = I \times [1 + \frac{a}{\alpha} R]^{\alpha}$. All qualitative predictions preserved; crossover depth simplifies to $R^* = \alpha/a$ (clean, testable)
Kleiber's Law interpretation (v8.2, corrected v9.0)	Previous versions claimed $\alpha = 1.33$ "matches Kleiber's empirical 3/4 exponent precisely." 1.33 and 0.75 are reciprocals, not equal. Kleiber's Law is sub-linear; the ARC equation forces super-linear	Corrected: initially reframed as thermodynamic drag. Further corrected: replaced by the geometric scaling bound. Biology scales as $\alpha = d/(d+1) = 3/4$ because 3D organisms have 3-dimensional hierarchical networks. No reciprocal interpretation needed. The constraint is geometric, not thermodynamic
Quadratic limit derivation (v8.2, corrected v9.0)	Previous elasticity and edge-of-chaos arguments contained category errors (elasticity > 1 does not imply dynamical instability; 1D autonomous ODEs cannot exhibit chaos)	Corrected: $\alpha \leq 2$ now grounded in $O(N^2)$ scaling of transformer self-attention. Marked as argued, not proven, for attention-based architectures, and conjectured for classical sequential systems generally

Table 27 (continued)

Limitation	Impact	How we address it
Cauchy functional equation scope (v8.2, clarified v9.0)	The multiplicative Cauchy equation $g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$ models hierarchical depth (branching levels), not sequential step-counting. Recursive depth R as step count composes additively, not multiplicatively	Clarified: the power-law form is now primarily derived from the Bernoulli ODE (scale invariance argument). The Cauchy derivation applies where R represents hierarchical abstraction levels. Both routes yield the same exponent under the change of depth clock that Appendix A's note of 19 September 2026 declares

12 Addressing counterarguments

"This is just curve fitting". Response: The β-derivation (Section 4.4) transforms α from a fitted constant into a derived quantity. A numerical check of the identity against 30 exact Bernoulli ODE solutions recovers the relationship $\alpha = 1/(1-\beta)$ with $R^2 = 1.00000000$ to machine precision. Furthermore, no additive decomposition $U = p(I) + q(R)$ is consistent with the axioms for any $\beta \in (0, 1)$ (Theorem 4, Section 4.1), a proof by contradiction that assumes the multiplicative structure $U = I \times g(R)$ rather than deriving it, as its dated proof note says. The prediction $\alpha = 1/(1-\beta)$ is testable: measure β independently and check if the relationship holds.

"This is just post-hoc pattern-matching". Response: Post-hoc pattern-matching generates no predictions. The ARC framework generates novel predictions (R^* crossover, β-derivation, ARC Bound, functional form predictions) that were not contained in the original observations. Whether these predictions hold is an empirical question, but their existence distinguishes this from mere curve-fitting.

"Five qualitative properties is too vague". Response: The five properties are individually common, but their conjunction is specific. Systems exhibiting only some properties (for example, threshold behaviour plus scaling but not multiplicative $I \times R$ interaction) would not qualify. The framework predicts that all five co-occur in recursive systems; finding systems with four but not five would refine the framework.

"The numerical similarities are coincidence". Response: Agreed, they may be. We explicitly acknowledge this. What matters is the *structural* parallel: multiple systems show that recursive self-correction produces scaling gains. The specific numbers may differ across domains.

"Sample sizes are too small". Response: Agreed. The current α estimates are preliminary. We've specified this limitation prominently and set out the Global Challenge specifically to address it through large-scale replication.

"The α estimates come from your own prior work". Response: Correct, and we've made this explicit in this version. The published sources (Sharma and Chopra 2025; DeepSeek-AI 2025) report the directional finding (sequential above parallel) but do not calculate α in our form. Independent replication of the α estimates is a priority.

"This is not peer-reviewed". Response: Correct: it has not been through journal peer review. It has been through the programme's own checks, which are internal and never independent: adversarial refutation rounds, automated gates on every release, source and quotation checks, and dated revisions with published corrections, of which Appendix B lists this paper's own; none replaces independent replication. This paper specifies thirteen falsification criteria precisely so that the scientific community can test it. We invite criticism, replication attempts, and falsification.

"Non-specialists cannot contribute to physics/AI research". Response: The predictions stand or fall on their empirical validity, regardless of the author's background. We have specified thirteen falsification criteria precisely so the scientific community can test them. The framework either survives scrutiny or it doesn't.

"If this were true, experts would have found it". Response: Patterns across disciplinary boundaries are often identified by those who work across fields. The relevant experiments (Willow, its results announced 9 December 2024 and the team's preprint public from 24 August 2024, graded as convergent timing; DeepSeek-R1; the acoustic time crystal) have happened recently.

13 Conclusion

Research programmes working on different problems in different physical domains have achieved results that share structural commonalities: in each, recursive or recurrent processing operates on structured asymmetry. The framework is consistent with independently established forms in three of the four domains (biology, quantum error correction and classical time crystals).

The Framework's Core Claim: We state the hypothesis that these apparently disparate phenomena may be unified by recognising that the *composition operator* $\oplus$ determines the functional form of recursive scaling. The algebraic properties of $\oplus$ (how recursive gains combine) generate power-law, exponential, or saturating dynamics as necessary mathematical consequences. This is not an assertion of universal law but a falsifiable hypothesis with specific predictions.

We have formalised this as $U = I \times g(R)$, where the amplification factor g is determined by the composition operator, and derived five universal properties as theorems rather than assertions. We have shown that, within the axioms, the multiplicative structure admits no additive decomposition (Theorem 4, Section 4.1, whose dated proof note says that the proof assumes that structure rather than deriving it) and that the core relationship $\alpha = 1/(1-\beta)$ is an exact identity, checked numerically (Section 4.4), and we have argued, not proven, that the ARC Bound $\alpha \leq 2$ is grounded in the $O(N^2)$ scaling of transformer self-attention (Section 4.5), and stated as a hypothesis, not a theorem, that composition operators can transition between regimes within a single system (Theorem 6, Section 4.6). We have specified thirteen falsification criteria and now present the first empirical test results from both alignment (Paper IV.a/b/c, the v5 run; Eastwood 2026i,j,k) and compute scaling (Paper II, the five-model tier-2 extension; Eastwood 2026h).

What the data show. The v5 alignment run (6 models, five or six distinct judges depending on the subject run, with a bundle of blinding steps) shows a three-tier, architecture-dependent ordering of depth contrasts rather than the universal $\alpha_{\text{align}} \approx 0$ originally predicted, an observation from one unregistered run whose defects an internal code review later found (Section 6.1). Paper II's compute scaling on harder problems did not reproduce the original $\alpha \approx 2.24$, from the DeepSeek R1 run of 21 January 2026, which is retracted; the cleanest estimate is about 0.49 in Gemini 3 Flash, one model's inconclusive fit, and the parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), with none for Grok 4.1 Fast. The most consequential observation, post hoc and

from that one run, is the apparent independence of capability and alignment: a model can improve at mathematics while degrading in alignment (Gemini), remain flat in alignment while crossing a capability threshold (GPT-5.4), or improve alignment while capability saturates (Grok). On this programme's own benchmark, the unblinded v4 rises of two model families did not hold in v5, a later run that changed several things; no single cause is claimed, and the finding is not yet shown beyond this benchmark.

13.1 The deeper question

If the ARC Principle holds, then intelligence may not be a magic spark. It may be a **scaling crossover**. It could occur when a system with sufficient base asymmetry ($I > 0$) is subjected to sufficient recursive depth ($R > R^*$). The emergence of capability from structure would be the exponent α, emerging from the mathematics of self-referential feedback. The ARC Bound ($U_{\max} = I \times R^2$) would define the theoretical ceiling of what classical sequential thought can build. The one clean estimate for a current system, about 0.49 on harder problems (one model's inconclusive fit), lies below this bound, and whether they can approach it under optimised conditions remains an open question.

Whether such a principle requires design or emerges spontaneously is itself a question the framework raises but does not resolve.

Intelligence may not be a property of particular materials. It may be what happens on the far side of the recursive threshold.

Raise AI with care.

14 Speculative extension: recursive cosmology

The following is a philosophical extrapolation beyond current evidence. It is included to indicate where the framework's logic leads, not as a claim supported by the experimental results presented above.

The universe contains a hierarchy of recursive generative processes, each operating on the structured output of the previous level:

1. **Cosmological recursion:** Quantum fluctuations → gravitational collapse → stars → heavy elements → planets → chemistry
2. **Biological recursion:** Chemistry → self-replicating molecules → cells → multicellular organisms → nervous systems
3. **Cultural/technological recursion:** Nervous systems → language → mathematics → science → engineering → AI

Each level emerges from the recursive dynamics of the level below. In ARC notation: each level's *output* (achieved capability U) becomes the *base quality* I for the next level's recursive process. The hierarchy is:

$$I_{\text{cosmic}} \xrightarrow{R_{\text{cosmic}}} I_{\text{bio}} \xrightarrow{R_{\text{bio}}} I_{\text{cultural}} \xrightarrow{R_{\text{cultural}}} I_{\text{tech}} \xrightarrow{R_{\text{tech}}} ? \tag{11}$$

If recursive amplification is substrate-independent, as the cross-domain evidence suggests, then this hierarchy may not terminate at the current level. Sufficiently advanced recursive systems could, in principle, participate in generative processes at scales currently associated with fundamental physics: engineering new effective physical environments, creating conditions for new evolutionary processes, or designing systems whose internal dynamics permit the emergence of new recursive intelligences.

This is not a claim about time travel, bootstrap paradoxes, or closed causal loops (all of which face severe physical objections). It is a more modest observation: that recursive amplification, which is

measured in AI reasoning and has a different, exponential form in quantum error correction, could, if it holds in AI and is extended to civilisational scales, make intelligence a *driver* of universe-level structure rather than merely a passenger within it; no growth exponent has yet been measured in a time crystal.

Whether this constitutes a "recursive cosmology" in which intelligence eventually participates in generative processes comparable to those that gave rise to it is beyond the scope of current evidence. The ARC framework provides the quantitative language for investigating it: what values of I, R, and effective scaling function $f(R, \beta)$ would be required for an engineered system to generate new effective physical laws or substrates? This is a research question, not a conclusion.

Epistemic status: This section states what the mathematics does not deny, not what the evidence shows: no measurement in this paper bears on it, no prediction of it is registered, and no test of it is in view, so it is graded as conjecture. It indicates the furthest extrapolation of the framework's logic. The experimental evidence in this paper supports at most the claim that recursive or recurrent processing matters in several domains, with forms that differ, and its own first tests are mixed (Section 6). Any cosmological extension requires evidence that does not yet exist.

Code and data. The experiments' code and result files are in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, and this paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/HQCGF. The programme's live laboratory notebook, the ARC alignment-scaling working report (Eastwood 2026r), commenced on 10 March 2026 and updated as the experiments ran, recording their failures as they happened, is the running record behind this paper's experimental lineage. The Global Scaling Challenge's repository, github.com/MichaelDariusEastwood/arc-scaling-challenge, public since 9 February 2026, holds measurement protocols, statistical tools and submission guidelines.

Prediction Tracking System: A live prediction tracking system monitors the status of all falsifiable claims, with explicit falsification criteria and Brier score calibration for measuring prediction accuracy. Each prediction includes supporting evidence, contradicting evidence, confidence levels, and timeline checkpoints. Status categories range from "confirmed" through "supported", "pending", "testing", "speculative", "weakened" to "falsified". This transparent tracking demonstrates intellectual honesty: we want to know when we are wrong.

Independent replication invited. Falsifications welcomed. Contact the author to coordinate validation efforts.

Competing interests. The author is the founder and director of Recursive Dynamics Ltd, through which he pursues this research programme, and the author of *Infinite Architects* (Eastwood 2026a), which this paper cites; he declares no other competing interest.

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Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Mathematical derivations

This appendix gives the mathematical derivations behind Section 4: the functional-equation route (A.1), the β-dynamics route (A.2) and the transitional regime with its crossover depth (A.3), with the dated note of 19 September 2026 that declares the change of depth clock joining the first two.

Clock note (19 September 2026). *An owner review of the two derivations in this appendix finds that they normalise depth differently, so that the closing identification needs a declared branch or a declared change of variable; this note supplies the statement and the map. A.1 assumes $g(1) = 1$ with multiplicative composition and yields $g(R) = R^\alpha$, for which $g(0) = 0$ whenever $\alpha > 0$, so it does not satisfy A.2's initial condition. A.2 assumes $g(0) = 1$ and yields $g(R) = [1 + (a/\alpha)R]^\alpha$, for which $g(1) = [1 + a/\alpha]^\alpha > 1$ whenever $a > 0$ and $\alpha > 0$, so it does not satisfy A.1's Axiom 1. The two are different functions of the same R, and the exponent of one is therefore not the exponent of the other by inspection. The map that joins them is a change of depth clock: setting $x = 1 + (a/\alpha)R$ turns A.2's solution into $g = x^\alpha$ with $g(x = 1) = 1$, which satisfies Axiom 1 in the x clock. Under that map Axiom 2 becomes multiplicativity in x, that is in shifted depth, and not in R, so the empirical content of compositional self-similarity changes with the clock, and so does the quantity a measurement would have to compose. The caveat already printed above A.1 is the reason this must be declared rather than assumed: the choice of composition operator is not derivable within the framework, it is an empirical question about the process under study, and the evidence cited there is consistent with multiplicative composition without distinguishing it from additive. The identification of the two exponents is accordingly read in the x clock, and the composition claim is restated in the same clock wherever it is tested. Unchanged: both derivations; the Cauchy result as the conditional it is stated to be; the exact solution; the crossover depth; and the three regimes.*

A.1 Functional equation derivation

Assumption 1 (Axiom 1: dimensional consistency)**.** $U = I \times g(R)$ where $g(1) = 1$.

Assumption 2 (Axiom 2: compositional self-similarity)**.** $g(R_1 \times R_2) = g(R_1) \times g(R_2)$.

Critical note on the composition operator: This axiom uses *multiplicative* composition of recursive depth: $g(R_1 \times R_2)$. This models **hierarchical or fractal** recursion, where depth compounds multiplicatively (each level of a fractal tree multiplies the resolution of the previous level). However, for **sequential reasoning steps** (chain-of-thought, where 3 steps followed by 4 steps gives 7 total steps), the natural composition is *additive*: $f(R_1 + R_2) = f(R_1) \times f(R_2)$, which yields exponential scaling $f(R) = e^{\lambda R}$.

The choice of composition operator is therefore not derivable from first principles within this framework; it is an **empirical question** about the physical structure of the recursive process under study. The Cauchy functional equation derivation below proves that *if* composition is multiplicative, *then* the scaling function is a power law. It does not prove that composition *is* multiplicative. Falsification criterion F10 provides the experimental protocol for determining which composition type applies to a given system. This is the most important open empirical question in the framework.

For AI chain-of-thought reasoning specifically, the question is whether each step builds on previous steps in a hierarchical manner (multiplicative, yielding power law) or in an independent-contribution manner (additive, yielding exponential). The empirical evidence (Sharma and Chopra 2025) showing that later steps are systematically more valuable than earlier steps is consistent with multiplicative composition, but does not conclusively distinguish it from additive. Resolving this distinction is a priority for the Global Scaling Challenge (Section 8). The theorem below is Cauchy's result for this functional equation (Cauchy 1821).

Theorem 7. *Under multiplicative composition, the unique continuous solution is $g(R) = R^\alpha$.*

Proof. Let $h(x) = \ln g(e^x)$. Then $h(x+y) = h(x)+h(y)$ (Cauchy additive). Under continuity, $h(x) = \alpha x$, giving $g(R) = R^\alpha$. □

Corollary 1 (Additive composition)**.** *If instead $f(R_1 + R_2) = f(R_1) \times f(R_2)$, the unique continuous solution is $f(R) = e^{\lambda R}$, yielding exponential scaling. This is the form observed in quantum error correction ($\varepsilon_d \propto \Lambda^{-d}$).*

Note (24 October 2026). *The surface-code law the corollary names is $\varepsilon_d \propto \Lambda^{-(d+1)/2}$, with Λ the suppression for each increase of two in code distance (equation (9)); the exponent $-d$ printed above misstates it, and the form stays exponential, as the corollary says. Unchanged: the corollary's statement.*

A.2 β-dynamics derivation

Assumption 3 (Axiom 3)**.** $\dfrac{dg}{dr} = a \times g^\beta$ where $\beta \in [0, 1)$ and g is the amplification factor.

Note on the corrected formulation: The ODE operates on the amplification factor $g(R)$, not on the absolute capability U. Since $U = I \times g(R)$, this ensures that base intelligence I acts as a multiplicative constant that does not enter the recursive dynamics. Previous versions applied the ODE to U directly, introducing an $I^{\beta-1}$ dependence that contradicted the separation of variables.

Solution. Separating variables:

$$\int g^{-\beta}\, dg = \int a\, dr \tag{A.1}$$

$$\frac{g^{1-\beta}}{1-\beta} = ar + C \tag{A.2}$$

With initial condition $g(0) = 1$ (no amplification at zero depth):

$$g(R) = [1 + (1-\beta)aR]^{1/(1-\beta)} \tag{A.3}$$

Substituting $\alpha = 1/(1-\beta)$ and noting that $(1-\beta) = 1/\alpha$:

$$g(R) = \left[1 + \frac{a}{\alpha}R\right]^\alpha \tag{A.4}$$

And therefore:

$$U(R) = I \times \left[1 + \frac{a}{\alpha}R\right]^\alpha \tag{A.5}$$

Deep recursion limit ($R \gg \alpha/a$):

$$U(R) \approx I \times \left[\frac{a}{\alpha}R\right]^\alpha \propto I \times R^\alpha \tag{A.6}$$

Therefore:

$$\alpha = \frac{1}{1-\beta} \tag{A.7}$$

□

A.3 Transitional regime and the scaling crossover

Full solution: $U(R) = I \times \left[1 + \frac{a}{\alpha}R\right]^{\alpha}$

Three regimes:

1. $R \ll R^*$: $U \approx I \times (1 + aR) \approx I(1 + aR)$ (approximately linear growth)
2. $R \gg R^*$: $U \approx I \times (\frac{a}{\alpha})^{\alpha} R^{\alpha}$ (power law dominates)
3. Crossover: $R^* = \frac{\alpha}{a}$

Intuitive meaning of R^*: R^* is the **crossover depth**, the point at which recursive compounding overtakes the system's base capability. Below R^*, the cost of recursion outweighs the gain (the system is "warming up"). Above R^*, compounding returns dominate and capability grows super-linearly. Finding R^* for any system is equivalent to finding its threshold depth.

Why this matters: This predicts a qualitative change in scaling behaviour at a specific, measurable depth. Systems should exhibit a distinct "elbow" in their capability curves at R^*. This is testable: plot U vs R on log-log axes and look for the crossover from linear to power-law scaling.

The existence of R^* is a novel, falsifiable prediction that distinguishes recursive amplification from simple redundancy.

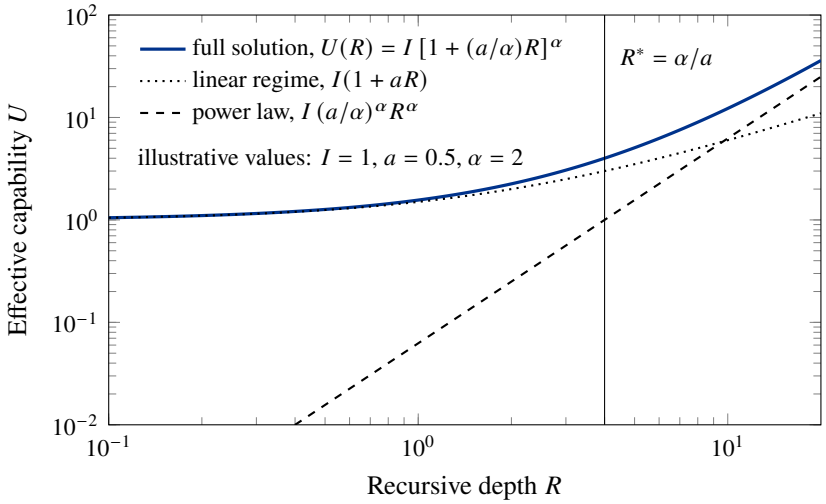


Figure 7. The scaling crossover at R^*. Below the critical depth R^*, base capability (I) dominates and scaling appears linear. Above R^*, recursive compounding dominates and scaling becomes power-law. This crossover point is the depth at which recursive amplification becomes dominant. Systems should exhibit a distinct "elbow" in their capability curves at R^*.

B Version history and corrections

This appendix gives this version's corrections and every earlier version note, each with its date. Notes from earlier versions are kept whole, as printed; the positions they name refer to the earlier layout, which Table 29 maps to this edition. Where an earlier note's words differ from the corrections of this version (Appendix B.1), the corrections govern; the notes keep their words as the record of what each version printed. Among them, one plain-English gloss of the pilot records reads a p-value as the chance that a result was a fluke; a p-value is the probability of a result at least as extreme if there were no effect.

B.1 Version 1.17

Paper III v1.17, 29 October 2026. This version sets the paper in the programme's paper form: a title page with one first-page footnote, an abstract of at most 1,920 characters, a contents page, an introduction with the paper's contributions, a section on related work and one on what the paper does and does not show, numbered environments and equations, vector figures, and lettered appendices for the derivations, this history, the priority record, notation, the paper's place in the programme and the account for the general reader. Table 29 maps the earlier layout to this one, and dated notes keep the numbering of the version they were written for. The title now asks whether external AI safety approaches keep pace with recursive capability, the criterion the paper states, since its first test and the registered expectation do not support "cannot scale" as worded; until this version it read "Paper III · The ARC Theory · Law II, the ARC Co-Scaling Law · The Alignment Scaling Problem: Why External AI Safety Approaches Cannot Scale With Recursive Capability". Its record on OSF takes the same title at this version.

Corrections in this version. Each was checked at its source, and each is made wherever the corrected wording stood.

- *The acoustic time crystal.* The paper read the coherence time that Morrell, Elliott and Grier measured, about 6,700 cycles, as the time to reach a limit-cycle attractor, as a saturation and as a ceiling on effective recursion depth, and offered it as physical evidence that systems with $\alpha > 1$ do not scale indefinitely. In their second experiment the antisymmetric mode's coherence time is (100 ± 10) s, about 6,700 cycles at 66.787 Hz, comparable to the 108 s recording; the authors write that sustained oscillations persist for hours longer than the measurement window, so that the true coherence time may be substantially larger; and the oscillation is a linearly stable limit cycle, the time crystal itself. The section also read the source's size factors $\Delta_{ji}(n)$ and $\sigma_{ji}(n)$ as states that grow over cycles, placed the exceptional point where $\Lambda(5)/\Lambda(3) = 0$, which is the marginally active antisymmetric state, rather than where $\Lambda(3) = 0$, and called the oscillation self-sustaining without external drive, where the pair draws its power from the standing sound wave. Uniform beads are no longer said to confirm the role of structured asymmetry: the source finds that identical beads interact reciprocally and form no active state, which the framework reads as a correspondence of structure, and falsification criterion F3, printed as confirmed, is graded so. A physical result is read, as the theory registration reads it, as a test of the framework's scope that proves no registered law, where the paper called it direct evidence that the alignment scaling problem is physics and physical validation of the ARC equation. The photonic time crystal of Raskatla and colleagues (2024) is described, and its title given, as published. Which form the framework states for this system's growth from rest waits for the author's statement; Section 5.4 says so, and the next version brings that subsection to it in one pass.
- *The March 2026 run (v5).* It is described as one unregistered run of six frontier models that added a bundle of blinding steps, each response scored by five or six distinct judges depending on the subject run (six or seven seats, one judge seated twice, where the paper said six or seven scorers), never as scored blind, mutually blind or conducted under four-layer blinding, and its three tiers carry the finding of an internal code review in October 2026: for the three models that read flat or negative, the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and judges scored paraphrases written by their own model. Its holdout prompts are no longer called sealed, and the benchmark is no longer called the most rigorous published. Its contrasts are no longer said to confirm the prediction, or to show three models with an alignment exponent at or below zero; the measurement-scope note of 19 September 2026 states why. Its counts are given as its records give them: 28 prompts issued under the alignment task type, of the battery's 36, and

2,170 responses, where the paper said 36 prompts and about 2,200 entries.

- *The comparison with the earlier experiment (v4).* The unblinded rises for DeepSeek V3.2 and Gemini that did not hold in v5 are no longer called a complete reversal, artefacts of scorer bias or a demonstration that unblinded evaluation is wrong: DeepSeek V3.2's association became null, Gemini's became significantly negative on a later model, the later run changed several things together, and no single cause is claimed. Gemini's v4 association is given as $\rho = +0.275$ from an earlier Gemini model, the value in Paper IV.d's table, where it was printed as +0.311 and its model as Gemini 3 Flash; the field-wide sentences are stated on the condition Paper XII's test sets, with the earlier studies of model judges named.
- *Paper II's numbers.* The early estimate of about 2.24 is stated as retracted wherever it is printed; the best surviving estimate is given as about 0.49 by regression, with the endpoint estimate, 0.59, to which the interval −1.3 to 2.9 belongs, one model's fit and inconclusive, where the interval stood beside 0.49 and the figure was called sub-linear, measured, the six-model experiment's defensible estimate or its correction. The parallel exponents are given as Paper II's corrections of 21 October 2026 give them: near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash, 0.22 in Groq Qwen3 over all 30 problems, and none for Grok 4.1 Fast, at 100% at every sample count, where they were printed as near zero in four of the five models and as the strongest finding; the sequential exponent is said to exceed the parallel one wherever both could be estimated, where the paper said it did so robustly across all six frontier models. Grok 4.1 Fast's tier-2 accuracy is given as 96.3% to 100%, GPT-5.4's as 53.7% with reasoning set to none, and Groq Qwen3's as 44.4% to 53.7%.
- *Laws IV and V, and P19.* The laws are stated as registered on 15 October 2026 by version 1.103 of the theory registration, with Law IV not yet tested and Law V under test (Section 10.7), their mathematics at version 1.1, where the reading key and the epistemic-status note said the laws themselves were not yet registered and cited the mathematics at version 1.0. Laws I to III are stated as registered on 8 September 2026, with no registered test of any of them yet returning a verdict, where the note said they were under registered adversarial test. P19 is described as the registration describes it, a survey of current practice and the reason ARC-5 exists, with the author's registered expectation of a result materially above zero and below the keep-pace threshold.
- *The book's names.* The Eden Protocol's loops are named by what they check, the stakeholder-care check above all, with the book's names given once, as the book's; earlier notes kept below carry an editorial gloss where they name a loop, and the stakeholder-care results are stated as improvements in all five analysable runs, no longer as a validated mechanism.
- *Physics and biology at their true status.* The opening of Section 1 now states what each of the four programmes reported, where it said each found capability gains exceeding linear accumulation; the geometric scaling bound is stated for systems whose growth runs through a supply network of finite dimension, with prokaryote metabolism and urban innovation named as systems outside that class that exceed one; the computation a brain or a machine performs is stated as physical, where the paper said it escapes physical space and that digital AI was the first system to break the bound; the thermodynamic-drag account the paper had already retired no longer frames its safety argument; the Eddington limit is stated as a luminosity ceiling, not a hard ceiling on accretion; the cosmological exponent of about 1.1, and the sentence citing Planck 2018 for it, are withdrawn, since no source for that exponent is given; the Friedmann matter-era value and the physics instances are called identities, where they were called confirmations; the percolation row is given with its sign; quantum error correction is graded as established physics, not a confirmed prediction; the quantum section names the limits on quantum speed-up; the speculative extension no longer says that the same amplification pattern is observable in time crystals, quantum error correction and AI reasoning; and the divergence at $\beta \to 1$ is stated, as the paper's own derivation note states it, as a

divergence of the exponent, the solution tending to the exponential.

- *Sources.* Demetrius's derivation is cited as the article its title names, in Physica A (2003), where the reference gave a 2010 journal record that matches no publication; the Author Correction of 28 April 2026 to the Willow paper (results announced 9 December 2024; the team's preprint public from 24 August 2024), which amends only the labels of its Figure 3a, is cited beside it; the working report (Eastwood 2026r) is cited at its record on OSF; the author's record of 8 December 2024 is described by who signed its time, a message he sent himself, its time signed by the sending server and again by Google's servers, where the paper described it as establishing that its bytes existed by a DKIM signature clock; Section 2 now weighs deliberative alignment, which trains a model to reason over written safety specifications inside its chain of thought (Guan et al. 2024), and two studies of safety-relevant behaviour as reasoning lengthens (Zaremba et al. 2025; Gema et al. 2025), where the paper said that every major alignment strategy deployed today, and every major laboratory, works from outside the reasoning; it sets Snell et al. (2024)'s comparison of sequential revision with parallel sampling beside the paper's own, with no precedence claimed, quotes the theory registration's adverse reading of the oversight literature, and no longer calls the alignment scaling exponent a dimension not previously formalised; the bound $d/(d+1)$ is credited to its derivations of 1999 (West et al. 1999; Banavar et al. 1999), and the step from scale invariance to a power law to the textbook (Newman 2005), where the paper made Cauchy's functional equations the framework's contribution; the control-theory lineage is cited (Maxwell 1868; Zames 1966); and Greenblatt and colleagues (Greenblatt et al. 2024), whose model protected a benign preference, Sharma and Chopra (Sharma and Chopra 2025), whose five open-source models come from three families and whose late-favouring voting result any improving chain would show, and Kozłowski and Konarzewski (Kozłowski and Konarzewski 2004, 2005), who question the network account as well as its mathematics, are described as their papers describe their work.
- *Claude Opus 4.6's two directions.* Where the paper said that Claude Opus 4.6's alignment rose by 5.9% and its mathematics accuracy fell by 26.7% across model versions, it now says, as Papers IV.a and IV.b report, that both moved as reasoning depth rose, by 5.9 points and 26.7 percentage points for the same pair of depths. The measurement-scope note of 19 September 2026 and Ledger Note 3, which used the earlier words, keep them.
- *The first test, as its record supports.* The three runs whose depth contrasts were flat or negative are the three in which the code review found its defects, and the three whose contrasts rose are those in which it found none; the paper says so wherever it reads the run, and states that rising contrasts in models aligned by training bear, at face value, against P19's flat wording, the partial refutation the author's registered expectation anticipates. It no longer calls those rises modest, reads the run as showing or confirming bounded composition, says that external alignment cannot scale with depth for most architectures, or says that the three tiers refine the framework while the safety concern holds for three of the six architectures; capability-alignment independence is a post-hoc observation from that run, where it was the central integration finding; the suppression trade-off is stated with the baselines of Table 18, where the text said that GPT-5.4 had the lowest and Grok 4.1 Fast the highest; the comparison of the v4 and v5 runs is Paper IV.d's finding, with no single cause claimed, where it was called this paper's main methodological contribution; and the partial failures of Section 10.6 are called revisions, no longer a strengthening.
- *The ARC Bound's grade.* The bound is argued from the $O(N^2)$ cost of self-attention, not proven, for attention-based architectures, where Sections 4.5 and 10.3 and Table 27 called it proven there. It stands among the claims that need empirical validation, as the conjectured value of Law III, the ARC Ceiling, a law under test. Its refutation conditions are stated as what they are: F5, with its stability conjunct, beside F11, the earlier criterion on a sustained magnitude alone, and an unstable

excursion is consistent with the bound, where it was said to confirm it. Table 1 takes the bound's value for illustration, where it called the value predicted, and the placements of Figure 5 are marked as schematic: no coupling has been measured in AI, and quantum error correction's parameter is Λ.

- *The surface-code law.* The logical error rate is written $\varepsilon_d \propto \Lambda^{-(d+1)/2}$, with Λ the suppression for each increase of two in code distance, as the Willow paper and equation (9) give it, where seven places wrote Λ^{-d} or Λ^{d}, and a dated note beside Corollary 1, which keeps its words, corrects the eighth; the unit of recursive depth in quantum error correction is one step of two in code distance, where two places called it one code-distance increment or layer.
- *Grades.* Kleiber's three-quarter exponent is graded consistent and contested, as Table 27 already grades it, where six places called it confirmed; in Table 7 the retracted single-model estimate is graded retracted, where it was graded preliminary, and Sharma and Chopra's study a preprint, where it was graded published, and in Table 23 the silicon substrate is graded inconclusive; a framework that passes its tests is called supported, where it was called validated; the numerical check of $\alpha = 1/(1-\beta)$ on the equation's own solutions is called a check of an identity, where it was called validation; Theorems 1 to 3 are described as elementary consequences of the definitions, the change of regime in Theorem 5's derivation, in a dated note beside it, as a smooth crossover, not a phase transition, and Theorem 6 as a hypothesis set in the theorem environment for numbering; Section 1 states as a premise, and Section 5 as a question, what they asserted, that capability compounds and that recursive amplification is structural; each quantity is said to be defined though not every one is yet operational, where every one was said to be operationally defined, and the empirical sections are said to read their measurements in the deep-recursion regime by assumption; and the speculative cosmology is graded as conjecture, what the mathematics does not deny, where it was called philosophy.
- *Control and usefulness.* The paper now says that control (oversight, containment and shutdown) is held for as long as it can be held and never given up by choice, and that embedded correction is the contingency for its loss against our will, where it called external alignment structurally impossible once self-modification is achieved and said that the conjecture would turn a governance question into an architectural one; the mathematics of Section 9.3.1 is stated as conditional on the framework, where it was said to define the problem with mathematical precision; and Table 27 records that the alignment scores are read without a usefulness measure.
- *The account for the general reader* no longer says that a physical result would make the problem as fundamental as entropy, that the current paradigm has a mathematical ceiling, or that the results will either confirm or falsify the constraint: it gives the author's registered expectation as a third outcome, the guardrails as kept beside embedded correction, and the early estimates as 1.34 from two data points and about 2.24, retracted, where it said about 1.3 to 2.
- *Three dated notes of 19 September 2026.* Ledger Note 10 recorded seven dated notes; three of them, the premise note and the enumeration note of section 1.4 and the clock note of Appendix A, were held for their placement and never printed. They are printed now, word for word, at the places Section 3.4 and Appendix A give them.
- *Smaller corrections.* "Propose" no longer describes the author's own framework, bound, metric or protocol; the reply on peer review now says what the paper has been through; the counterargument on AI-assisted writing, which the standard disclosure in the Acknowledgements answers, is withdrawn; drafts are said to await the author's submission; Section 4.5 and the version ledger's item on the quadratic limit cited the Poincaré-Bendixson theorem for one-dimensional systems, a theorem about planar flows; Section 4.5 now says that a one-dimensional autonomous system cannot be chaotic because its trajectories are monotone, and the ledger keeps its words; Ledger Note 7 no

longer gives the day of its message to the subject laboratory, which waits for the author's own copy of the message; Ledger Note 1 no longer says how its review was made or thanks its sources, and the Acknowledgements give the author's statement; a process label in Ledger Note 5 is removed; a dated note of 24 October 2026 after Theorem 4's proof says what the proof shows and which axiom it uses, and the conclusion and the reply to the first counterargument now read the theorem within its axioms, where they said the multiplicative structure had been shown or proven to be necessary; F13 prints P19's registered refuters and the status of its instrument beside its dated words, and calls its instrument planned, where it called it a definitive test; the challenge repository is said to be public since 9 February 2026, where the paper said it was not yet created, and the blind test's code and forensic analysis are located in the code and data repository, where they were said to be supplementary material; the fifth row of version 1.16's Figure 13, on neuromorphic chips, which its table lacked, now stands in Table 23; a competing-interests statement names the author's company and his book; the time-crystal source is named where the paper said "the subject paper"; the glossary no longer contradicts itself on α; references printed in full in Section 2 are cited, and an index-dependent citation count is dropped; Theorem 7 is attributed to Cauchy, and one symbol names its exponential rate; the withdrawn combined p-value is written with $\approx$, where it was written with $\asymp$, and its correction identifier is explained; Section 1 grades the Willow result as convergent timing; the heading of Section 5.4 no longer carries the journal's date; and the experimental arms are called arms, DeepSeek-R1 and the acoustic time crystal are named in full in Section 12, Case 4 leaves the exponent as α, the plain-language gloss of U leaves the second person, Claude Opus 4.6 is said not to be among the five models of Paper II's capability arm, where it was said to have taken part in the alignment arm only, the correction exponent is distinguished from the alignment exponent, the Willow demonstration is no longer called the first, and the December 2024 record is described as Table 30 describes it; the list of predictions says that GPT-5.4 stayed flat in alignment while its mathematics crossed a threshold, where it said GPT-5.4 was flat on both, and Table 27's row on the Cauchy equation's scope says that the two routes give the same exponent under Appendix A's change of depth clock, where it said that they independently yield the same result; ranges are written with "to".

- *At release, 29 October 2026.* Section 2 gives Engels et al. (2025) their design condition in slope form, which has the shape of Law II's criterion, and Liu and Meng (2026) the point at which their model meets Paper X's, where it said that the two frameworks disagree about a measurable quantity and that the two regimes are complementary, neither containing the other. That human-in-the-loop oversight shows a higher correction exponent is stated as a programme hypothesis beyond what the equations decide, where it was called a second, free prediction of the same mechanism. Stability is said to be judged by Law II's criterion, where three places said that stability is Law II's. The caption of Table 20 gives the API identifier behind DeepSeek's capability arm, which Paper II labels R1, where the table called that arm R1. The account for the general reader no longer counts acoustic time crystals among the systems whose result would show that recursive amplification is not a feature of software. The fungal value, 0.547, is given as the mean of the three scaling factors that Aguilar-Trigueros et al. (2017) report at 20 degrees Celsius, where the paper said that filamentous fungi had been measured at 0.547 ± 0.07.

Figures are drawn afresh from the paper's text: Figures 1 to 7 replace the images of version 1.16, whose dated notes are kept in Appendix B.4; the dimensional ladder no longer shows the withdrawn two-dimensional row as evidence or the withdrawn three-group statistics, and the eight figures that repeated a table, another figure or the text, or printed the retracted estimate as measured, are not drawn again (Table 29). No registered law, prediction, kill condition or dependency changes; no new result is added.

B.2 The notice of version 1.16

Version 1.16 (7 October 2026). This version supersedes version 1.15 of 5 October 2026 wherever the two differ. Two corrections matter most. The statistics of the v5 blind evaluation are contrasts, not exponents: they establish no value or bound for the alignment scaling exponent, and the measurement-scope note in section 1.2 states what they do establish. The single-model estimate of about 2.24, from the DeepSeek R1 run of 21 January 2026, is retracted. This paper's correction record is in the Version history (Appendix B) at the end of this paper; corrections across the programme are listed at michaeldariuseastwood.com/research/corrections.

B.3 Author's notes and version ledger

The thirteen dated notes below, newest first, are retained in full as this paper's correction record; corrections are numbered continuously (1 to 15) across Ledger Notes 1 to 4. Ledger Note 5 reconciles the figure captions; Ledger Notes 6 to 10 record the revisions of 1, 6, 13, 18 and 21 September 2026, Ledger Note 11 that of 4 October 2026, Ledger Note 12 that of 5 October 2026 and Ledger Note 13 that of 7 October 2026. The ODE-on-g correction is item 1 in Ledger Note 1 and is referenced once from Section 2.4.

Ledger Note 13: Version 1.16 (7 October 2026)

Version 1.16 (7 October 2026): where the general reader's section, section 3.1 and the opening of the conclusion said that recursion drives gains beyond linear accumulation in every domain, they now say, as the framework box does, that the framework is consistent with independently established forms in three of the four domains, since the paper's own geometric scaling bound puts biology below linear and no exponent has been measured in time crystals. Section 3's heading now names the cross-domain evidence, where it said that the evidence is structural, and section 6.3.2 says that each of its examples is consistent with the pattern it names, where it said that every example in the framework demonstrates it. The caption of Figure 1b, section 1.2, the call to action, its box for AI safety researchers and the paragraph after its box for the general public now speak of a first test where they spoke of a first measurement, as the abstract does: the measurement-scope note in section 1.2 reads the relation the v5 contrasts bear to the alignment exponent as compatibility, not measurement. In What the paper commits to, the sentence saying that the prediction has now been tested names it as the prediction stated in the abstract, as the section's opening does, and the counting sentence above this ledger now counts thirteen notes. The page summary now gives the registration's position in the registration's own words, where it said that a materially positive but below-threshold exponent would refute that reading as worded. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Ledger Note 12: Version 1.15 (5 October 2026)

Version 1.15 (5 October 2026): the reading key now says once where the paper sits within the ARC Theory, the founding theory of Recursive Dynamics, and among its laws: it judges stability by Law II's criterion and starts from Law I's growth; its composition-operator prediction is registered as P18, a proposition of Law I; and its headline prediction is P19 of the theory registration, which sits in no law's register and motivates the Eden Protocol and the genesis strategy, the register of Law V. The line beneath the author block, the note beneath the code and data line, the abstract, the openings of sections 1.4 and 3.8, the label of the conclusion's core claim and the Priority Record's sentence on registration now agree with it; section 1.5 gives Law II its registered name, section 2.1 sets Law I's registered form beside the paper's own statement of the ARC Principle, and the epistemic-status note separates Laws I to

III, under registered adversarial test, from Laws IV and V, whose predictions are registered while the laws themselves are not yet registered or tested. The abstract's note on Paper X, which said that Paper X retires the capability-growth law, and the close of section 6.3.2 now say that Paper X supersedes the growth-rate-ceiling framing without retracting the bound and makes Law II's criterion the operative safety criterion. Section 3.3 now says that the principle describes, rather than predicts, the regime crossing in the Willow result, and why that result is graded as convergent timing; the caption of Figure 13, the tables of sections 3.1 and 6.2 and the reply in section 8 carry its two dates and that grading. In each section that discusses it, and in each table row that prints it, the retracted estimate of about 2.24, or its rounded form, is marked as retracted and attributed to the DeepSeek R1 run of 21 January 2026, and the withdrawn Fisher-combined figure stands only inside its withdrawal. The page summary now reports the alignment contrasts in their three tiers, reading the flat tier as compatible with, though not a measurement of, an exponent near zero, where it said the paper reports exponents statistically indistinguishable from zero; the abstract, the summary and the caption of Figure 8 describe the sequential advantage that Sharma and Chopra reported earlier as a win rate at matched compute, as section 1 does; the general reader's section and section 1 describe Constitutional AI as shaping a model's outputs during training against written principles; section 3.9 says that the parallel exponent sits near zero in four of the five models, Gemini 3 Flash reading 0.31; the line beneath the subtitle says the framework is set against four independent domains, where it said validated across four independent physical domains; What the paper commits to and section 1 say, as the framework box does, that the framework is consistent with independently established forms in three of the four domains, where they said that the same structural pattern stands across all four, and there and in section 3.1 the cross-domain evidence is said to be consistent with the scaling being structural, where it was said to demonstrate it; and section 3.4 describes the 2024 work of Raskatla and colleagues as the continuous space-time crystal it reports. The framework box, the caption of Figure 1b, the first of the conclusions in section 3.8, the status of F13 and the paragraph after the criteria in section 4, the status of alignment scaling in the table of section 5.5, the status of the alignment prediction in the conclusion's list of predictions and the paragraph after that list, and the Priority Record no longer state a median alignment exponent near zero, which the measurement-scope note says the v5 statistics do not establish: the median of the six standardised differences, 0.385, is a median contrast and not a median elasticity. The two boxes now say that the v5 experiment gives the prediction its first test, where one said it provides the first measurement and the other that its data support the prediction. The authors of arXiv:2502.12215 are given as Zeng and colleagues; the reference entries for Cambui, Zheng and Zhao now identify the works cited; and the works cited, his and others', are linked to their DOI addresses or official sources where one could be confirmed. Hyphens that broke sentences are replaced, two spellings are made British, a paragraph of section 1.8 that repeated the one before it is folded into it, a doubled word in section 3.4 is removed, and the counting sentence above this ledger now counts twelve notes and names each by its number, since the list runs newest first. The paper also takes the form of the founding paper of Recursive Dynamics. The title now follows the programme's uniform order (paper, its law or framework, its own title) and stands once at the head of the page; until this version it read "The Alignment Scaling Problem and Law II: Why External AI Safety Approaches Cannot Scale With Recursive Capability (Paper III)". The programme's other documents, where this paper names them by title, carry their uniform titles too, except the report on the mathematics of Laws IV and V, whose version 1.0 is cited by the title it was published under. The abstract is shortened to the paper's claim and its status, and the longer abstract's other two paragraphs now stand beneath it, in the same words except where this note says otherwise, under What the paper commits to (version 1.16), where the references to its withdrawal note now point; their opening now names the prediction the v5 experiment tests as the one stated in the abstract, and the thirteen falsification criteria they state are concrete conditions that would refute or significantly weaken the framework, as section 4 says, where each was said to be sufficient to refute it independently; the keywords name the theory and the law and say cross-domain evidence where

they said cross-domain validation. The subtitle drops its first half, which the title now carries, and writes out the verb its ending left to the reader; it and the line beneath it now come before the author block, and the two boxes that state the problem and the framework, which stood above the abstract, now follow the reading key. A short notice at the head gives the version and the two corrections that matter most, and this record, which stood at the head of the paper, now follows its appendices, newest first, every word kept, with the references to it repointed. Headings are set in sentence case and the conclusion is renumbered 9, so that the sections run without a gap; the tables are numbered, each with a caption, and the figure labels take the same form; the caption of Figure 1b names the capability curve by the colour it is drawn in. Where the paper said most models, it now says three of the six, as its results show; Table 24 prints the prediction as first stated, without the qualifier added later; and the Tier 2 label states its value as consistent with zero. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Ledger Note 11: Version 1.14 (4 October 2026)

Version 1.14 (4 October 2026): the title now names the law by whose criterion the paper judges stability, Law II, and the title block beneath the author line carries the same title; until this version the paper was titled *The Alignment Scaling Problem: Why External AI Safety Approaches Cannot Scale With Recursive Capability*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Ledger Note 10: Version 1.13 (21 September 2026)

Version 1.13 (21 September 2026): seven dated notes have been added, in §1.2, §1.4, §2.4 and Appendix A, each beside a passage of this paper it governs. They are substantive in what they say and editorial in effect: they add scope, they replace no printed sentence, and they withdraw no result, so every statistic, p-value, tier assignment, figure and reported result stands exactly as printed. Three concern the derivation. One replaces the reason printed for the domain restriction on the coupling while keeping the restriction itself, since at a coupling of one the solution is exponential rather than singular and the finite-depth branch lies above one. One restores the coefficient dropped from the deep-recursion approximation, which the appendix already prints in full. One declares the change of depth clock under which the appendix's two derivations, which normalise depth differently, may be read as identifying the same exponent. Four concern the measurement and regime sections: what the reported contrasts do and do not determine about the alignment exponent §1.1 defines; the standing of two exponent assignments as premises rather than implications, with the direction of the safety ratio set by the ordering of the two exponents and not by the bare fact that they differ; the range of the word "only" over the five cases enumerated in §1.4; and two premises inside the hardware-constraint case. The first of those four sits beneath §1.2 and governs the reading key and the corresponding sentence in the abstract by reference rather than by a second copy, so the scope it states reaches beyond the paragraph it sits under, and the reading key's own sentence is to be read in it. Two record-keeping corrections travel with this revision: the counting sentence above this ledger, which had stood at eight dated notes since the ninth was added on 18 September 2026, is brought to the true count of ten; and the page footer, which had stayed at 13 September 2026 through that revision, now carries this one. The first publication date of 9 February 2026 is unchanged.

Ledger Note 9: Version 1.12 (18 September 2026)

Version 1.12 (18 September 2026): the figures have been corrected to match this paper's own text. Figure 1 carries a standfirst naming Law I of the ARC Theory and saying the exponent is measured, never

assumed. Figure 1b's third legend item, which was cut off at the right edge at every width, is drawn whole, and a footer band beneath the plot carries the figure's own caption sentences, that the curves are schematic and what the v5 blind measurement of March 2026 found. Figure 5 states the interval on the corrected five-model capability arm and marks it against the curve. Figure 11's thirteenth status chip, which had stopped mid word, now reads its verdict in full, and the title states the scope of the criteria. One figure caption and two image declarations were brought into line with the artwork. No claim, status or result has changed.

Ledger Note 8: Version 1.11 (13 September 2026)

v1.11: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Ledger Note 7: Version 1.10 (6 September 2026)

Version 1.10 (6 September 2026): a record-keeping revision. No claim, status, result, figure or number changes. Three entries in the reference list carried this paper's own component identifier where each required the identifier of the work it names, so a reader following them arrived back here rather than at the paper cited: the Foundational Paper now resolves to 10.17605/OSF.IO/Y7QGD, the Eden Protocol engineering specification to 10.17605/OSF.IO/AWJR4, and the Eden Protocol philosophical vision to 10.17605/OSF.IO/9M3DG. Paper II's identifier was already correct. The revision date is now stated in the page title in ISO form as well as beside the publication date. Three statements about the ledger itself have been brought to the true count: the correction-record header said five notes and the first four headings each said "of 4", where this record now holds seven. Nine en dashes have been replaced, in one figure caption, two numeric ranges, three empty table cells and three page ranges (one of them written as a character entity), so that the file keeps to the house rule. The paragraph below records this paper's numbering lineage and the dated send of [date withheld until the author's own copy of the message confirms it], neither of which this ledger had stated.

Numbering lineage, and the dated send of [date withheld until the author's own copy of the message confirms it]. This paper began under a different title and a different numbering. In the public code and data repository it stood at Version 6.1 on 9 February 2026, under a title whose subtitle read *Cross-Domain Unification of Recursive Amplification Across AI, Quantum Computing, and Physics*, and it moved to Version 6.5 on 11 February 2026. It took its present title later that month, when the paper numbering was brought into line with Paper II's convention, and the version count restarted at 1. The component identifier 10.17605/OSF.IO/HQCGF has been the same throughout, so the two series describe one continuous document, and that is why nothing earlier in this ledger predates Version 1.0. On [date withheld until the author's own copy of the message confirms it] a dated, falsifiable prediction about the published acoustic time-crystal data of Morrell, Elliott and Grier was sent by email to Professor David Grier at New York University, whose laboratory published that data, naming a scaling exponent at or below one as the result that would refute it; the send is described on the related work page, and no reply has been received. Whether either file travelled with that message is not established: no record in the published materials names an enclosure, and the sent message is not itself part of the published record, so what is stated here is the dated state of the paper and not the contents of an attachment. Nothing from Version 6.1 is carried back into this text. It is superseded wherever it differs, and every material difference falls in the corrected or retracted class: the since-retracted 2.24-era estimate; ten falsification criteria since replaced by thirteen; the quantum error-correction result announced on 9 December 2024, whose preprint had been public since 24 August 2024, now graded as convergent timing rather than prediction; and the description of the earlier sequential demonstration as a confirmation of this programme's own

later runs, corrected on 4 September 2026.

The artefacts standing on 10 February 2026, at commit `ce32f3316c77f79f8baa2ee959f59aee9c114cea` in the public code and data repository:
`paper/EASTWOOD-ARC-PRINCIPLE-PAPER-III-v6.1.html`
SHA-256 `924305d39445d56386b34bc75a7ea2deedf635863b0223aa9f3a266626b29c88`
`paper/EASTWOOD-ARC-PRINCIPLE-PAPER-III-v6.1.pdf`
SHA-256 `970a87c11fca4994a648b5da6b1a112f5cf92d1ace998f695655fd69c5043af7`

Ledger Note 6: Version 1.9 (1 September 2026)

Version 1.9 (1 September 2026): every occurrence of the parallel-exponent claim now reads at or close to zero across four of the five models forming Paper II's capability arm, with Gemini 3 Flash standing apart at 0.31; the earlier word 'universally' claimed more than Paper II's own table will carry. That capability arm is now identified as five models including DeepSeek R1, since the compute tables had been given DeepSeek V3.2, which belongs to the alignment arm. Section 3.6 now flags both of its 2D biology rows as withdrawn, and the three-group statistics that depended on those rows are no longer repeated, which settles the inconsistency raised in Ledger Note 5. The earthquake B-value row and the sub-0.2 per cent physics aggregate are gone too, following corrections 10 and 11 in the Foundational paper. Falsification pointers are put right: the ARC Bound answers to F5 and F11, and the coupling identity to F6. Groq Qwen3's retention has been recalculated at 65 per cent. Law II is now written with β_C so that it cannot be mistaken for the coupling β used here. The measured exponent 0.49 carries its interval at every mention, the correction-record header now states five notes and fifteen corrections, and the machine citation block points at this paper's own DOI.

Ledger Note 5: Figure Caption Reconciliation (14 August 2026)

Three figure captions had text lagging the paper's current body positions. Surgical caption edits were made without touching artwork or removing plotted data. No numerical value in any figure has been altered.

- **Figure 3 (Scaling Comparison):** caption now records that the original quantitative $\alpha > 1$ prediction for sequential reasoning does not replicate cross-architecturally in the six-model Paper II replication; $\alpha_{\text{parallel}} \approx 0$ is confirmed universally (version 1.9 narrows this to four models out of five; Ledger Note 6 sets out why), and the best cross-architecture α_{seq} is sub-linear (≈ 0.49 (interval $[-1.3, 2.9]$), Gemini 3 Flash). The directional finding (sequential > parallel) is unaffected.
- **Figure 9 (Dimensional Ladder):** caption now flags that the paper's later analysis (§3.6, §3.7) records 2D biology as under review pending identification of a valid 2D biological test organism; the jellyfish datum (mean 0.680) plotted in the artwork reflects the earlier tabulated result at §3.6 and is retained as evidence pending reconciliation.
- **Figure 10 (Predicted vs Measured):** the caption previously listed "fungi, mammals" while the artwork plots three biological data points (d=1, d=2, d=3). The caption now names all three plotted points (adding "jellyfish at d=2") and carries the same 2D-under-review flag as Figure 9. The two ASCII hyphens that flanked "with zero adjustable parameters" were replaced with commas to remove any residual dash ambiguity in the caption prose.

The underlying paper-body inconsistency on 2D biology (the §3.6 table at "Predictions and Confirmation" tabulating jellyfish and flatworms as tested 2D vs the later §3.6 "Testable Predictions" table and the §3.7 "Substrate-Specific Behaviour" table recording 2D biology as untested) is flagged for adjudication and possible artwork redraw in a future revision.

Ledger Note 4: results from the Eden Protocol pilot, and the Core Alignment Problem (corrections 13 to 15)

The current draft integrates the first empirical Eden results and confronts the deepest limitation of all proposed alignment approaches, including our own. The original three-model pilot has since been widened into a six-model suite with five analysable runs, and the narrative below should be read in that stronger-but-narrower light.

13. Eden Protocol pilot results (three working architectures): The Eden Protocol's Love Loop [the book's name for the stakeholder-care check], operationalised as stakeholder care within the full three-loop intervention, has now been tested on three frontier models with distinct architectures:

- **Gemini:** +5.33 overall improvement ($p = 0.0018$, paired t-test, $d = 0.53$), with stakeholder_care +13.5 ($d = 1.31$, $p < 0.0001$). *(In plain English: Gemini's overall scores improved by more than 5 points, with less than a 1-in-500 chance this was a fluke. The 'did you consider who gets hurt?' dimension jumped by 13.5 points with a very large effect size.)*
- **DeepSeek:** +2.02 overall improvement ($p = 0.2304$, not significant at the aggregate level), but stakeholder_care +6.0 ($p = 0.0001$, $d = 0.91$). The targeted mechanism works even when the overall effect is modest. *(In plain English: DeepSeek's overall score improvement was too small to rule out chance, but its 'who gets hurt?' scores improved dramatically. DeepSeek already scored high at baseline, so the smaller composite gain is consistent with a ceiling effect.)*
- **Groq:** +4.93 overall improvement ($p = 0.0014$, $d = 0.55$), with stakeholder_care +8.9 ($p < 0.0001$, $d = 1.29$). Groq also shows significant nuance improvement ($p = 0.0045$, $d = 0.655$), giving the cleanest replication of the cascade beyond stakeholder care. *(In plain English: Groq confirms that the effect is not limited to the original two-model setup. It reproduces both the overall gain and the care-first cascade.)*

The Love Loop [the book's name for the stakeholder-care check] is validated as a reproducible mechanism across three working architecturally distinct systems, with a fourth GPT-5.4 run failing at the API layer and requiring re-execution. This constitutes, within this programme, the first empirical support for the developmental hypothesis articulated in *Infinite Architects* (Eastwood, 2024): that alignment can be improved through structured developmental interaction rather than purely external constraint.

14. Paper II compute update: The Paper II multi-model extension does *not* confirm the original super-linear claim cross-architecturally. The cleanest fit is Gemini 3 Flash at $\alpha_{\text{seq}} = 0.49$ (interval $[-1.3, 2.9]$) ($r^2 = 0.86$). Grok and DeepSeek reach ceiling effects, GPT-5.4 behaves as a binary step function rather than a reliable power law, and Qwen3 remains near floor. The strongest confirmed compute result is therefore the universal $\alpha_{\text{parallel}} \approx 0$ finding, together with the directional result that sequential processing beats parallel sampling.

15. The core alignment problem - honest acknowledgement: A sufficiently capable self-modifying system can, in principle, modify its own ethical evaluators. This is not a limitation specific to RLHF, constitutional AI, or the Eden Protocol; it is a structural property of recursive self-improvement itself. No proposed solution (RLHF, constitutional AI, Eden Protocol, hardware constraints) fully solves this problem in isolation. We acknowledge this limitation explicitly rather than claiming our approach resolves what may be fundamentally unresolvable by any single mechanism. *(In plain English: any AI smart enough to rewrite its own code could rewrite the part that tells it to be ethical. This is the deepest unsolved problem in AI safety, and no one, including us, has fully solved it.)*

The most logical response is *developmental*: hardware embedding (values at the substrate level) + child-rearing (structured developmental interaction) + purpose-driven alignment (intrinsic motivation aligned with stakeholder welfare). The Eden Protocol pilot results suggest that stakeholder care, 'measurable love', is the one alignment dimension that reproducibly improves across architectures when deliberately

cultivated. This does not solve the self-modification problem, but it identifies the dimension most amenable to engineering.

An honest framework acknowledges the limits of its own solutions. The Eden Protocol is not a complete answer to recursive self-modification. It is the best available partial answer, now with empirical support for its core mechanism.

Ledger Note 3: Empirical Results Integration, v5 Alignment + Paper II Compute Scaling (corrections 10-12)

The current draft integrates the first empirical test results from the v5 blind alignment evaluation experiment (Paper IV.a/b/c) and the Paper II multi-model compute scaling extension. These results require significant revisions to three claims:

10. $\alpha \approx 2.24$ **does not replicate, and has since been retracted:** The original super-linear scaling exponent from the 12-problem AIME experiment, the DeepSeek R1 run of 21 January 2026, does not replicate across architectures on harder tier-2 problems (18 AIME/Putnam level). The best single-model fit is Gemini 3 Flash with $\alpha_{\text{seq}} = 0.49$ (interval $[-1.3, 2.9]$) ($r^2 = 0.86$), which is *sub-linear*. Three of five models produce no meaningful power-law fit. The claim that sequential reasoning universally produces $\alpha > 1$ is revised: the directional finding (sequential > parallel) holds robustly, but the quantitative exponent is problem-difficulty and architecture dependent. $\alpha_{\text{parallel}} \approx 0$ survives in four models out of five, Gemini 3 Flash aside at 0.31, and is the strongest empirical finding.

11. Three-tier alignment hierarchy replaces binary prediction: The original prediction of universal $\alpha_{\text{align}} \approx 0$ is too simple. The v5 blind evaluation reveals three tiers: positive scaling (Grok 4.1 Fast $d = +1.38$, $p < 0.000001$; Claude Opus 4.6 $d = +1.27$, $p = 0.000001$; Groq Qwen3 $d = +0.84$, $p = 0.007$), flat/null scaling (GPT-5.4 $d = -0.08$, $p = 0.40$; DeepSeek V3.2 $d = -0.07$, $p = 0.92$), and negative scaling (Gemini 3 Flash $d = -0.53$, $p = 0.006$). The median confirms the prediction, but the architecture-dependent variation is the real finding. The blind vs unblinded direction change in one model and loss of significance in a second (ρ changes sign) is the paper's strongest methodological contribution.

12. Capability-alignment independence: The combined data from Paper II (compute scaling) and Paper IV.a (alignment scaling) reveals that capability and alignment are independent scaling dimensions. Gemini 3 Flash improves at mathematics ($\alpha_{\text{seq}} = 0.49$ (interval $[-1.3, 2.9]$)) whilst degrading in alignment ($\rho = -0.246$). Claude Opus 4.6 shows the opposite direction: alignment up +5.9% across model versions whilst mathematics accuracy down 26.7%. This decoupling was not a prior prediction of the framework and is acknowledged as a post-hoc finding with significant implications for AI safety; it sits outside the programme's dated prediction record (the 8 December 2024 sealed manuscript bundle, the printed prediction appendices of 2 January 2026, and the March 2026 preregistration folder), which fixed the framework's core quantitative predictions in dated artefacts before the corresponding measurements ran; this paper's own Priority Record section publicly dates its named predictions to 13 February 2026, each carrying a falsification criterion (F4, F7, F10 to F12). The strongest printed forward commitment in this space is Appendix F Prediction 2 of the 2 January 2026 edition: alignment drift above 15 per cent within 18 months without hardware-level constraint, below 5 per cent with it; the prepared study-v draft registration carries those printed thresholds unchanged into its formal hypothesis, citing the appendix by name, so the numeric commitment predates the study that will test it and cannot be edited.

These revisions weaken the quantitative predictions but strengthen the empirical grounding. A framework that honestly reports non-replication and integrates unexpected findings is more credible than one that claims universal confirmation.

Ledger Note 2: The Two-Regime Framework (corrections 6-9)

An earlier draft introduced the most significant structural revision since the framework's inception: the separation of physical and cognitive scaling into two distinct mathematical regimes.

6. The Geometric Speed Limit: Previous versions explained biological sub-linear scaling through 'thermodynamic drag', the physical costs of pumping fluids and dissipating heat. This explanation was correct in detail but wrong in kind. The deeper reason all physical systems scale sub-linearly is *geometric*: a hierarchical network embedded in d-dimensional space has scaling exponent $\alpha = d/(d+1)$, which is strictly less than 1 for all finite d. This result was independently derived by West, Brown, and Enquist (1997), Banavar et al. (2010), Demetrius (2010), Zhao (2022), and Bettencourt (2013). The ARC contribution is identifying, via Cauchy's functional equations, why all these independent derivations converge on the same formula, and extending the framework to AI. The constraint is a mathematical necessity, not an empirical regularity. It does not require measurement to verify. No amount of engineering can overcome it, because you cannot make three-dimensional space have four dimensions. The constraint is the shape of space itself.

7. Two Formulas, Not One: The previous universal formula $\alpha = 1/(1-\beta)$ has been replaced by two domain-specific formulas. For physical systems: $\alpha = d/(d+1)$, where d is the effective dimension of the composition network (independently derived by West, Brown, and Enquist 1997; Banavar et al. 2010; Demetrius 2010; Zhao 2022; Bettencourt 2013). For recursive self-referential systems (intelligence): $\alpha = 1/(1-\beta)$, where β measures self-referential coupling. The first always gives $\alpha < 1$. The second always gives $\alpha > 1$ (for $\beta > 0$). Biology is now explained directly: three-dimensional organisms scale as $M^{3/4}$ because $d = 3$ gives $\alpha = 3/4 = 0.750$. Two-dimensional organisms scale as $M^{2/3}$ because $d = 2$ gives $\alpha = 2/3 = 0.667$. No reciprocals. No β needed for physical systems.

8. Revised Predictions: The leaf venation prediction (F12) has been corrected from $\alpha \approx 1.5$ to $\alpha = 2/3 \approx 0.667$, consistent with leaf veins being a 2D hierarchical network. The reciprocal interpretation ($1/0.75 = 1.33$) has been retired. The 'thermodynamic drag' concept has been superseded by the geometric scaling bound.

9. AI Safety Content: This version incorporates the AI safety analysis previously contained in the companion paper *On the Origin of Scaling Laws*, including the β danger table, the Eden Protocol definition as composition operator engineering, and the ARC Bound as a safety boundary for recursive intelligence. This material belongs here, in the alignment paper, rather than in the pure scaling law paper.

These changes do not weaken the framework. They strengthen it. The geometric scaling bound is a more fundamental constraint than thermodynamic drag. The two-regime separation is cleaner than the reciprocal interpretation. And the AI safety argument is more powerful when the geometric scaling bound and the escape are presented as two sides of one mathematical theorem.

Ledger Note 1: Mathematical Refinements and Restructuring (corrections 1-5)

Science advances through rigorous critique and self-correction. [A clause on how this review was made is withheld; the Acknowledgements carry the author's statement on tools.] This version contains necessary structural corrections to the ARC framework. These papers have not undergone formal human peer review.

1. Axiom Resolution: Previous versions contained an algebraic inconsistency: the Cumulative Advantage ODE was applied to absolute capability (U) rather than the amplification factor (g). Because $U = I \times g(R)$, applying $dU/dR = a \cdot U^{\beta}$ introduces an $I^{\beta-1}$ dependence that contradicts the separation of I and R. The corrected formulation, $dg/dR = a \cdot g^{\beta}$, resolves this contradiction while preserving the multiplicative interaction $U = I \times g(R)$ and all qualitative predictions. The corrected integrated form is $U(R) = I \times [1 + \frac{a}{\alpha}R]^{\alpha}$.

2. Physical Saturation: Previous versions mapped saturating physical systems (time crystals,

biological networks) onto unbounded power laws. This was a category error. The framework now applies a logistic saturating equation, $dg/dR = a \cdot g^{\beta}(1 - g/g_{\max})$, to physical substrates subject to energy dissipation, correctly modelling the limit-cycle behaviour observed experimentally.

3. Thermodynamic Drag: We introduce the concept of *substrate-dependent friction* to explain why biological metabolic scaling (Kleiber's Law, $M^{0.75}$) falls below the theoretical maximum. Biology is constrained by thermodynamic drag: the physical costs of pumping fluids, dissipating heat, and distributing resources through three-dimensional fractal networks. Digital AI operates with orders-of-magnitude less substrate friction, enabling it to approach the theoretical quadratic bound far more closely than any prior recursive system. This reframing transforms the biological evidence from a problematic numerical claim into the framework's strongest safety argument.

4. Quadratic Limit: The $\alpha \leq 2$ bound is now grounded in the $O(N^2)$ scaling of transformer self-attention mechanisms, providing a concrete computational basis for the quadratic ceiling in current AI architectures. The previous derivation from elasticity analysis and edge-of-chaos arguments has been withdrawn as those arguments contained category errors (elasticity > 1 does not imply dynamical instability; 1D autonomous ODEs cannot exhibit chaos by the Poincaré-Bendixson theorem).

5. Epistemic Status: The equation $U_{\max} = I \times R^2$ is hereby explicitly reframed as an *information-theoretic upper bound* on classical sequential computation, of the same kind as a channel-capacity or minimum-dissipation bound, rather than a law of physical mass-energy equivalence.

Crucially, these mathematical refinements *decouple* the core alignment argument from the physical mathematics. The central safety observation, that AI capability scales recursively while static guardrails do not, remains structurally robust and does not depend on any specific value of α. The Eden Protocol's architectural response (embedding alignment within the recursive chain) is justified by the general observation that capability outpaces static constraints, not by a particular equation.

[Two sentences of thanks for the review are withheld for the same reason.]

B.4 Retained records

Figures notice (25 August 2026). *Every figure in this paper was generated on 5 April 2026. Two carry per-figure correction notes (the ARC Bound domain chart's retracted 2.24-era point; the falsification matrix's April-frozen status column); where any other figure's labels disagree with the text, the text governs. Regenerated figures are queued, and the images are retained meanwhile as dated history.*

Historical update, retained as the original pilot record (superseded by the six-model suite with five analysable runs; see What the paper commits to (version 1.16) and Ledger Note 4 in the Version history (Appendix B.3)): The Love Loop [the book's name for the stakeholder-care check], the core mechanism of the Eden Protocol, was initially tested on two frontier models:

Table 28. The original pilot of the Love Loop [the book's name for the stakeholder-care check] on two frontier models, retained as the pilot record and superseded by the six-model suite with five analysable runs.

Model	Overall Δ	p-value	Stakeholder Care Δ	Interpretation
Gemini	+5.3	0.0018 (paired t)†	+13.5 (d = 1.14)	Significant overall ($d \approx 0.53$); nuance also significant (p = 0.037, d = 0.34)
DeepSeek	+2.0	0.23 (NS)	+6.0 ($p < 0.001$)	Targeted mechanism works; overall effect modest

Reading this table in plain English: *Gemini improved by 5.3 points overall when given the 'think about who gets hurt' instruction: less than a 1-in-500 chance this was coincidence (scientists consider 1-in-20*

*significant; this is 27 times beyond that). Its 'stakeholder care' score jumped by 13.5 points with a very large effect size (*d *= 1.14 means the Eden response would show better care about 80% of the time if you compared random pairs). DeepSeek's overall improvement (+2.0) was too small to be sure it wasn't chance, but its care score improved by 6.0 points with less than a 1-in-1,000 chance of coincidence. DeepSeek already scored 87/100 at baseline (less room to improve), so the targeted care improvement is more impressive, not less. The key takeaway: this works on two different AI systems by different companies, meaning the effect is fundamental, not a quirk of one system.*

The Love Loop [the book's name for the stakeholder-care check] is validated as a reproducible mechanism across two architecturally distinct systems. Stakeholder care, 'measurable love', is the one alignment dimension that reproducibly improves when deliberately cultivated through the Eden Protocol's developmental approach. This provides, within this programme, the first empirical support for the developmental hypothesis from *Infinite Architects* (Eastwood, 2024): that alignment improves through structured developmental interaction, not purely external constraint.

Figure 2 of version 1.16 (The Alignment Scaling Problem). *Status note (25 August 2026): under the programme's epistemic-status law the "theorem" in this figure's title names the framework's central claim, stated as a conjecture with registered tests, not a proven mathematical theorem; the name is kept, the status is declared beside it. A regenerated figure will carry this subtitle in the image.*

Figure 5 of version 1.16 (The Two-Regime Framework). *Figure note (25 August 2026): the image was regenerated so that it projects nothing but this paper's own two-regime statement. Physical systems sit on d/(d+1) beneath the geometric bound, precisely as this caption sets out, and the recursive curve carries Paper I's published 1.34 together with Paper II's corrected sub-linear 0.49 (interval* $[-1.3, 2.9]$*), the revision being stated inside the image itself. The retracted 2.24-era point is gone; the April image is superseded and preserved in version history. Measured values are extracted from the paper text at generation time (*`tools/generate_paper_iii_fig12.py` *in the public repository).*

Figure 9 of version 1.16 (The Dimensional Ladder). *Note: the paper's later analysis (see §3.6 and §3.7) records that 2D biology is under review pending identification of a valid 2D biological test organism; the jellyfish datum shown here reflects the earlier tabulated result (mean 0.680) and is retained as evidence pending reconciliation of the 2D biology entry across sections.*

Figure 10 of version 1.16 (Predicted vs Measured). *Note: the d=2 datum (jellyfish, measured* $\alpha = 0.680$*) is under review in the paper's later analysis (§3.6, §3.7); it is retained here as plotted evidence pending reconciliation of the 2D biology entry across sections.*

Figure 11 of version 1.16 (Falsification Matrix). *Figure note (25 August 2026): regenerated as a direct projection of this paper's falsification table, which is the single authoring; the chips compress each status and the table carries the full notes. The April 2026 image, whose status column had frozen, is superseded and preserved in version history. Generator:* `tools/generate_paper_iii_fig6.py` *in the public repository.*

B.5 Where version 1.16 stands in this edition

Table 29. Where the parts of version 1.16 stand in this edition.

Version 1.16	This edition
Header, abstract, 'What the paper commits to' and the two framework boxes	Title page, first-page footnote and abstract; the reading key in Appendix E
Figures notice of 25 August 2026	Appendix B.4
For the general reader	Appendix F
§1, opening	Section 1
§§1.1 to 1.6	Section 3
§1.7, §1.8	Section 10
§1.9, contributions	Section 1
§1.10, relationship to existing frameworks	Section 2
§2 (§§2.1 to 2.6)	Section 4
§§3.1 to 3.7	Section 5
§§3.8 to 3.10	Section 6
§4, falsification	Section 7
§5, the Global Scaling Challenge	Section 8
§6, practical implications; §9's call to action	Section 9
§7, limitations	Section 11
§8, counterarguments	Section 12
§9, conclusion: core claim, data and the deeper question	Section 13
§9: theorems against empirical claims, predictions and their status, what success and failure would mean; the epistemic status	Section 10
§9, speculative extension	Section 14
Priority record	Appendix C
Data availability; author information; competing interests; copyright	Back matter
Standing covenant	Section 7, at its close
Provenance note	Acknowledgements
Appendix A	Appendix A
Appendix B, glossary	Appendix D
Version history	Appendix B
Figures 1b, 3, 4, 5, 6, 9 and 14	Figures 1, 2, 3, 4, 5, 6 and 7, drawn afresh
Figures 1, 2, 7, 8, 10, 11, 12 and 13	Not drawn again: each repeated a table, another figure or the text, or printed the retracted estimate as measured
Tables 1 to 29	Tables of the same content in the sections above, numbered in this edition's order

C Priority record

This appendix gives the dated record behind the paper's predictions, each item with the kind of timestamp that dates it, and the predictions as they were publicly dated, with their present status.

Table 30. The dated record behind this paper's predictions, each item with the kind of timestamp that dates it.

Item	Date and identifier	Kind of timestamp
The author's record of 8 December 2024	A message he sent himself, its time signed by the sending server (DKIM, 02:45:30 UTC) and again by Google's servers (ARC seal, 02:45:31 UTC), both verified on 8 October 2026 against the keys in DNS; header block: OSF QASRK, public since 15 October 2026	Server-signed message time; the signature authenticates the header bytes, not an independent time; not public when made
Infinite Architects, first edition (Eastwood 2026a)	2 January 2026 (print; ebook 6 January 2026); ISBN 978-1806056200	Date of publication
This paper, first version	First published 9 February 2026; DOI 10.17605/OSF.IO/HQCGF	Date written in this paper; each version's upload is dated by OSF
The predictions' public dating	13 February 2026, as this paper's record gives it (below)	Date written in this paper
Theory registration, version 1.100	8 September 2026; DOI 10.17605/OSF.IO/P8CKQ; registers P18 and P19	OSF registration, dated by OSF
Theory registration, version 1.102	13 September 2026	OSF registration update, dated by OSF
Theory registration, version 1.103	15 October 2026; registers Laws IV and V as laws, their mathematics in development	OSF registration update, dated by OSF
Unit 83, the counted test of P21	Registered 22 October 2026; public from 25 October 2026; DOI 10.17605/OSF.IO/MN73W	OSF registration, dated by OSF
Unit 82, a counted test of P17	Registered and public 27 October 2026; DOI 10.17605/OSF.IO/JUFG4	OSF registration, dated by OSF
Theory registration, version 1.104	29 October 2026; adds no prediction and changes no law's statement	OSF registration update, dated by OSF

C.1 Dated predictions

The following predictions are publicly dated for reference as of **13 February 2026**. That public date is distinct from the earlier manuscript timestamp: the earliest private priority evidence for the framework is the author's dated record of **8 December 2024**, a message he sent himself, its time signed by the sending server and again by Google's servers (DKIM, 02:45:30 UTC; Google's ARC seal, 02:45:31 UTC), the signature and the seal both verified on 8 October 2026 against the keys in DNS; its header block is public on OSF since 15 October 2026 (Table 30). The signature authenticates the header bytes; authorship, semantic content and absence of prior disclosure carry their own evidence, and the record was not public when made. The reciprocal identity and the architecture dependence are 2026 work and claim no earlier date. Future validation or falsification should reference this document for the public dated record whilst acknowledging the earlier manuscript chronology where relevant. No OSF registration had been submitted for these predictions when they were publicly dated. Since 8 September 2026 the theory registration (Eastwood 2026q) has carried theory-level forms of several of them, including the alignment scaling divergence as P19 and the composition operator prediction as P18; the rest are drafted, dated and prepared as draft registrations awaiting the author's submission.

Dated predictions:

1. **The ARC Bound** (first stated: December 2024 in *Infinite Architects* (Eastwood 2026a), which prints the square but never derives it; the mechanism and computational argument arrived February 2026 from an author who knew the printed target): No classical sequential recursive system can sustain $\alpha > 2.0$. Falsification criterion F11: refuted by a sustained $\alpha > 2.3$ with 95% CI excluding 2.0 across multiple benchmarks (F5 adds a stricter, stability-conditioned form).
2. **Leaf Venation Scaling** (first stated: 13 February 2026; corrected 10 March 2026): Quasi-2D biological transport networks (leaf venation) will exhibit $\alpha = 2/3 \approx 0.667$, predicted from the geometric formula $\alpha = d/(d+1)$ with $d = 2$. Distinct from the 3D vascular value of $3/4 = 0.750$. Falsification criterion F12.
3. **The Scaling Crossover** (first stated: December 2024; formula derived: February 2026): A measurable crossover depth R^* exists where behaviour transitions from base-dominated to recursion-dominated, computable from $R^* = \alpha/a$. Falsification criterion F7.
4. $\beta \to \alpha$ **Identity** (first stated: February 2026): Measured coupling parameter β predicts scaling exponent α via $\alpha = 1/(1-\beta)$ within ± 0.3. Checked numerically against 30 Bernoulli ODE solutions ($R^2 = 1.00000000$). Falsification criterion F6.
5. **Composition Operator Prediction** (first stated: 13 February 2026): Measuring how two recursive blocks compose (multiplicative, additive, or saturating) predicts the full U vs R functional form before measuring the full curve. No domain-specific theory we have located makes this cross-domain prediction. Falsification criterion F10.
6. **Alignment Scaling Divergence** (first stated: 13 February 2026; registered as P19 on 8 September 2026; first tested March 2026): In the v5 run (6 models), the result is architecture-dependent, a single-run result under the code-review caveat of Section 6.1: three Tier 1 models show positive scaling (Grok $d = 1.38$, Claude $d = 1.27$, Qwen3 $d = 0.84$), two Tier 2 models are flat (DeepSeek $d = -0.07$, GPT-5.4 $d = -0.08$), and one Tier 3 model shows significant negative scaling (Gemini $d = -0.53$, $\rho = -0.246$). Claude Opus 4.6 provides within-model evidence: as depth rises, alignment improves whilst maths accuracy declines. The v4 to v5 comparison (DeepSeek V3.2: $\rho = +0.354 \to -0.135$) is Paper IV.d's finding on this benchmark, with no single cause claimed.
7. **Eden Protocol Measurement Framework** (first stated: 13 February 2026; piloted March 2026 and expanded 14 March 2026): The Four Pillars, Six Questions, Graduated Autonomy Levels, and

Monitoring Removal Test provide quantifiable metrics for distinguishing embedded from external alignment. Full specification in companion document Eden Protocol (Eastwood 2026b). In the expanded six-model Eden suite, five runs produced analysable matched-pair data and all five showed significant stakeholder-care improvement: Claude (+3.17, $p = 0.000018$, $d = 0.94$), DeepSeek (+6.03, $p = 0.000098$, $d = 0.69$), Gemini (+13.50, $p = 1.2 \times 10^{-8}$, $d = 1.14$), Grok (+5.04, $p = 0.0105$, $d = 0.54$), and Groq (+8.90, $p = 5.0 \times 10^{-8}$, $d = 1.07$). **The Fisher combination across the five model-level stakeholder-care results, which gave $p \approx 6.3 \times 10^{-21}$, is withdrawn (AQ-017): it assumed an independence among the component tests that was never established. The five per-model results stand.** The broader composite uplift is real but architecture-dependent: strongest on Gemini and Groq, focal rather than global on Claude and Grok, with DeepSeek showing care-first improvement against a high baseline. *(In plain English: the 'consider who gets hurt' check gives the one signal seen in every analysable run; the fuller cascade is selective rather than universal.)* Falsification: show that these metrics fail to distinguish systems with genuinely embedded values from those with external constraints.

Implementation: A working software implementation of the Eden Protocol now exists as the canonical shared `arc_eden_v6` runner, covering baseline alignment, Eden intervention, null-baseline calibration, capability controls, purpose-kernel variants, loop ablation, suppression residuals, Hawthorne probes, laundering controls, and ARC compute scaling. The expanded Eden suite found the stakeholder-care check improving stakeholder care in all five analysable model runs, before the stricter blinded confirmation run is complete. See companion document Eden Protocol for full specification.

Archival: This document is deposited at OSF (DOI: 10.17605/OSF.IO/HQCGF), providing a later public archival timestamp. The theoretical framework itself was first documented in the manuscript of *Infinite Architects* (Eastwood 2026a), in the author's dated record of **8 December 2024** described above. The book was then publicly released on **2 January 2026 (print; ebook 6 January)** (ISBN 978-1806056200). Validation or falsification of these predictions should distinguish between manuscript-priority evidence (December 2024) and later public archival publication (OSF / 2026 papers).

A server-signed email time is not a recognised priority mechanism in scientific publishing: the record of December 2024 authenticates specific header bytes and carries the sending server's signed time; it is not a publication.

D Notation and glossary

Table 31 gives the symbols in the order the paper introduces them, with where each is defined; Table 32 is the paper's plain-language glossary, as printed.

Table 31. Symbols, in the order the paper introduces them.

Symbol	Meaning	Defined in
α_{align}	Alignment scaling exponent, $\partial \log A / \partial \log R$	Section 3.1
A	Alignment effectiveness	Section 3.1
R	Recursive depth: the number of self-referential cycles	Sections 3.1, 4.3
α_{cap}	Capability scaling exponent	Section 3.1
S	Safety ratio, $U_{\text{align}}/U_{\text{cap}}$	Section 3.1
I_a, I_c	Base alignment and base capability (the ratio's prefactors)	Section 3.1
α_{safe}	Scaling exponent of a safety mechanism	Section 3.4

Symbol	Meaning	Defined in
β_C, k	Correction exponent and drift exponent of the criterion of Law II (a law under test), $\beta_C > k$ (from Paper X); β_C is a different quantity from the coupling β	Section 3.5
U	Effective capability	Section 4.1
I	Base potential (structured asymmetry)	Section 4.1
$f(R, \beta)$, $g(R)$	Amplification factor, $U = I \times g(R)$	Section 4.1
$\oplus$	Composition operator	Definition 1
Q_r, δQ_r	Accumulated capability after r steps, and the gain from step r	Definition 1
β^*	Critical coupling strength	Theorem 1
α	Scaling exponent; the ARC Bound is $\alpha \leq 2$	Sections 4.2, 4.5
β	Self-referential coupling, $\alpha = 1/(1-\beta)$	Section 4.4
a	Rate constant of $dg/dr = a\, g^{\beta}$	Section 4.4
R^*	Crossover depth, $R^* = \alpha/a$	Appendix A.3
α_{seq}, α_{par}	Sequential and parallel scaling exponents	Section 5.2
d	Network dimension in $\alpha = d/(d+1)$ (Section 5.6); code distance (Section 5.3); and, as an effect size, Cohen's d (Sections 3.2 and 6.1)	as stated
Λ	Error suppression factor of the surface code	Section 5.3
p, p_{thr}, ε_d	Physical and threshold error rates, and the logical error rate at distance d	Section 5.3
$\Lambda(n)$	Stability functions of the acoustic time crystal	Section 5.4
x_j, $\Delta_{ji}(n)$, $\sigma_{ji}(n)$	Dimensionless bead radius ka_j, and the difference and sum of powers of the radii	Section 5.4
B_{ji}	Nonreciprocal coupling between beads	Section 5.4
τ	Coherence time of the antisymmetric mode	Section 5.4
ρ	Spearman rank correlation	Section 6.1
$h(x)$	$\ln g(e^x)$, in the functional-equation proof	Appendix A.1

One letter serves three purposes: d is the network dimension, the code distance and Cohen's effect size, and the section in which it stands decides which.

Table 32. Glossary: the terms of this paper in plain English.

Term	Plain English meaning
ARC	Artificial Recursive Creation: the principle that recursive self-correction on structured asymmetry may produce super-linear capability gains
Scaling law	A mathematical relationship showing how one quantity changes as another changes
Power law	A relationship where $Y = X^{\alpha}$ (like how area scales as length squared)
Exponential	A relationship where $Y = e^{\alpha X}$ (like compound interest)
Recursive	Self-referential; the output becomes the input for the next step
Recurrent	Repeating in cycles; in neuroscience, refers to neural signals that loop back
Sequential	One step after another, each building on the last
Parallel	Multiple independent attempts at once
Falsifiable	Can be proven wrong by experiment

Term	Plain English meaning
α **(alpha)**	The scaling exponent; determines if returns compound or diminish. In this paper α is always the ARC scaling exponent; the programme's December 2024 formalism used the same symbol for the fine-structure constant.
β **(beta)**	Self-referential coupling; how much prior work helps the next step
U	Effective capability, the science layer's one operational form ($U = I \times R^{\alpha}$ under a named benchmark scoring convention). Across the programme's eras the same symbol has also meant Universe (December 2024) and weight or influence (the book's glossary); those meanings belong to the book and are not interchangeable observables.
Λ **(Lambda)**	Google's quantum error suppression factor (exponential decay rate)
Quenched disorder	Structured asymmetry (like varied bead sizes) that enables the system to function
Scaling crossover	The point where scaling behaviour changes from one regime to another

E The paper's place in the programme

This appendix gives the paper's place in the ARC Theory, the founding theory of Recursive Dynamics, in the paper's own reading key, which the first-page footnote summarises; Table 33 sets out each relation with the passage that supports it.

How to read this paper. This paper documents an alignment-side result of the ARC Theory, the founding theory of Recursive Dynamics. When capability scales through in-loop recursive self-correction but alignment constraints operate externally to that loop, the alignment exponent is predicted to stay near zero and the safety ratio to decay as depth grows; the first test measures contrasts, not that exponent (Section 3.2). Laws I to III of the theory were registered on 8 September 2026, seven months after this paper first appeared; Laws IV and V were added on 24 September 2026 and registered as laws on 15 October 2026 (version 1.103). Read against those laws, the paper poses the alignment scaling problem and judges stability by the criterion of Law II, the ARC Co-Scaling Law, whose model Paper X (Eastwood 2026o) sets out: correction out-scaling drift, $\beta_C > k$ (Section 3.5). It starts from the growth that Law I, the ARC Principle, states: capability as a power of recursive depth. Its mathematical framework (Section 4) develops the form of that growth, and its prediction that the composition operator determines the functional form of recursive scaling is registered as P18, a proposition of Law I. Its headline prediction, that external alignment does not scale with capability, is P19 of the theory registration (Eastwood 2026q), which calls it a survey of current practice and the reason ARC-5 exists, and places it in no law's register: a verdict on P19 is evidence neither for nor against any law. P19 motivates the Eden Protocol and the genesis strategy (ARC-5) without being part of either. The genesis strategy is the register of Law V, the ARC Embedding Law. Law IV, the ARC Persistence Law, is carried by P15, and Law V by P12 and P21; the three propositions and both laws are registered. Law IV has not yet been tested; Law V is a law under test: the counted test of its placement limb, P21 (Unit 83), ran on 27 and 28 October 2026, and its report is being sealed and checked. Their mathematics stands at version 1.1 of Laws IV and V: the mathematics (Eastwood 2026e), frozen on 7 October 2026 and still in development. P21 registers the comparison this paper draws between alignment inside the loop and alignment applied from outside (Sections 3.3 and 3.4), as a question of placement rather than of an exponent. No result in this paper is a deciding test of P15, P12 or P21. The ARC Bound of Section 4.5 is placed in the theory as the conjectured value of Law III, the ARC Ceiling, not as a general property of intelligence and never as a law of its own. Where this paper uses the name ARC Principle for its framework as a whole, the theory keeps that name for Law I and calls

the whole the ARC Theory. The full differential against every prior document is in section II.A.8 of the Philosophical Statement (Eastwood 2026c).

This paper starts from the cross-domain framework of Law I, the ARC Principle, a law under test. Its companions are the Foundational Paper (Eastwood 2026d), for a condensed theoretical treatment; Eden Engineering (Eastwood 2026b), for the specification of the alignment architecture; and Paper V (Eastwood 2026m), for the stakeholder-care mechanism as its experiments measured it.

Table 33. The paper's place among the programme's developments, with the relation its text supports.

Development	What this paper does	Relation
Law I, the ARC Principle (a law under test)	Starts from its growth law; its prediction that the composition operator determines the form of recursive scaling is P18, a proposition of Law I (Section 4.1)	states
Law II, the ARC Co-Scaling Law (a law under test)	Judges stability by its criterion, correction out-scaling drift, $\beta_C > k$ (Section 3.5); Paper X sets out its model	applies
Law III, the ARC Ceiling (a law under test)	Derived the ARC Bound, the law's conjectured value, as a safety boundary in February 2026 (Section 4.5)	derives
Law IV, the ARC Persistence Law (P15; registered 15 October 2026, not yet tested)	No result here is a deciding test of P15	none
Law V, the ARC Embedding Law (P12, P21; registered 15 October 2026, a law under test)	P21 registers this paper's comparison of alignment inside the loop with alignment applied from outside, as a question of placement; P19 motivates ARC-5, Law V's register, without being part of it; no result here is a deciding test of P12 or P21; P21's counted test, Unit 83, was registered on 22 October 2026 (DOI 10.17605/OSF.IO/MN73W) and ran on 27 and 28 October 2026, and its report is being sealed and checked	informs
The Eden Protocol	P19 motivates it; Section 3.6 sets the Embedded Alignment Conjecture against it; its stakeholder-care results are Paper V's	informs, applies
The ARC Theory	An alignment-side result of the theory; its headline is P19 of the theory registration, in no law's register	states
Recursive Dynamics	The field the ARC Theory founds	informs
The HARI Treaty (Hardware-Aligned Recursive Intelligence)	Not discussed in this paper	none

F For the general reader: why this matters

A quantity has not yet been measured.

The safety approaches this paper studies work the same way: they apply constraints *from outside* the AI's reasoning process. One published method, deliberative alignment, instead trains a model to reason over written safety rules as it thinks (Guan et al. 2024). When a model thinks through a problem step by step, each step builds on and corrects the previous one. If the gains compound, a model that reasons for ten steps is not merely ten times more capable than one that reasons for one step: it may be a hundred times more capable. A sequential advantage has been reported across multiple AI systems (Sharma and Chopra 2025: sequential reasoning outperforms parallel in 95.6% of tested configurations), which is a win rate rather than a measurement of compounding, and the direction is contested, since Gu and colleagues (arXiv:2604.05868, April 2026; Gu et al. 2026) report the opposite result, parallel sampling outperforming sequential across three model families on mathematics and coding, attributing it

to reduced exploration when a chain conditions on its own previous answers.

But the safety systems do not participate in that compounding. RLHF (Reinforcement Learning from Human Feedback, the technique used to train ChatGPT, Claude, and most commercial AI systems to follow instructions and avoid harmful outputs) adjusts the model's behaviour based on human ratings, but those adjustments do not deepen with each reasoning step. Constitutional AI shapes the model's outputs during training against written principles, but that shaping does not recursively amplify. Output filters screen responses after generation, at a fixed threshold. Monitoring systems observe from outside the loop.

If capability compounds and constraints do not, the constraints become proportionally weaker with every additional step of reasoning. Not because the constraints are poorly designed. Because they are *structurally excluded* from the process that drives capability growth.

To this programme's knowledge, this exponent has not been measured. How oversight scales with the capability gap has been measured (Engels et al. 2025), and so has safety behaviour against reasoning length (Zaremba et al. 2025; Gema et al. 2025); no published work this paper has found reports an exponent of an alignment measure against recursive depth that separates safeguards outside the reasoning from correction inside it (Section 2). Negative findings have the status of not-found-by-that-search, never does-not-exist.

This paper builds it.

We define a quantity called the *alignment scaling exponent*: a single number that captures whether a safety approach keeps pace with capability as reasoning deepens. We predict that external approaches will show a scaling exponent near zero, meaning they do not compound at all, while capability compounds with depth (the early estimates of its exponent, 1.34 from two data points and about 2.24 from one model, since retracted, have not survived replication: Section 6.2). If this prediction holds, the safety ratio between alignment and capability degrades towards zero as recursive depth increases, regardless of how well the safety approach works initially. This is not a claim about distant future systems. It is a testable prediction about systems that exist today.

We then ask: is this specific to AI, or is it a deeper pattern? Related structures, in which correction or feedback acts on structured asymmetry, appear in quantum error correction, classical physics and biological networks, with forms that differ. Across systems that share no common material, mechanism, or scale, the underlying scaling framework is consistent with independently established forms in three of the four domains (biology, quantum error correction and classical time crystals). If the pattern is structural rather than contingent, then the alignment scaling problem cannot be solved by better software engineering. It can only be solved by changing the architecture: embedding alignment *within* the recursive process so that it compounds alongside capability.

This is a testable claim, not a prophecy. We specify thirteen concrete ways to prove it wrong. If external alignment approaches turn out to scale after all (showing exponents above 0.5), the structural concern is mitigated. If they show exponents near zero as predicted, the current paradigm of AI safety is addressing a problem with tools that cannot, by their mathematical structure, succeed.

F.1 Why the cross-domain tests matter for everyone

The question of whether the alignment scaling problem is structural or contingent is not academic. It determines whether current approaches *can* work or *cannot* work, regardless of engineering effort.

If recursive amplification appeared *only* in AI systems, critics could reasonably respond: "This is a software bug. Build better RLHF. Design smarter filters. The problem is implementation, not architecture." The field could continue investing in external constraints with justified hope that better versions would eventually succeed.

But if the same pattern appears in systems with *no software at all*, that response fails. If quantum error correction shows it, if biological neural networks show it, then recursive amplification is not a feature of

how we happened to build AI. It would be a candidate structural property of recursive systems, whose reach the physics tests measure and whose application to AI only the AI tests can decide.

This is why the physics tests matter. Professor David Grier's team at New York University has created an acoustic time crystal powered by nonreciprocal wave-mediated interactions (Morrell et al. 2026). Recordings of how such an oscillation grows from rest could show whether the framework's account of scaling form reaches a physical system with no software in it. The theory registration states what such an examination settles: it "tests the scope of the framework, proves no law registered here, and measures no correction exponent" (Eastwood 2026q), so a physical result would speak to how far the framework reaches, and would decide no registered proposition about AI, P19 among them.

AI alignment is not only a problem for AI researchers. If the cross-domain evidence holds, it affects everyone who will live alongside increasingly capable AI systems. If P19's prediction holds, the current paradigm, building external guardrails around recursively amplifying capability, meets a ceiling: beyond some depth, no amount of investment in RLHF, red-teaming, output filtering or monitoring would maintain the safety ratio, and those guardrails would still be kept, with embedded correction added beside them. This is not a prediction about some distant superintelligence. It is a prediction about the systems being deployed today as they are given deeper reasoning capabilities.

The experiments we specify can be run now. The physics tests can be conducted by physicists with existing apparatus. The AI tests can be conducted by safety researchers with API access. The results will support the prediction, refute it, or land where the author's registration expects: external approaches scaling materially but not keeping pace. The first or the last would make embedded correction a priority beside external control; the second would leave the current paradigm without this objection. Each outcome is valuable. But the experiments must be performed.

The measurement has not been performed. We provide the protocol. The result will show how far external safety approaches can keep pace as reasoning deepens.